

CS311 Assignment 1 | 2026 Sem 1

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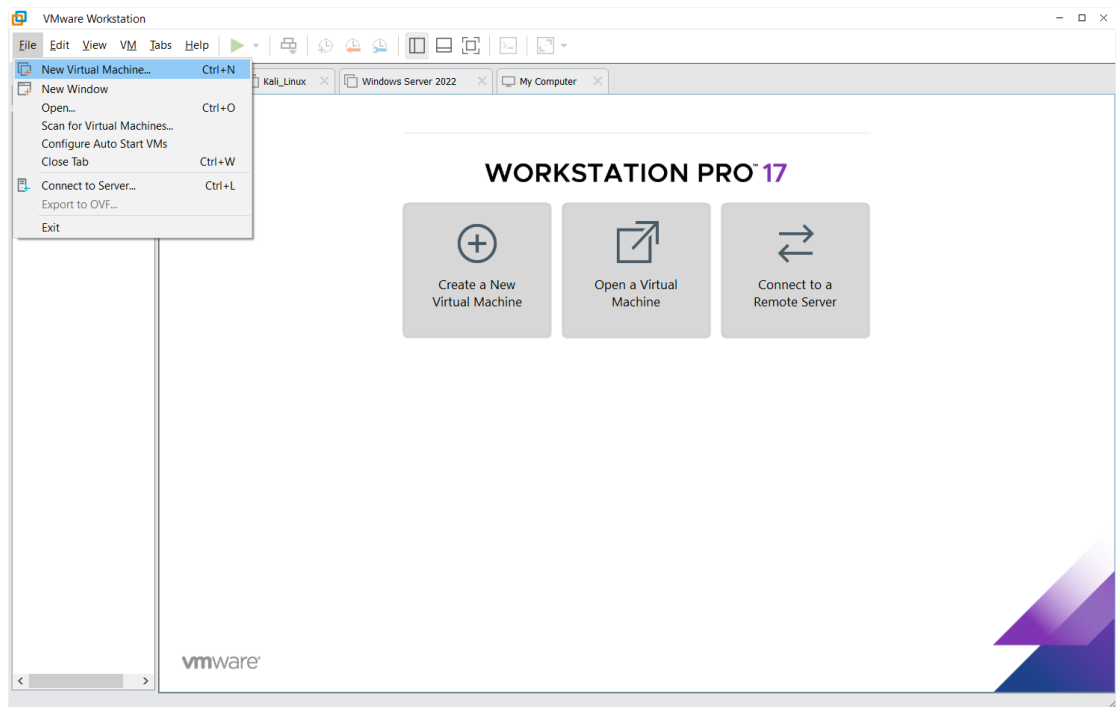
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Part A

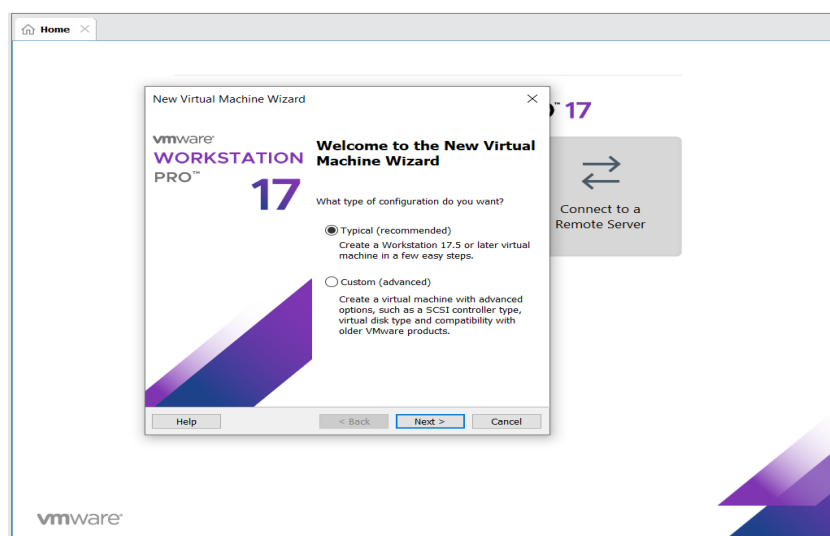
1. Installation of systems

a. Installation of Kali Linux

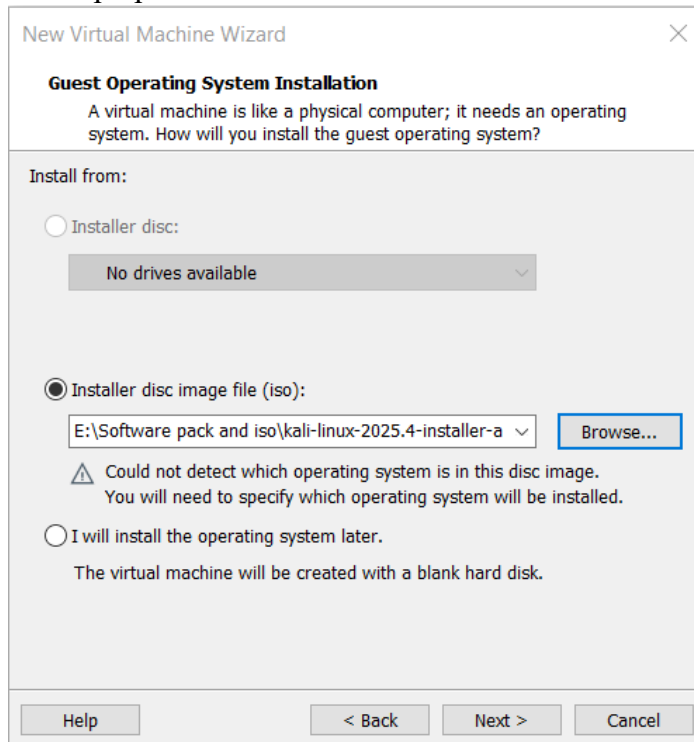
1) Creating a new virtual machine



2) Click on next



3) Select prepared ISO file



The screenshot shows the 'New Virtual Machine Wizard' window, specifically the 'Guest Operating System Installation' step. The window title is 'New Virtual Machine Wizard' with a close button (X) in the top right corner. Below the title bar, the section is titled 'Guest Operating System Installation' with a subtitle: 'A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?'. The main area is labeled 'Install from:' and contains three radio button options. The first option is 'Installer disc:', which is currently unselected; below it is a dropdown menu showing 'No drives available'. The second option is 'Installer disc image file (iso):', which is selected. Below this option is a text box containing the path 'E:\Software pack and iso\kali-linux-2025.4-installer-a' and a 'Browse...' button. A warning triangle icon is present next to the text: 'Could not detect which operating system is in this disc image. You will need to specify which operating system will be installed.' The third option is 'I will install the operating system later.', which is unselected; below it is the text: 'The virtual machine will be created with a blank hard disk.' At the bottom of the window are four buttons: 'Help', '< Back', 'Next >', and 'Cancel'.

New Virtual Machine Wizard

Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐ Installer disc:
No drives available

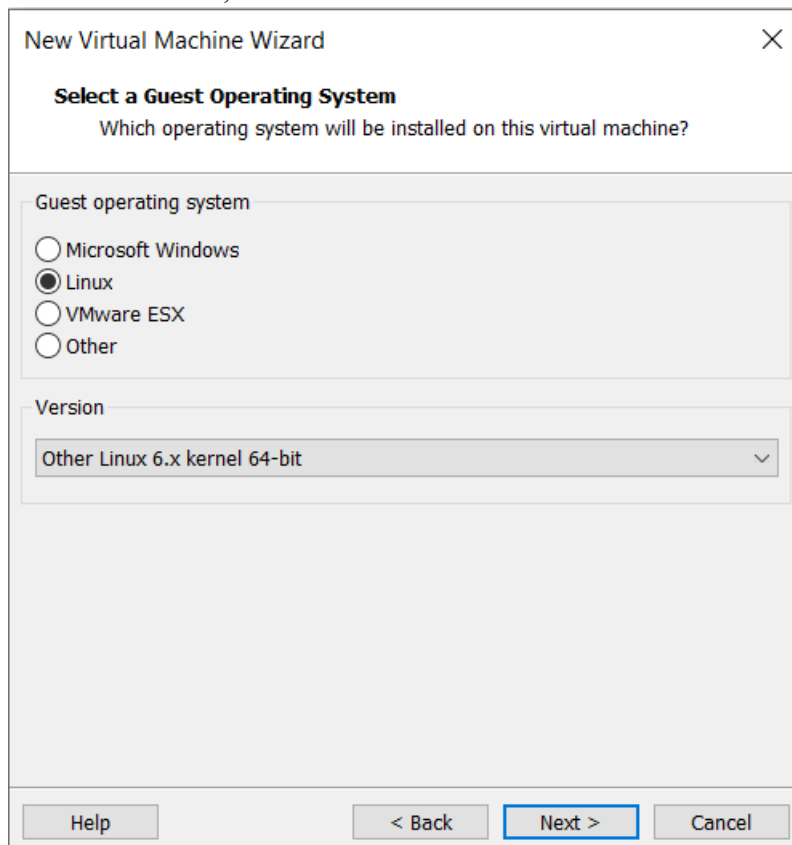
☒ Installer disc image file (iso):
E:\Software pack and iso\kali-linux-2025.4-installer-a Browse...

⚠ Could not detect which operating system is in this disc image.
You will need to specify which operating system will be installed.

☐ I will install the operating system later.
The virtual machine will be created with a blank hard disk.

Help < Back Next > Cancel

4) Set operating system as Linux and then choose the corresponding kernel version of the system prepared before. The kernel version of Kali Linux used here is 6.16, choose the Linux 6.x kernel



The screenshot shows the 'New Virtual Machine Wizard' window, specifically the 'Select a Guest Operating System' step. The window title is 'New Virtual Machine Wizard' with a close button (X) in the top right corner. Below the title bar, the section is titled 'Select a Guest Operating System' with a subtitle: 'Which operating system will be installed on this virtual machine?'. The main area is labeled 'Guest operating system' and contains four radio button options: 'Microsoft Windows', 'Linux' (which is selected), 'VMware ESX', and 'Other'. Below this section is a section labeled 'Version' with a dropdown menu showing 'Other Linux 6.x kernel 64-bit'. At the bottom of the window are four buttons: 'Help', '< Back', 'Next >', and 'Cancel'.

New Virtual Machine Wizard

Select a Guest Operating System
Which operating system will be installed on this virtual machine?

Guest operating system

☐ Microsoft Windows
☒ Linux
☐ VMware ESX
☐ Other

Version
Other Linux 6.x kernel 64-bit

Help < Back Next > Cancel

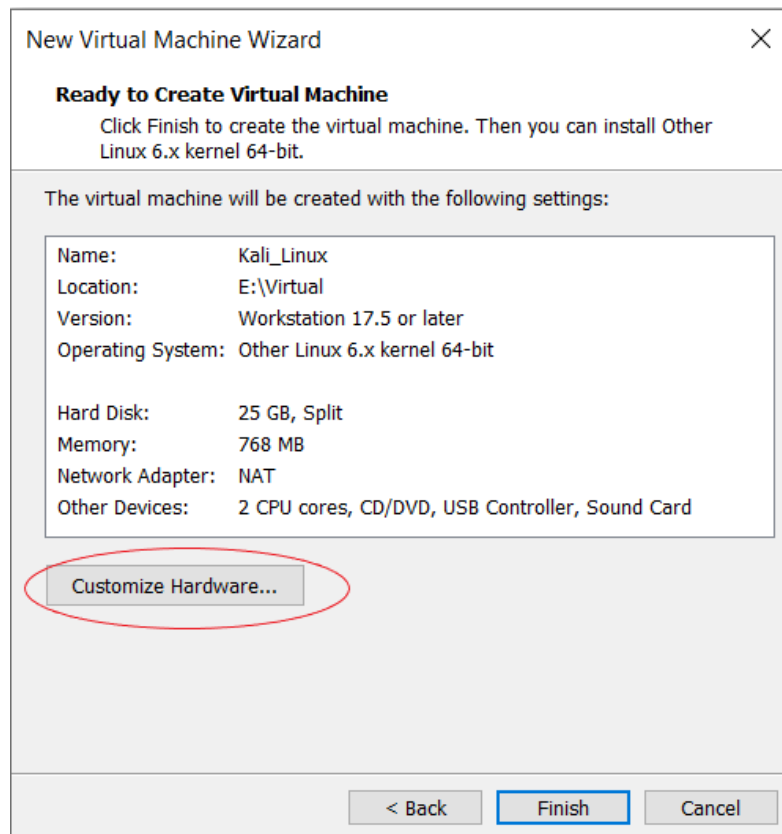
- 5) Name the virtual machine and configure the path to store it

The screenshot shows the 'New Virtual Machine Wizard' dialog box. The title bar says 'New Virtual Machine Wizard' with a close button. The main heading is 'Name the Virtual Machine' with the subtitle 'What name would you like to use for this virtual machine?'. Below this, there is a text box for 'Virtual machine name:' containing 'Kali_Linux'. Underneath is a 'Location:' section with a text box containing 'E:\Virtual' and a 'Browse...' button. A note states: 'The default location can be changed at Edit > Preferences.' At the bottom, there are three buttons: '< Back', 'Next >', and 'Cancel'.

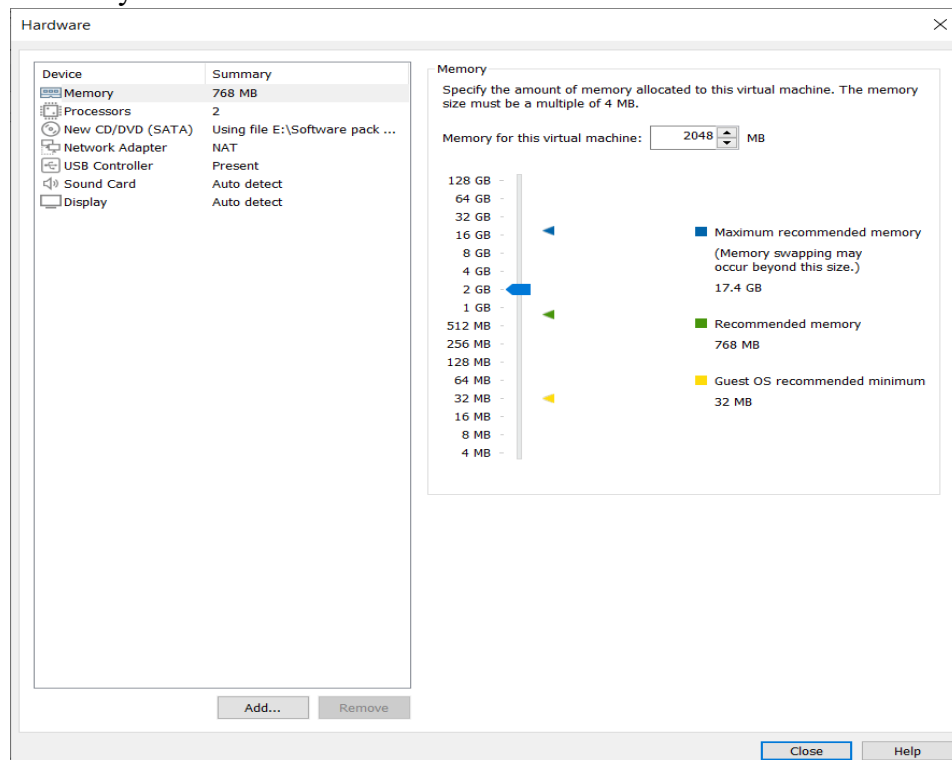
- 6) Configure the disk size, the minimum storage requirement for kali Linux is 25GB

The screenshot shows the 'New Virtual Machine Wizard' dialog box, Step 2: 'Specify Disk Capacity'. The title bar says 'New Virtual Machine Wizard' with a close button. The main heading is 'Specify Disk Capacity' with the subtitle 'How large do you want this disk to be?'. Below this, there is explanatory text: 'The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.' There is a 'Maximum disk size (GB):' label followed by a spinner box set to '25'. Below that, it says 'Recommended size for Other Linux 6.x kernel 64-bit: 8 GB'. There are two radio button options: 'Store virtual disk as a single file' (unselected) and 'Split virtual disk into multiple files' (selected). A note below the selected option says: 'Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.' At the bottom, there are four buttons: 'Help', '< Back', 'Next >', and 'Cancel'.

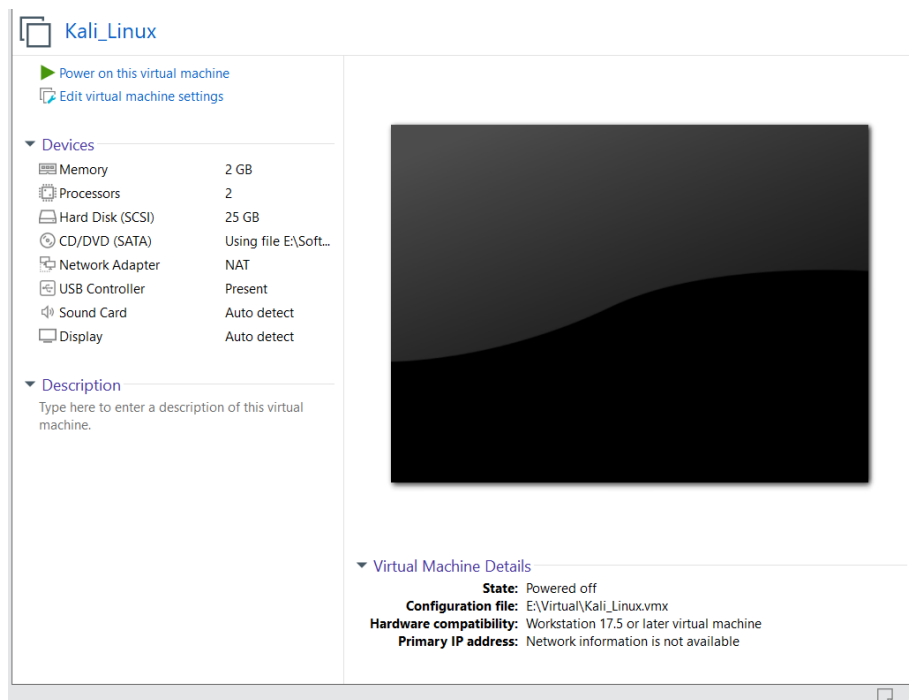
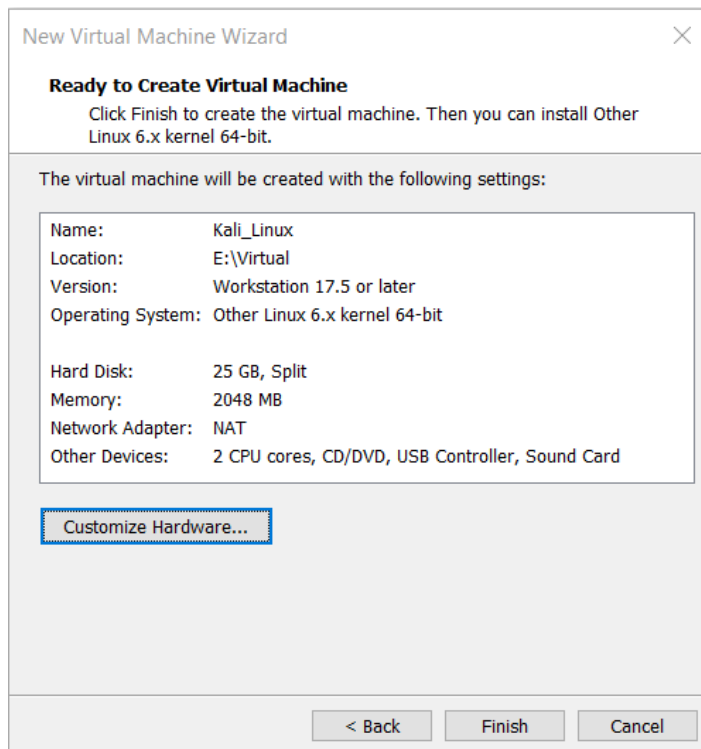
7) Click on customize hardware



8) Configure memory size. The minimum memory requirement for Kali Linux is 2GB. Leave the network adapter set to the default NAT since it is necessary to install Samba. Click on close.



9) Click on Finish to complete the virtual machine configuration



10) Turn on the virtual machine, choose graphical install by press enter



11) Select a preferred language and location



Select a language

Choose the language to be used for the installation process. The selected language will also be the default language for the installed system.

Language:

Chinese (Simplified)	- 中文(简体)
Chinese (Traditional)	- 中文(繁體)
Croatian	- Hrvatski
Czech	- Čeština
Danish	- Dansk
Dutch	- Nederlands
Dzongkha	- ཇོང་ཁ།
English	- English
Esperanto	- Esperanto
Estonian	- Eesti
Finnish	- Suomi
French	- Français
Galician	- Galego
Georgian	- ქართული
German	- Deutsch

[Screenshot](#) [Go Back](#) [Continue](#)



Select your location

The selected location will be used to set your time zone and also for example to help select the system locale. Normally this should be the country where you live.

This is a shortlist of locations based on the language you selected. Choose "other" if your location is not listed.

Country, territory or area:

India
Ireland
Israel
New Zealand
Nigeria
Philippines
Seychelles
Singapore
South Africa
United Kingdom
United States
Zambia
Zimbabwe
other

[Screenshot](#) [Go Back](#) [Continue](#)

- 12) Select a keymap that matches keyboard layout of physical keyboard. After Click on continue, there will be a loading time



Configure the keyboard

Keymap to use:

- American English
- Albanian
- Arabic
- Asturian
- Bangladesh
- Belarusian
- Bengali
- Belgian
- Berber (Latin)
- Bosnian
- Brazilian
- British English
- Bulgarian (BDS layout)
- Bulgarian (phonetic layout)
- Burmese
- Canadian French

[Screenshot](#) [Go Back](#) [Continue](#)

- 13) After loading, enter host name of the system



Configure the network

Please enter the hostname for this system.

The hostname is a single word that identifies your system to the network. If you don't know what your hostname should be, consult your network administrator. If you are setting up your own home network, you can make something up here.

Hostname:

[Screenshot](#) [Go Back](#) [Continue](#)

- 14) Left the Domain Name blank here, it is essential for local network communication. Communication between systems will be enabled by using static IP addressing (10.0.0.x/8) later



Configure the network

The domain name is the part of your Internet address to the right of your host name. It is often something that ends in .com, .net, .edu, or .org. If you are setting up a home network, you can make something up, but make sure you use the same domain name on all your computers.

Domain name:

Screenshot Go Back Continue

- 15) Enter username, according to the assignment requirement, set Student ID as account name



Set up users and passwords

Select a username for the new account. Your first name is a reasonable choice. The username should start with a lower-case letter, which can be followed by any combination of numbers and more lower-case letters.

Username for your account:

Screenshot Go Back Continue

16) Set password for user account



Set up users and passwords

Make sure to select a strong password that cannot be guessed.
Choose a password for the new user:

●●●●●●●●●●

☐ Show Password in Clear

Please enter the same user password again to verify you have typed it correctly.
Re-enter password to verify:

●●●●●●●●●●

☐ Show Password in Clear

[Screenshot](#) [Go Back](#) [Continue](#)

17) Choose time zone. Click on continue and wait for loading



Configure the clock

If the desired time zone is not listed, then please go back to the step "Choose language" and select a country that uses the desired time zone (the country where you live or are located).
Select your time zone:

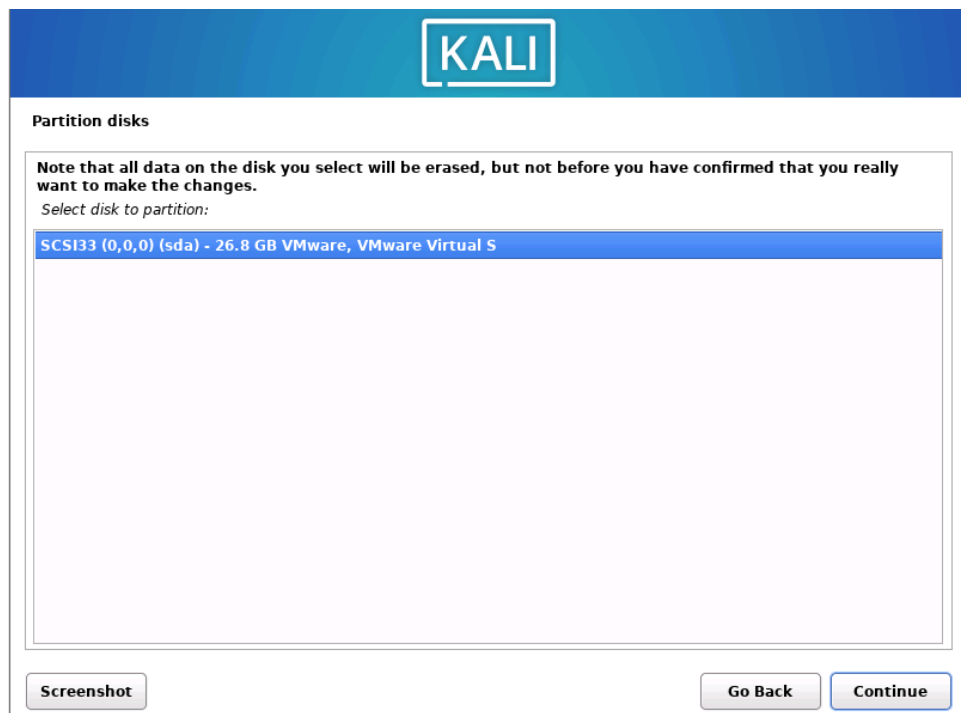
Eastern
Central
Mountain
Pacific
Alaska
Hawaii
Arizona
East Indiana
Samoa

[Screenshot](#) [Go Back](#) [Continue](#)

18) LVM is not necessary for this assignment, so select “Use entire disk” here



19) Click on continue



20) Select default option



Partition disks

Selected for partitioning:
SCSI33 (0,0,0) (sda) - VMware, VMware Virtual S: 26.8 GB

The disk can be partitioned using one of several different schemes. If you are unsure, choose the first one.
Partitioning scheme:

- All files in one partition (recommended for new users)**
- Separate /home partition
- Separate /home, /var, and /tmp partitions
- Separate /var and /srv, swap < 1GB (for servers)
- Small-disk (< 10GB) partitioning scheme

Screenshot **Help** **Go Back** **Continue**

21) Select write changes to disk to save partition setting



Partition disks

This is an overview of your currently configured partitions and mount points. Select a partition to modify its settings (file system, mount point, etc.), a free space to create partitions, or a device to initialize its partition table.

Guided partitioning

- Configure software RAID
- Configure the Logical Volume Manager
- Configure encrypted volumes
- Configure iSCSI volumes

▼ **SCSI33 (0,0,0) (sda) - 26.8 GB VMware, VMware Virtual S**

>	#1	primary	25.4 GB	f	ext4	/
>	#5	logical	1.4 GB	f	swap	swap

Undo changes to partitions

- Finish partitioning and write changes to disk**

Screenshot **Help** **Go Back** **Continue**

- 22) Choose yes to write changes into virtual disk unless configurations need to be modified. There will be another loading after this page.



Partition disks

If you continue, the changes listed below will be written to the disks. Otherwise, you will be able to make further changes manually.

The partition tables of the following devices are changed:
SCSI33 (0,0,0) (sda)

The following partitions are going to be formatted:
partition #1 of SCSI33 (0,0,0) (sda) as ext4
partition #5 of SCSI33 (0,0,0) (sda) as swap

Write the changes to disks?

☐ No
☒ Yes

- 23) Keep default options, click on continue and enter another loading time



Software selection

At the moment, only the core of the system is installed. The default selections below will install Kali Linux with its standard desktop environment and the default tools.

You can customize it by choosing a different desktop environment or a different collection of tools.
Choose software to install:

- ☒ Desktop environment [selecting this item has no effect]
 - ☒ ... Xfce (Kali's default desktop environment)
 - ☐ ... GNOME
 - ☐ ... KDE Plasma
- ☒ Collection of tools [selecting this item has no effect]
 - ☒ ... top10 -- the 10 most popular tools
 - ☒ ... default -- recommended tools (available in the live system)

- 24) There is no demand for dual-boot system, so choose yes to install GRUB and choose the default path to install GRUB.

KALI

Install the GRUB boot loader

It seems that this new installation is the only operating system on this computer. If so, it should be safe to install the GRUB boot loader to your primary drive (UEFI partition/boot record).

Warning: If your computer has another operating system that the installer failed to detect, this will make that operating system temporarily unbootable, though GRUB can be manually configured later to boot it.

Install the GRUB boot loader to your primary drive?

☐ No

☒ Yes

Screenshot

Go Back

Continue

KALI

Install the GRUB boot loader

You need to make the newly installed system bootable, by installing the GRUB boot loader on a bootable device. The usual way to do this is to install GRUB to your primary drive (UEFI partition/boot record). You may instead install GRUB to a different drive (or partition), or to removable media.

Device for boot loader installation:

Enter device manually

/dev/sda

Screenshot

Go Back

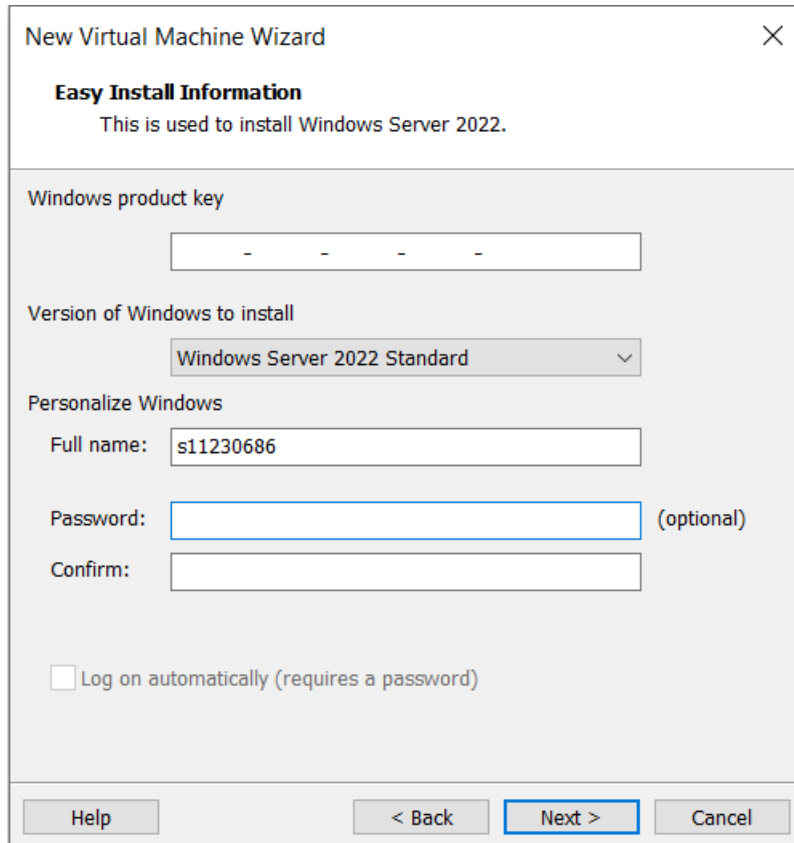
Continue

25) Once GRUB installation finish, the Kali Linux installation complete



b. Installation of Windows Server 2022 Desktop

- 1) Repeat step 1-3 above, select Windows Server 2022 ISO file prepared before
- 2) Configuration options here do not affect the windows system. Once the installation is complete, the system will prompt user to set a password. Just Click on next

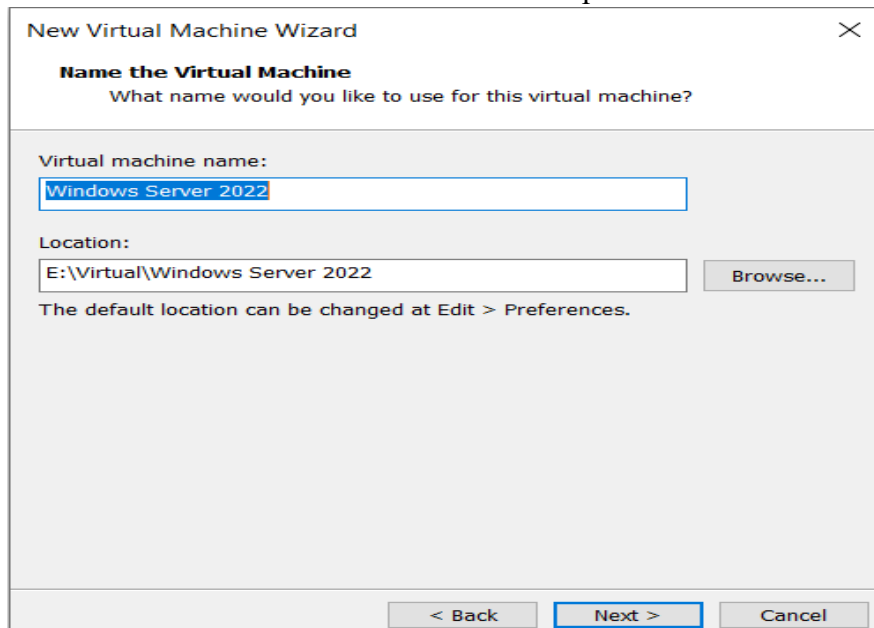


The screenshot shows the 'New Virtual Machine Wizard' window, specifically the 'Easy Install Information' step. The window title is 'New Virtual Machine Wizard' with a close button (X) in the top right corner. Below the title bar, the text 'Easy Install Information' is displayed, followed by 'This is used to install Windows Server 2022.' The main content area contains the following fields and options:

- Windows product key:** A text input field with four dashes (- - - -) as a placeholder.
- Version of Windows to install:** A dropdown menu showing 'Windows Server 2022 Standard'.
- Personalize Windows:**
 - Full name:** A text input field containing 's11230686'.
 - Password:** A text input field, followed by '(optional)'.
 - Confirm:** A text input field.
 - ☐ Log on automatically (requires a password)

At the bottom of the window, there are four buttons: 'Help', '< Back', 'Next >' (highlighted with a blue border), and 'Cancel'.

- 3) Set the virtual machine's name and where to put it

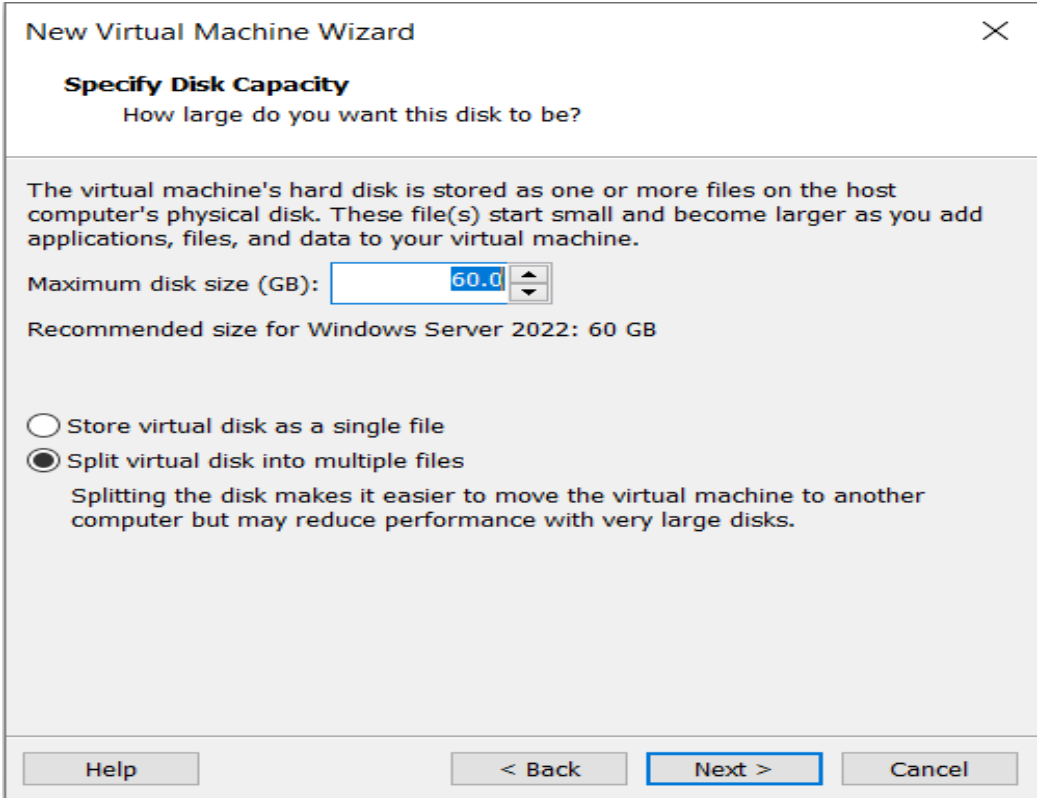


The screenshot shows the 'New Virtual Machine Wizard' window, specifically the 'Name the Virtual Machine' step. The window title is 'New Virtual Machine Wizard' with a close button (X) in the top right corner. Below the title bar, the text 'Name the Virtual Machine' is displayed, followed by 'What name would you like to use for this virtual machine?'. The main content area contains the following fields and options:

- Virtual machine name:** A text input field containing 'Windows Server 2022'.
- Location:** A text input field containing 'E:\Virtual\Windows Server 2022', followed by a 'Browse...' button.
- The default location can be changed at Edit > Preferences.**

At the bottom of the window, there are three buttons: '< Back', 'Next >' (highlighted with a blue border), and 'Cancel'.

- 4) Set disk size to 60GB to improve the performance. The minimum system requirement for storage is 32GB



New Virtual Machine Wizard

Specify Disk Capacity
How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB):

Recommended size for Windows Server 2022: 60 GB

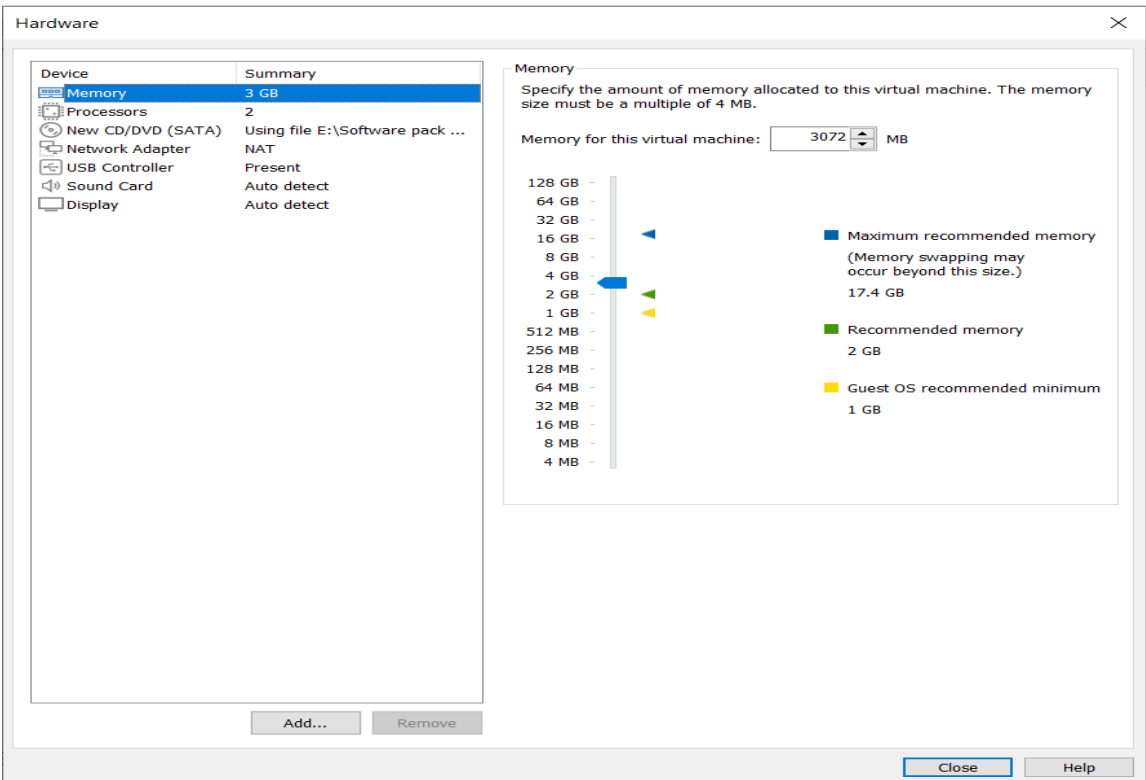
☐ Store virtual disk as a single file

☒ Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help < Back Next > Cancel

- 5) Set memory size to 3GB to improve performance. The minimum system requirement for RAM is 2GB



Hardware

Device	Summary
Memory	3 GB
Processors	2
New CD/DVD (SATA)	Using file E:\Software pack ...
Network Adapter	NAT
USB Controller	Present
Sound Card	Auto detect
Display	Auto detect

Add... Remove

Memory

Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.

Memory for this virtual machine: MB

128 GB -
64 GB -
32 GB -
16 GB -
8 GB -
4 GB -
2 GB -
1 GB -
512 MB -
256 MB -
128 MB -
64 MB -
32 MB -
16 MB -
8 MB -
4 MB -

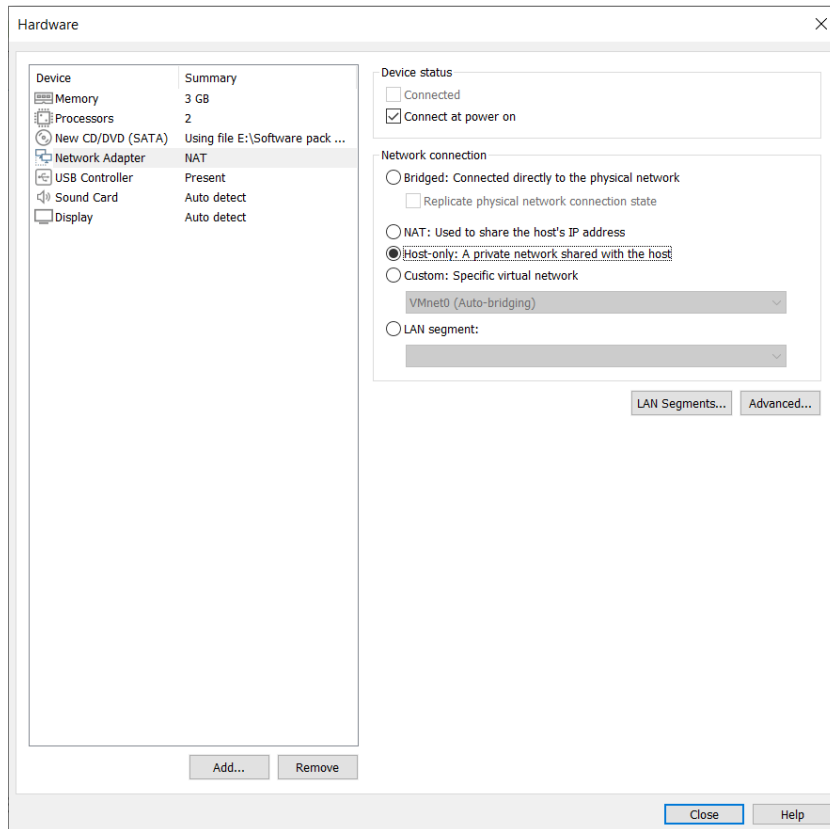
Maximum recommended memory
(Memory swapping may occur beyond this size.)
17.4 GB

Recommended memory
2 GB

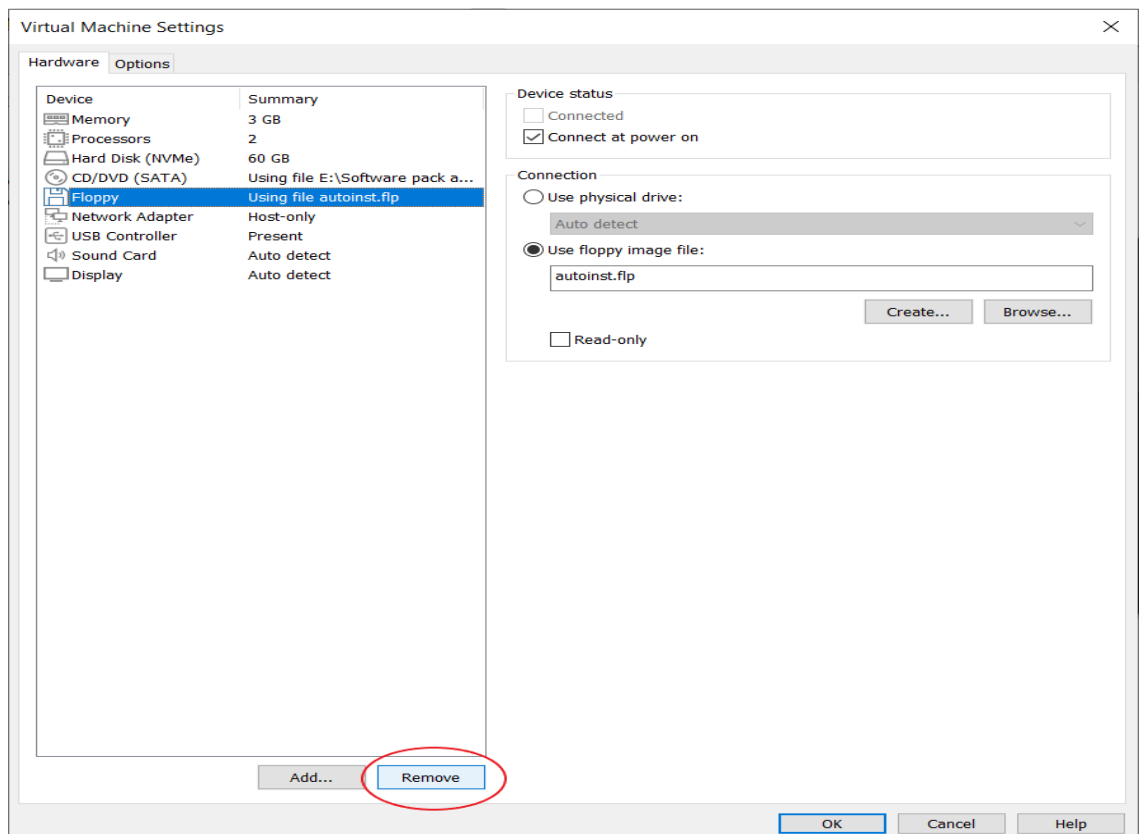
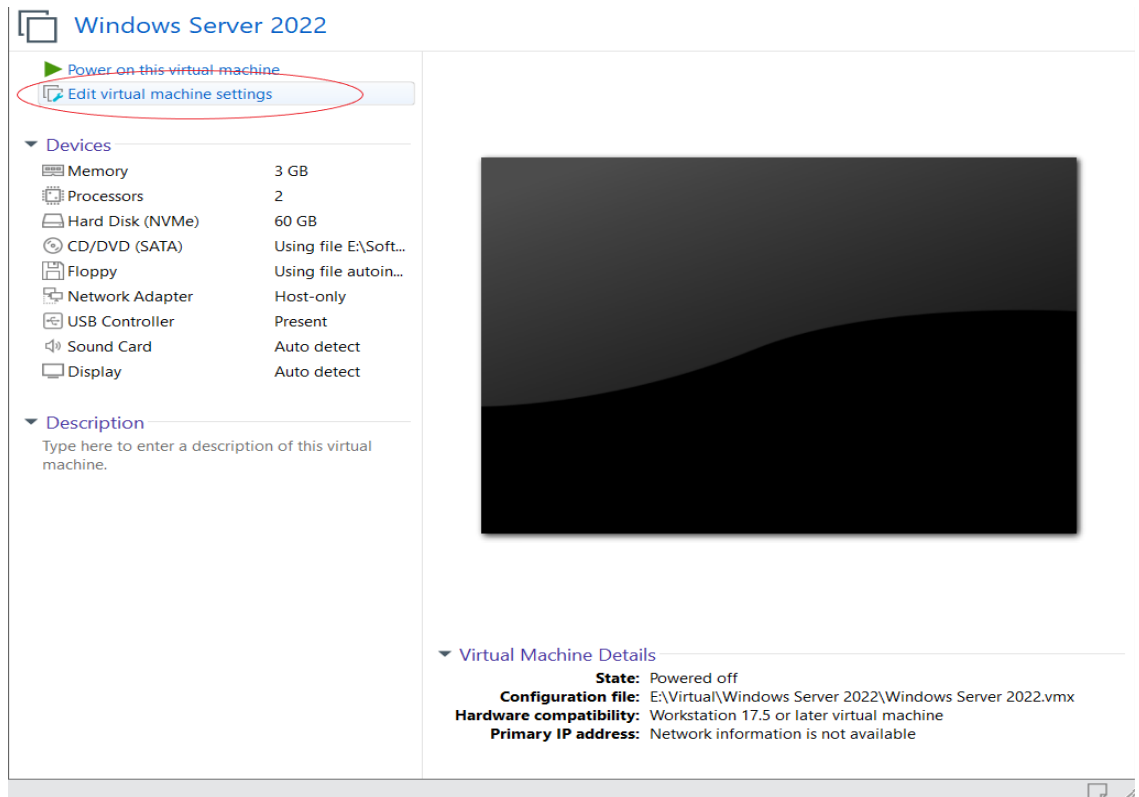
Guest OS recommended minimum
1 GB

Close Help

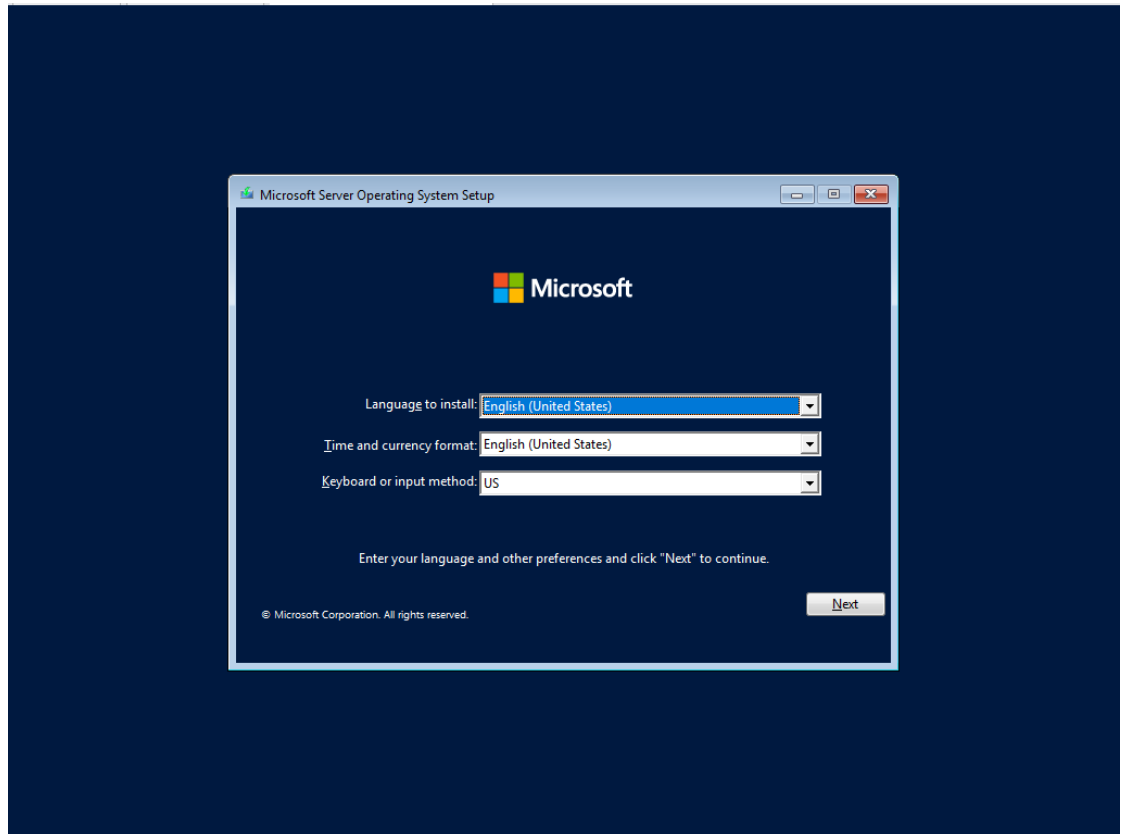
- 6) According to the assignment requirement, set adapter to Host-only. Click on Close.



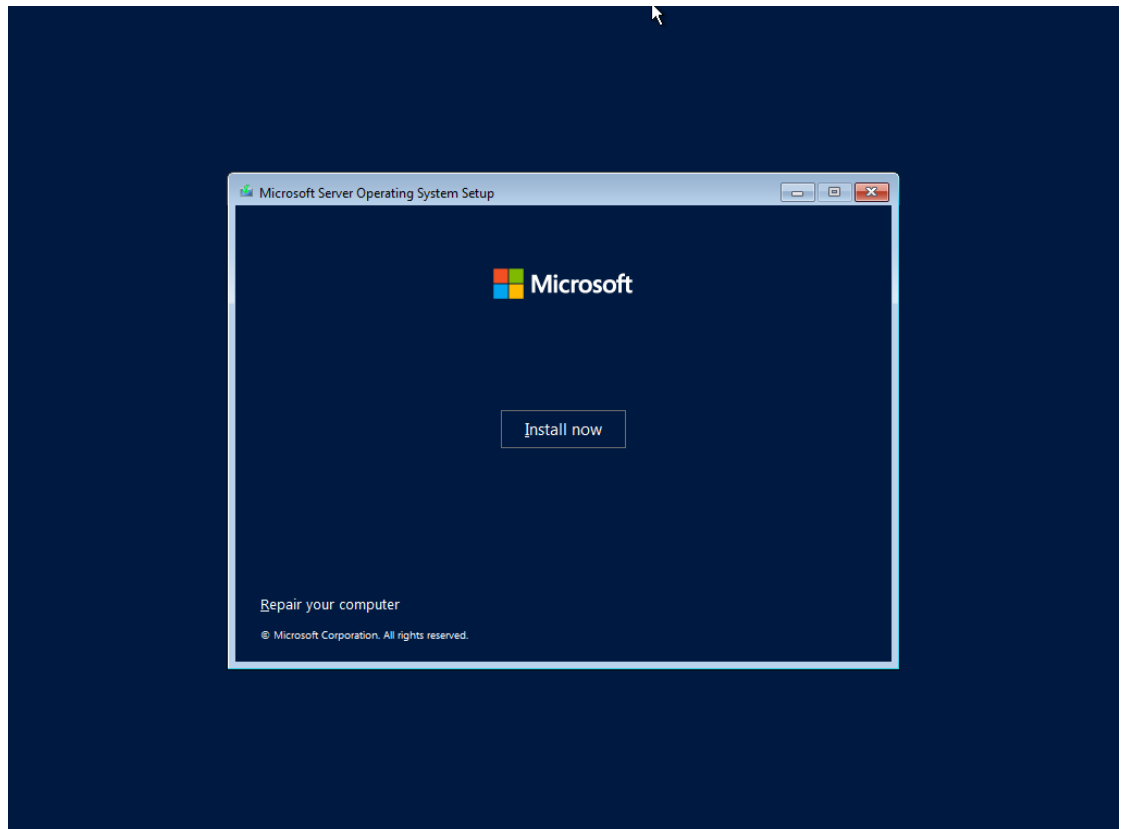
- 7) The virtual machine will automatically add a floppy disk; Remove it to avoid installation boot failure



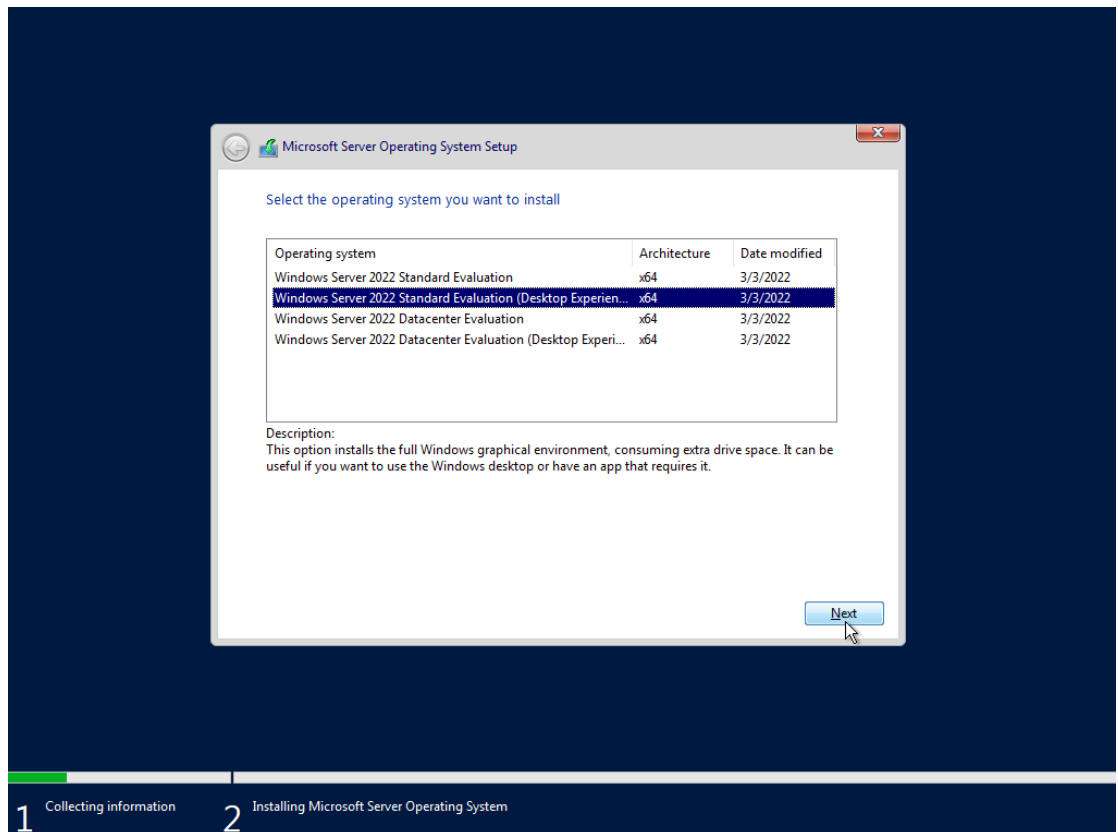
- 8) Turn on the virtual machine, when the following screen shows up, Click on Next



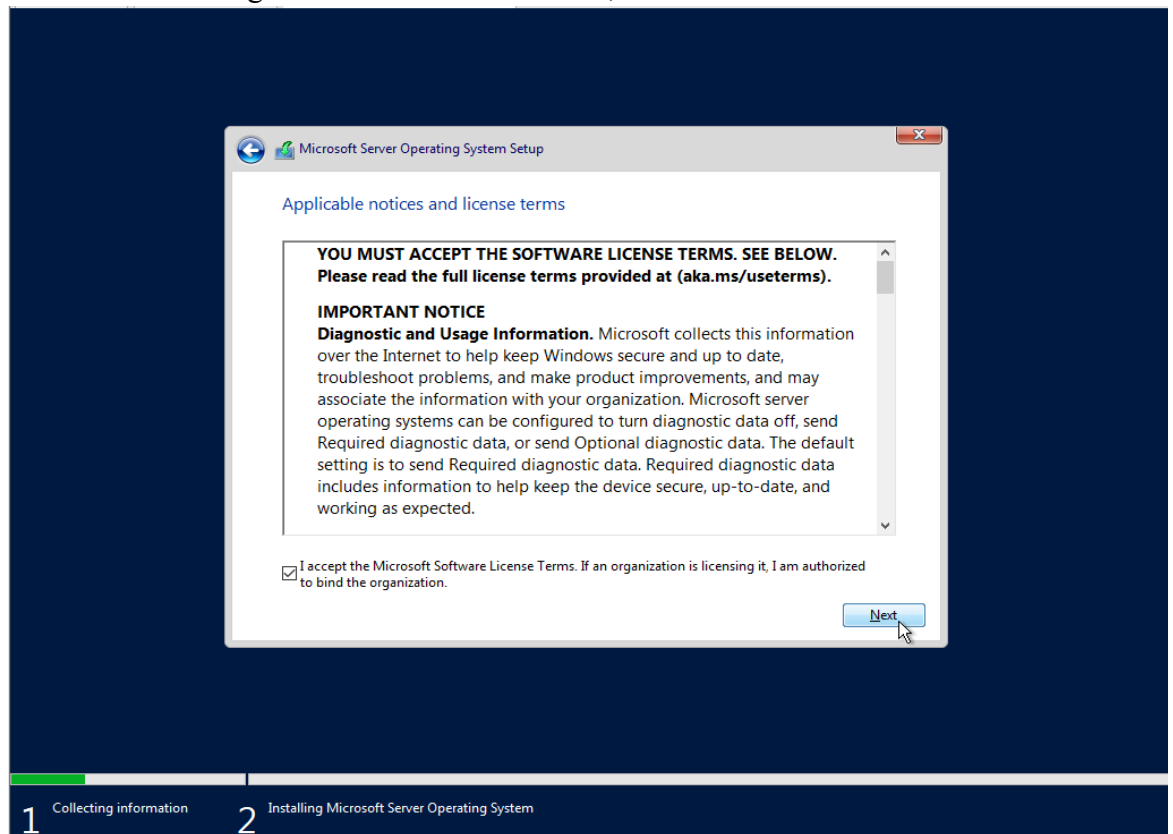
- 9) Click on Install Now



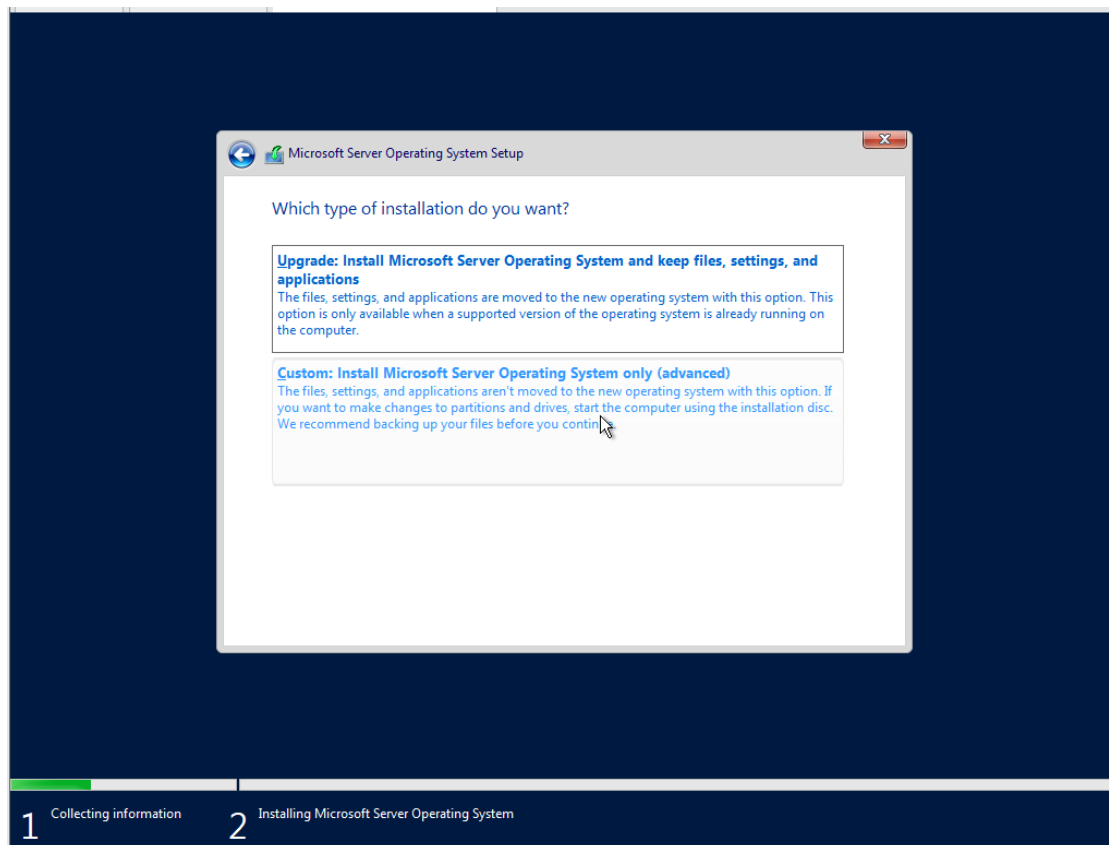
10) Select Standard Evaluation with Desktop, Click on Next



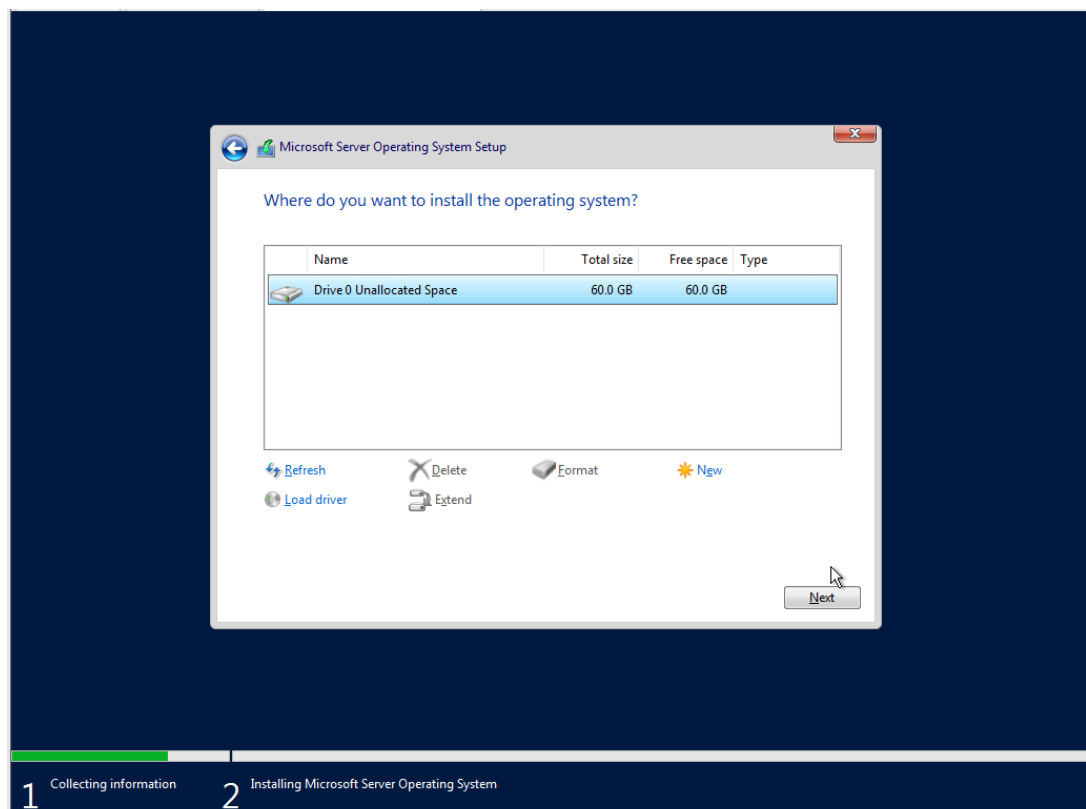
11) Check the box to agree to the Terms of Service, Click on Next



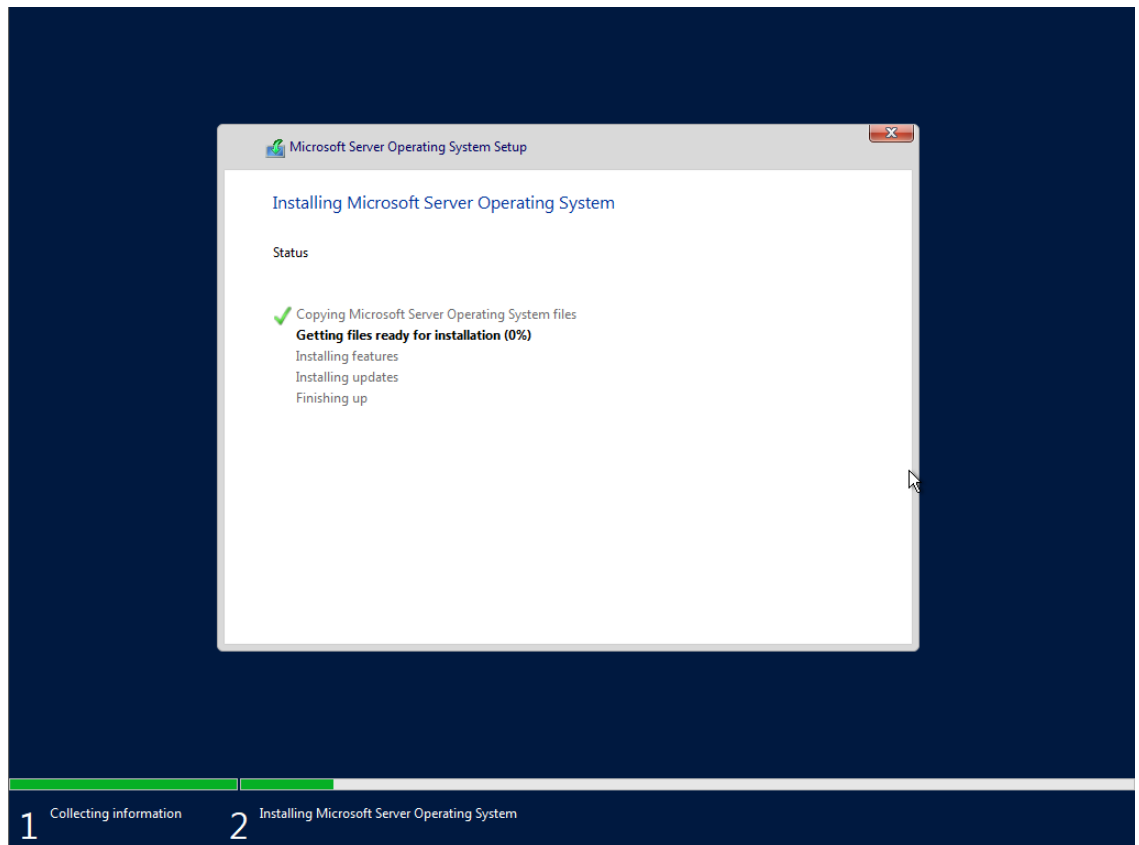
12) Choose Custom



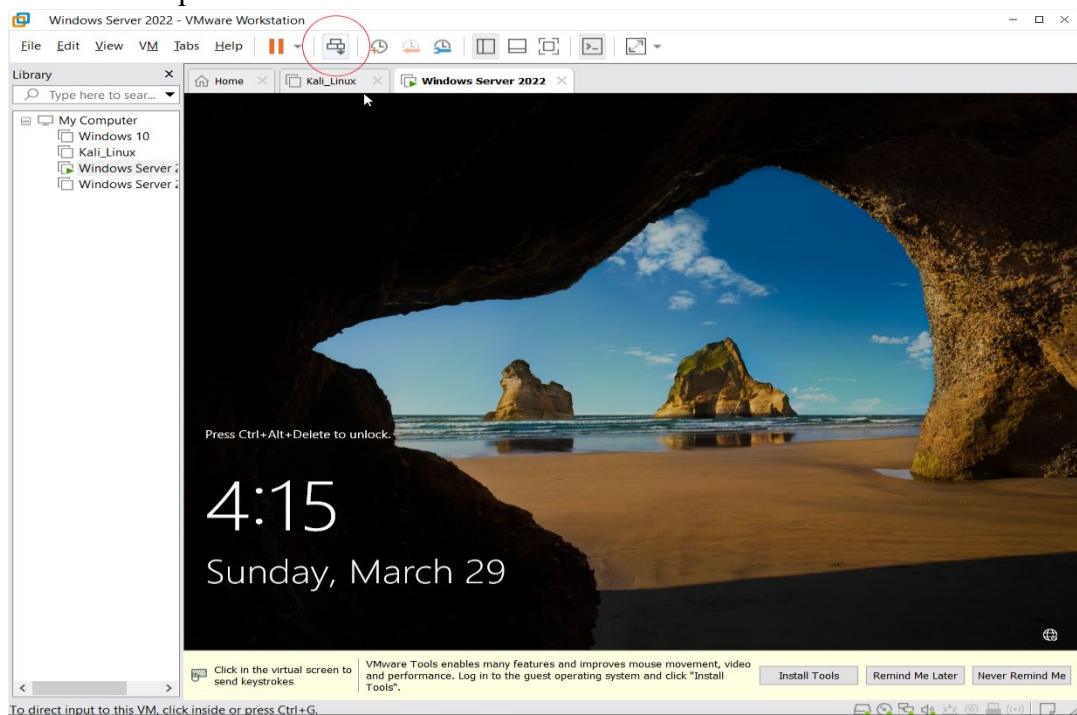
13) No disk partition for this virtual machine. Click on Next



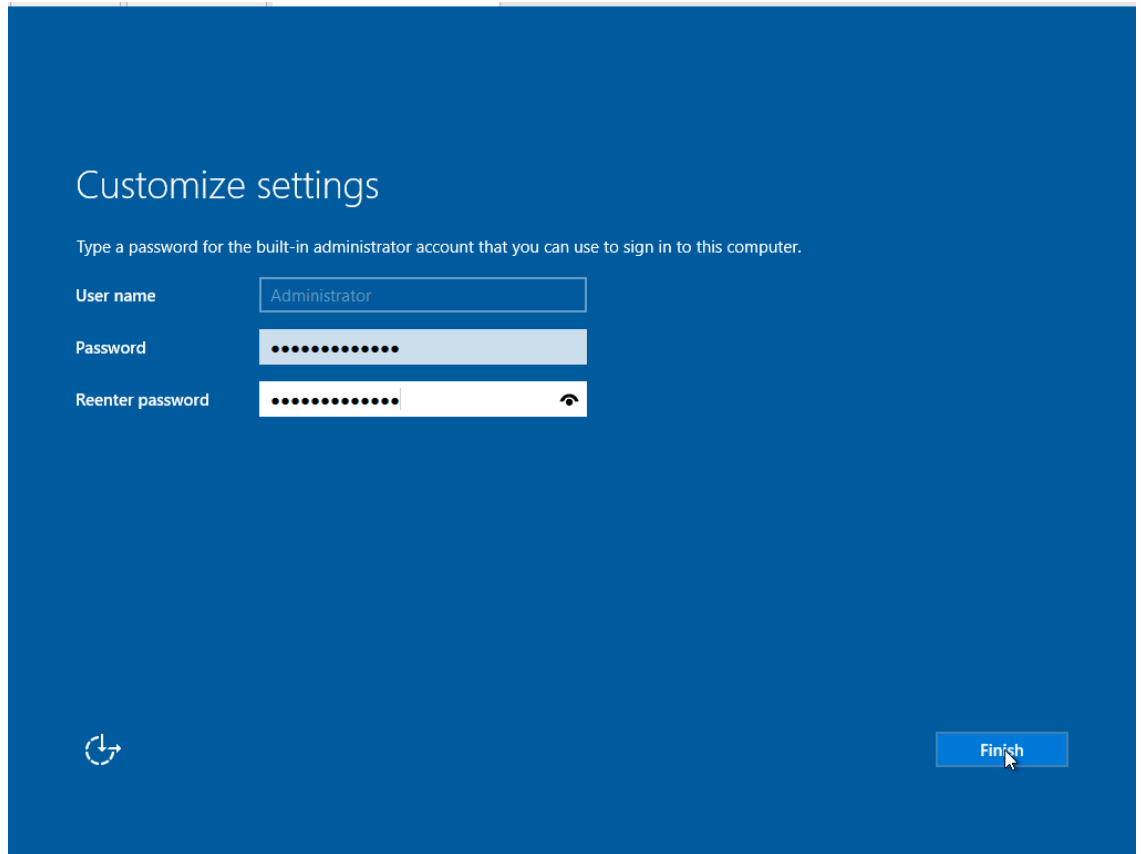
14) The system will automatically restart when installation complete



15) For VMware Workstation Pro, press Ctrl + Alt + Ins to enter the system interface or Click on the button shown below. Press Ctrl + Alt + Del will open the account options interface of the host machine



- 16) After the system restarts, set a password for the administrator account.
Installation complete.



The image shows the 'Customize settings' window in Windows. The title bar is blue. The main content area has a blue background. The title 'Customize settings' is in white. Below it, a subtitle in white says 'Type a password for the built-in administrator account that you can use to sign in to this computer.' There are three input fields: 'User name' with the text 'Administrator', 'Password' with 12 dots, and 'Reenter password' with 12 dots and a small eye icon to its right. At the bottom left is a circular arrow icon. At the bottom right is a blue button labeled 'Finish' with a mouse cursor over it.

Customize settings

Type a password for the built-in administrator account that you can use to sign in to this computer.

User name

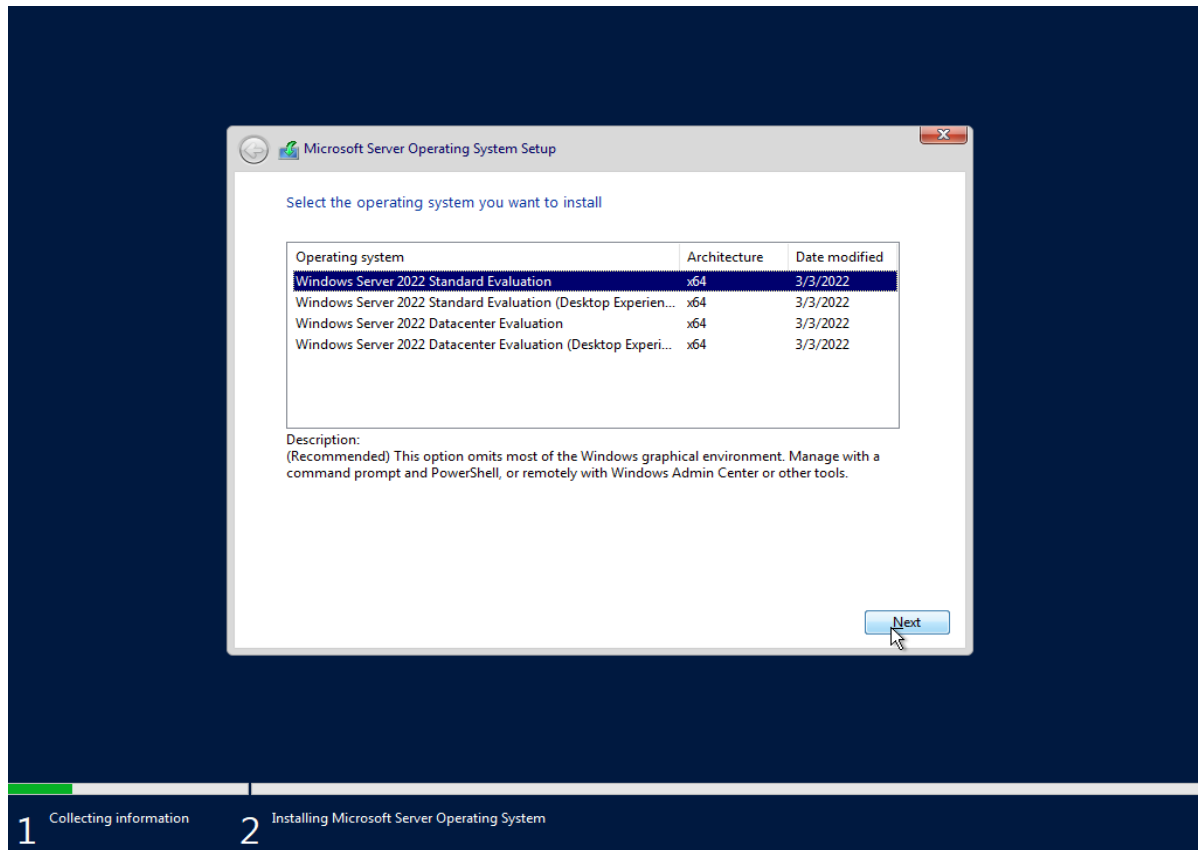
Password

Reenter password 

 Finish

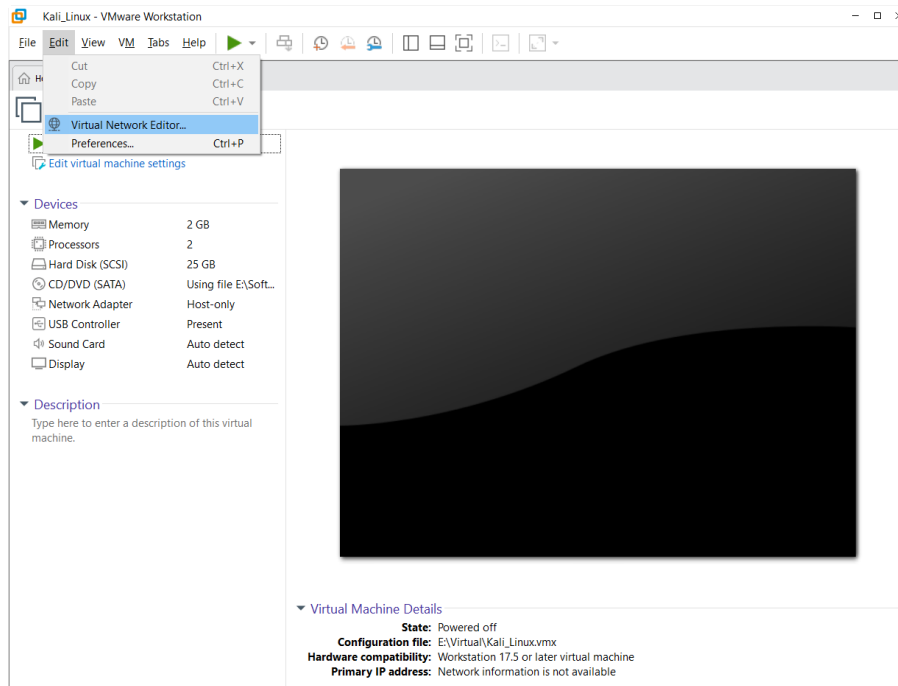
c. Installation of Windows Server 2022 Core

- 1) Follow the same steps as described above, except on the page where to select the operating system to install, choose the Standard Evaluation without the desktop experience

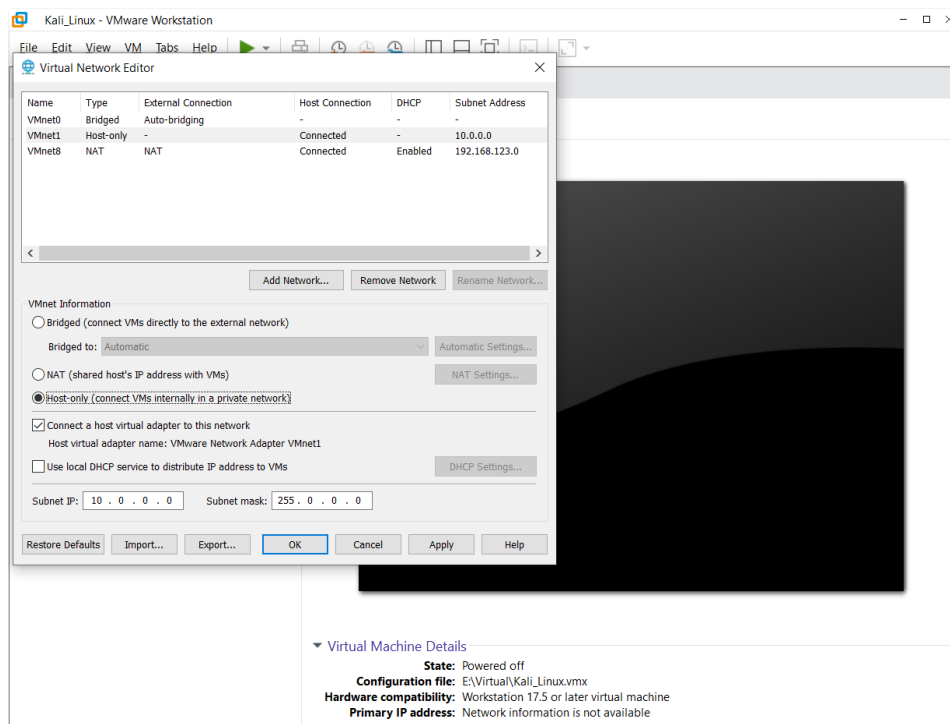


d. Change the Virtual Network Editor Setting

1) Go to Menu, click on Edit – Virtual Network Editor

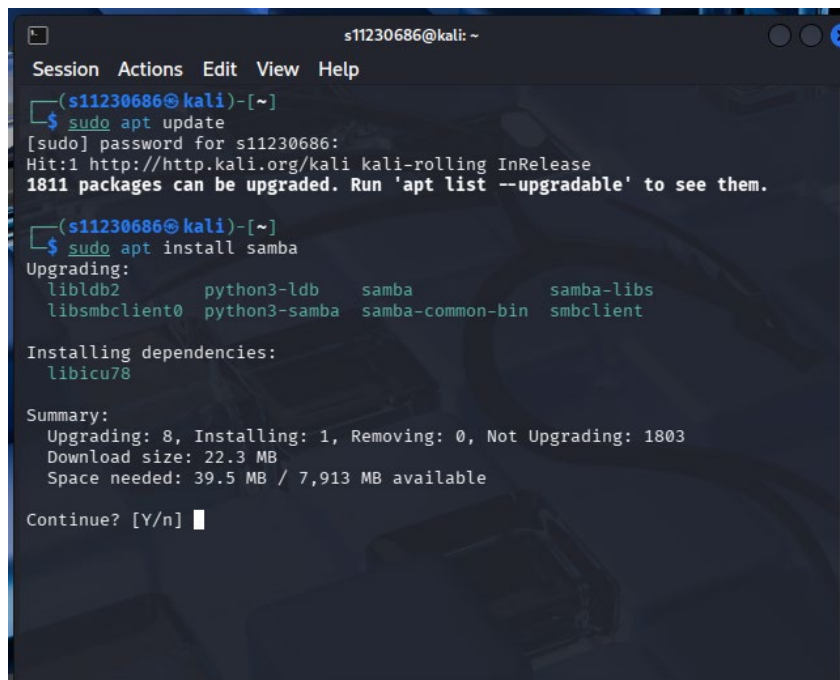


2) Click Vmnet1 and uncheck “Use the local DHCP service to distribute IP address to VMs.” Then change the subnet IP to 10.0.0.0 and the subnet mask to 255.0.0.0.



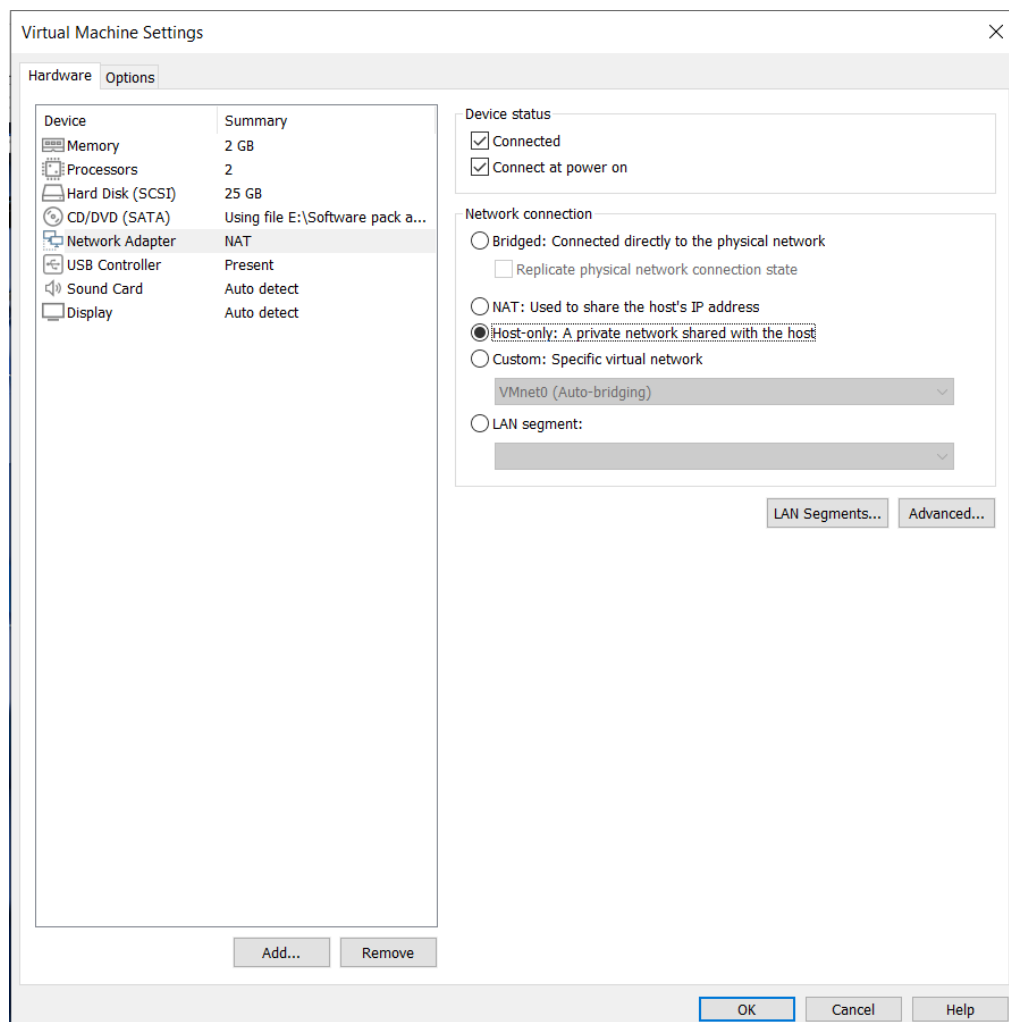
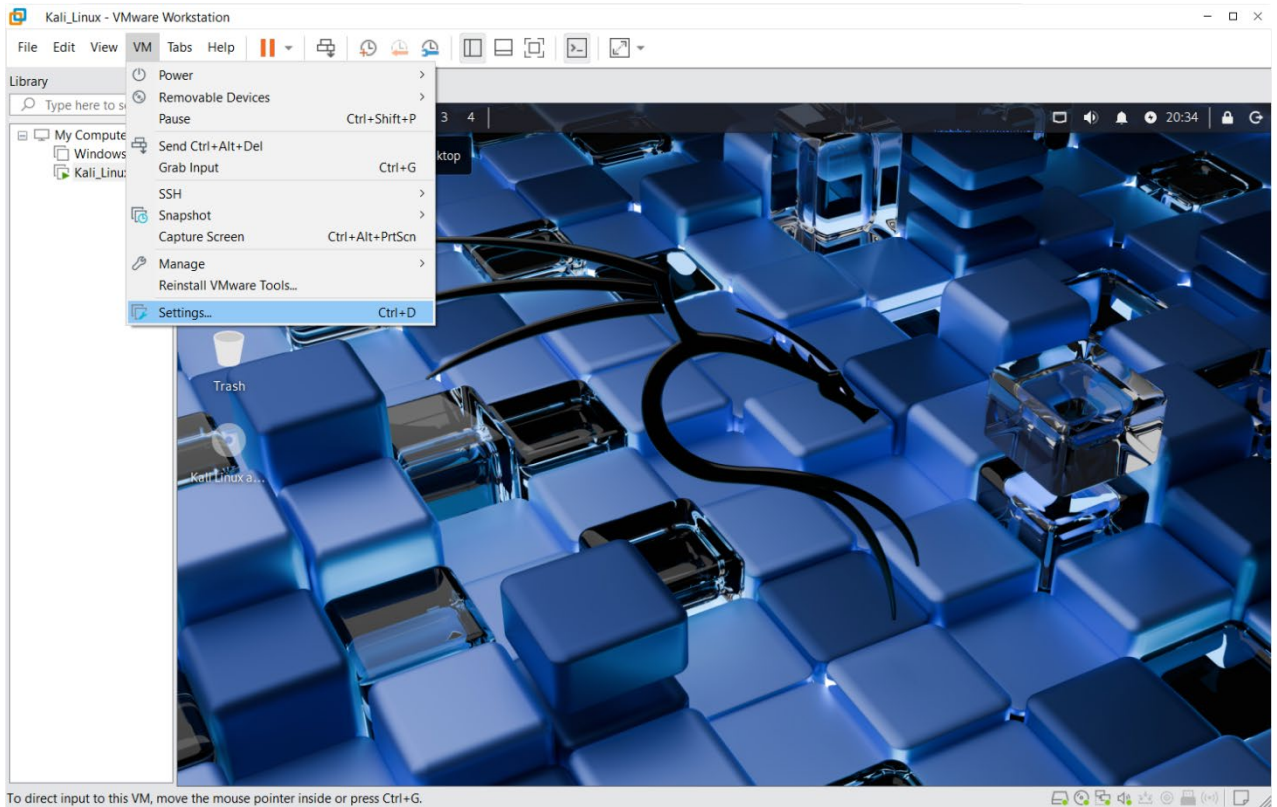
e. Assign Static IP to Kali Linux

- 1) Before assigning IP addresses, enter “sudo apt install samba” to install Samba, this is essential for file sharing

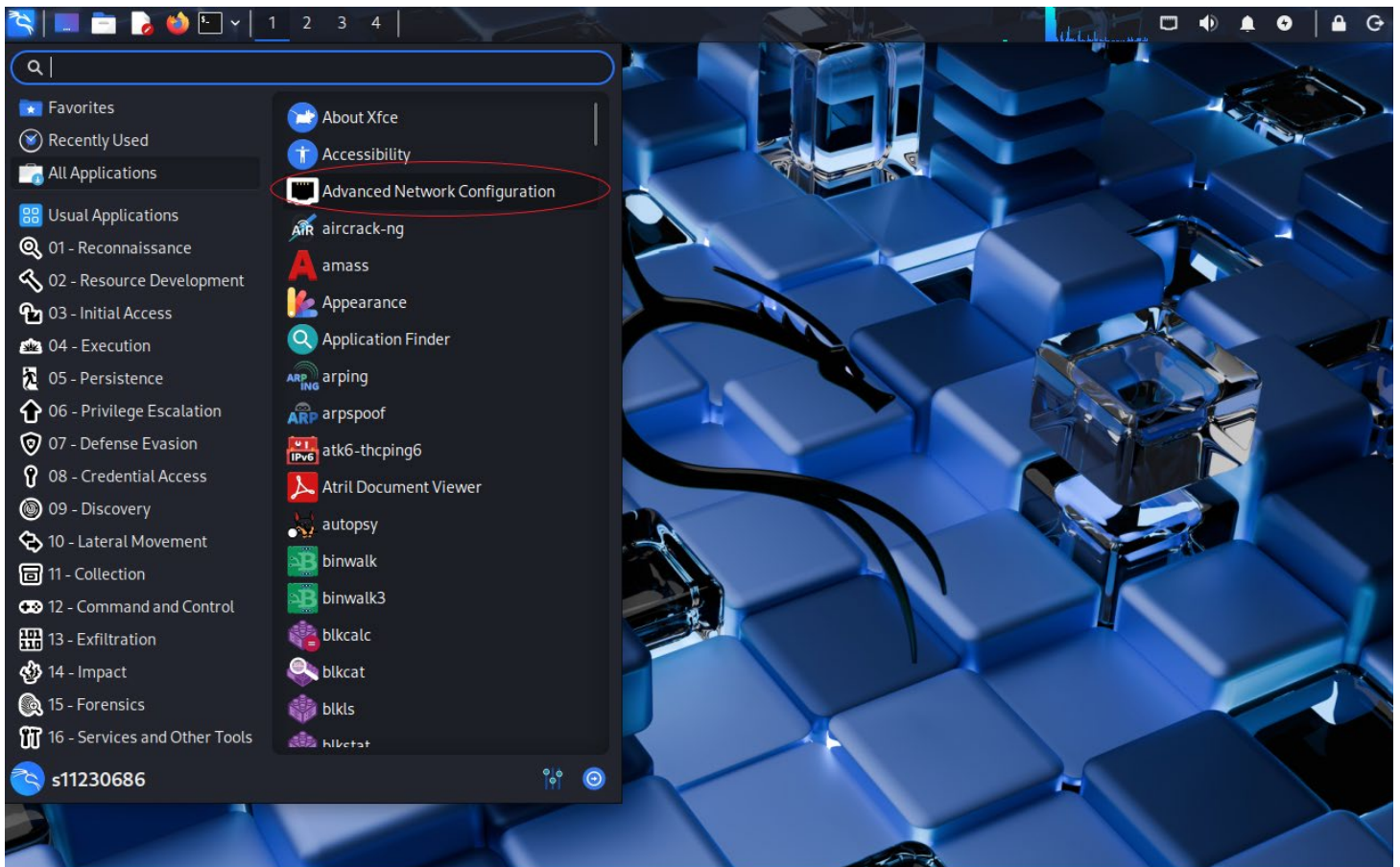
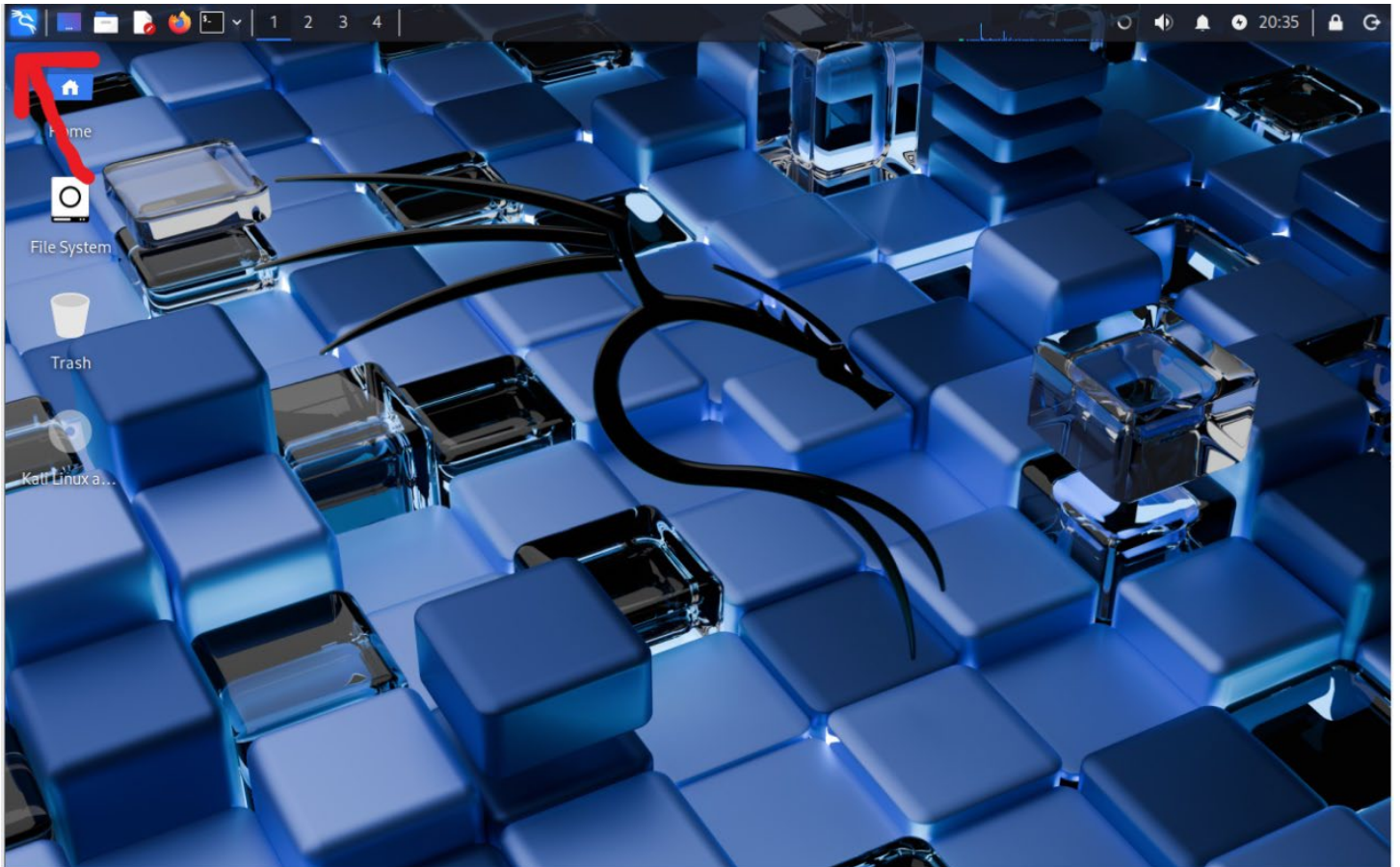


```
s11230686@kali: ~  
Session Actions Edit View Help  
(s11230686@kali)-[~]  
$ sudo apt update  
[sudo] password for s11230686:  
Hit:1 http://http.kali.org/kali kali-rolling InRelease  
1811 packages can be upgraded. Run 'apt list --upgradable' to see them.  
(s11230686@kali)-[~]  
$ sudo apt install samba  
Upgrading:  
libldb2 python3-ldb samba samba-libs  
libsmclient0 python3-samba samba-common-bin smbclient  
  
Installing dependencies:  
libicu78  
  
Summary:  
Upgrading: 8, Installing: 1, Removing: 0, Not Upgrading: 1803  
Download size: 22.3 MB  
Space needed: 39.5 MB / 7,913 MB available  
  
Continue? [Y/n]
```

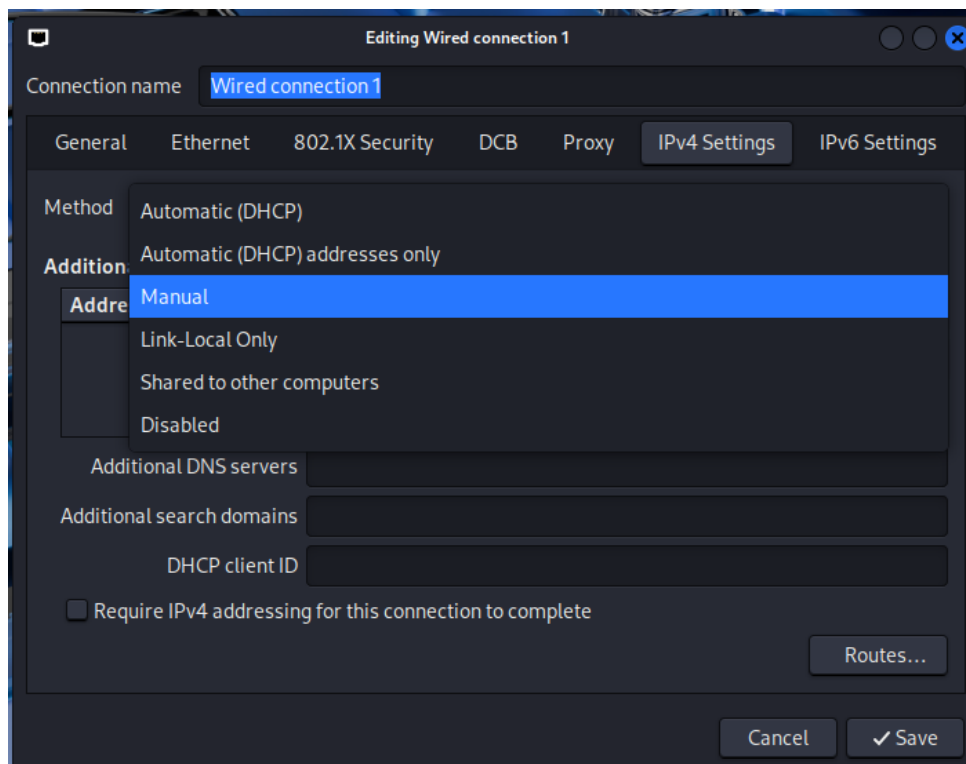
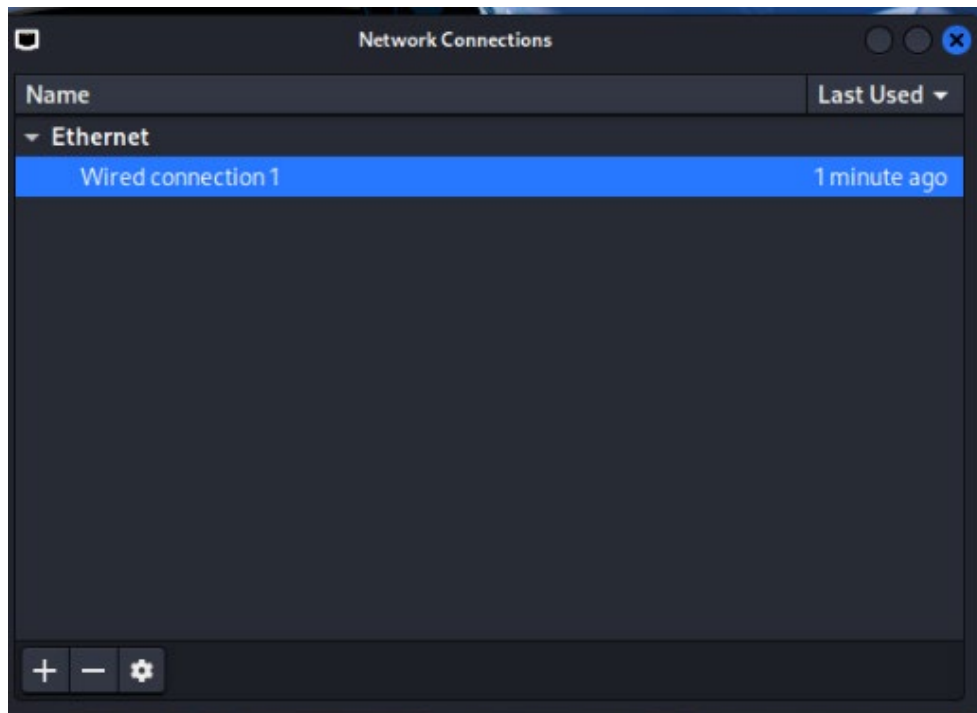
- 2) After Samba installation finishes. Go to VM- Settings, change adapter into Host-only adapter



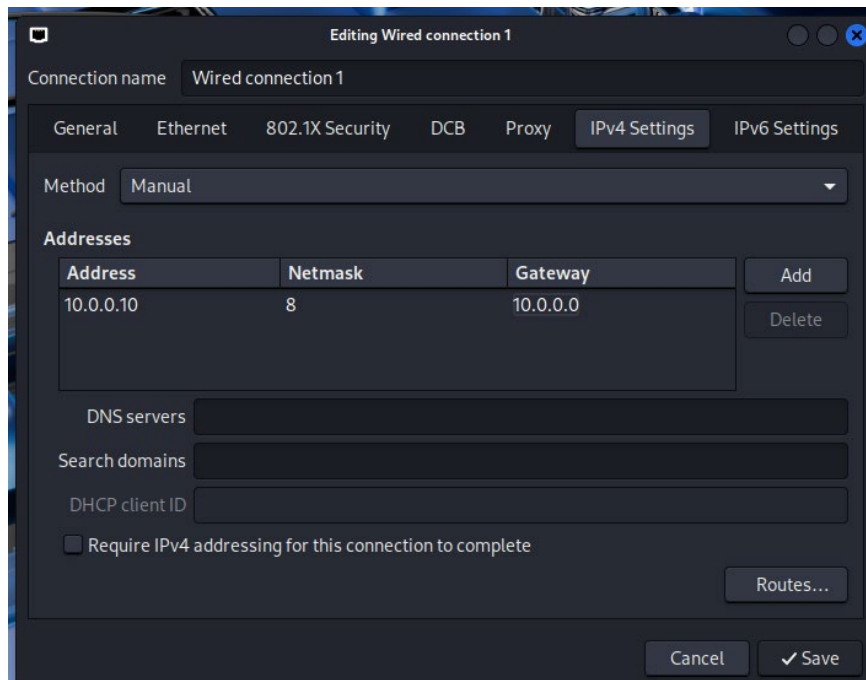
- 3) Click on system menu, go to All Applications – Advanced Network Configuration



- 4) Double click on Wired Connection, go to IPv4 Settings, change method to Manual

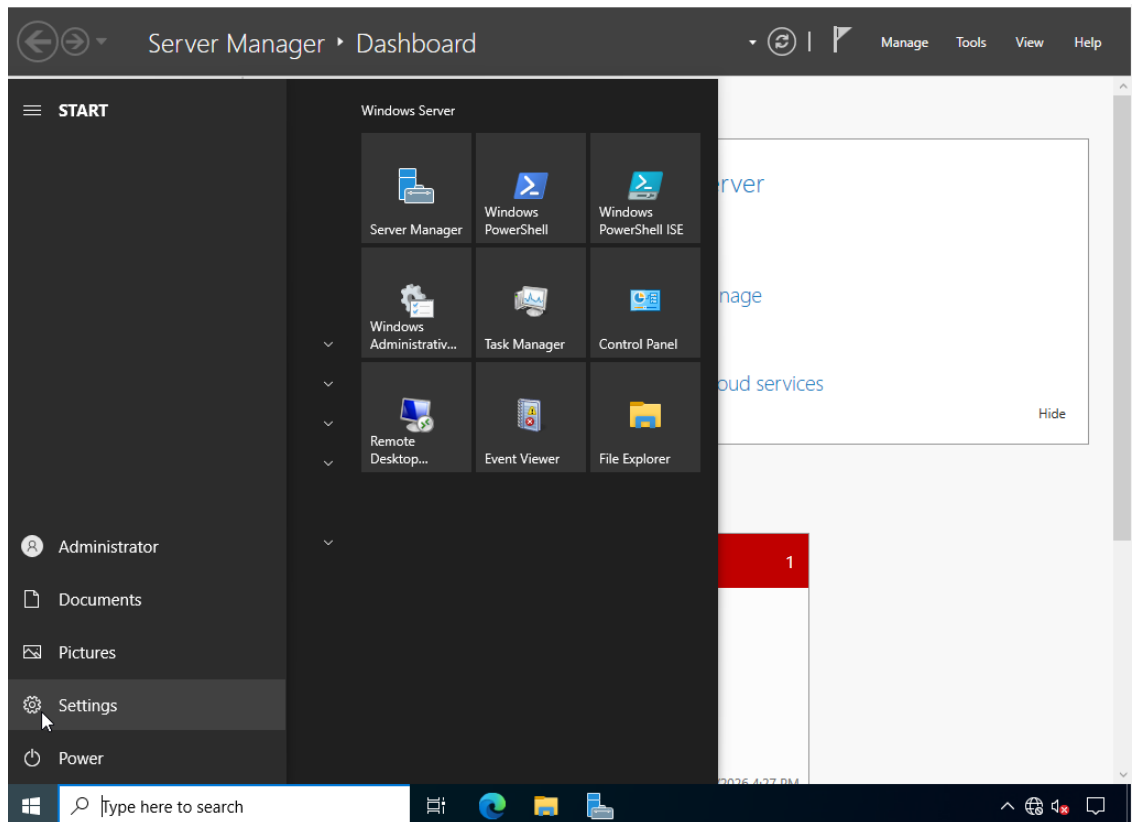


- 5) Add a new IP address: 10.0.0.10, Net Mask: 8, Gateway: 10.0.0.0, then click on Save

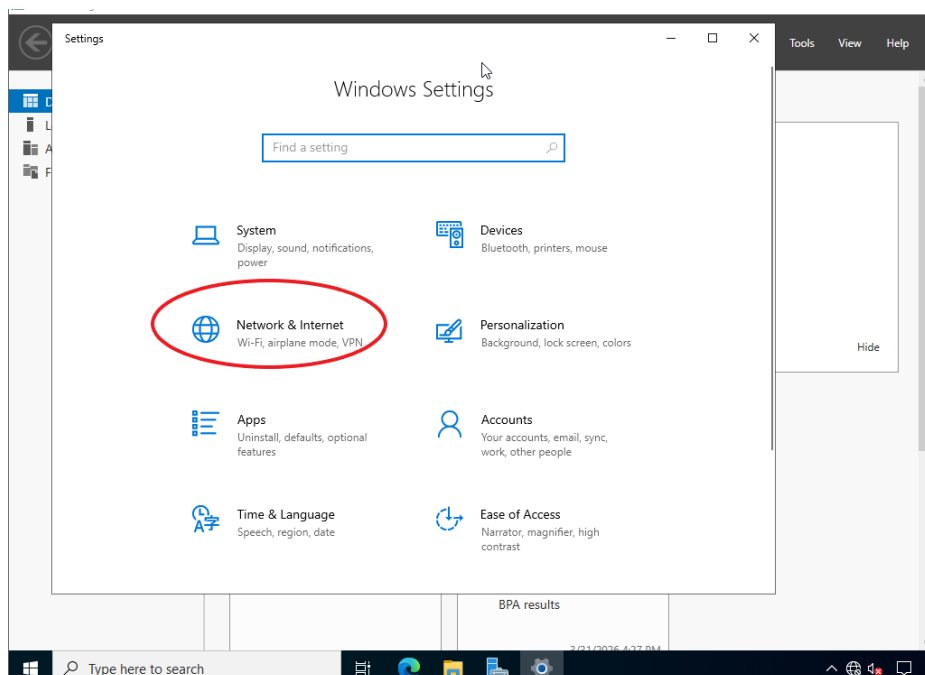


f. Assign Static IP to Windows Server Desktop

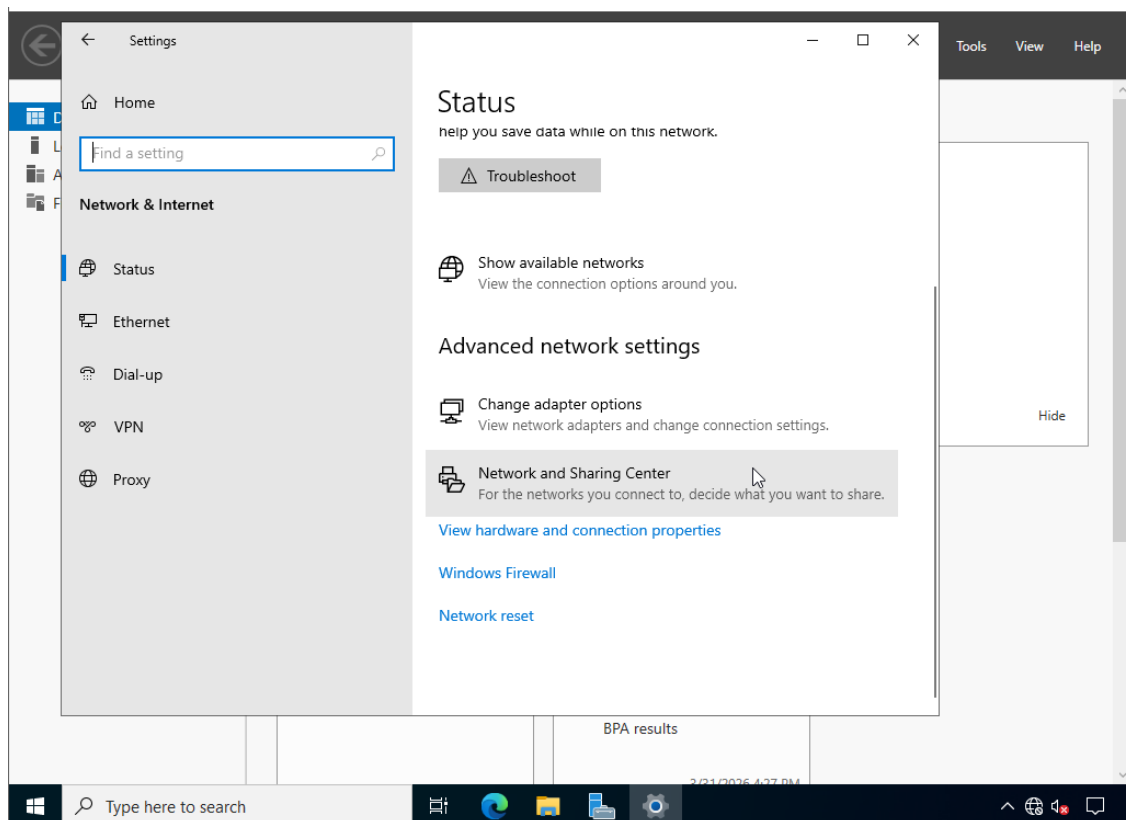
1) Click on Windows Menu – Settings



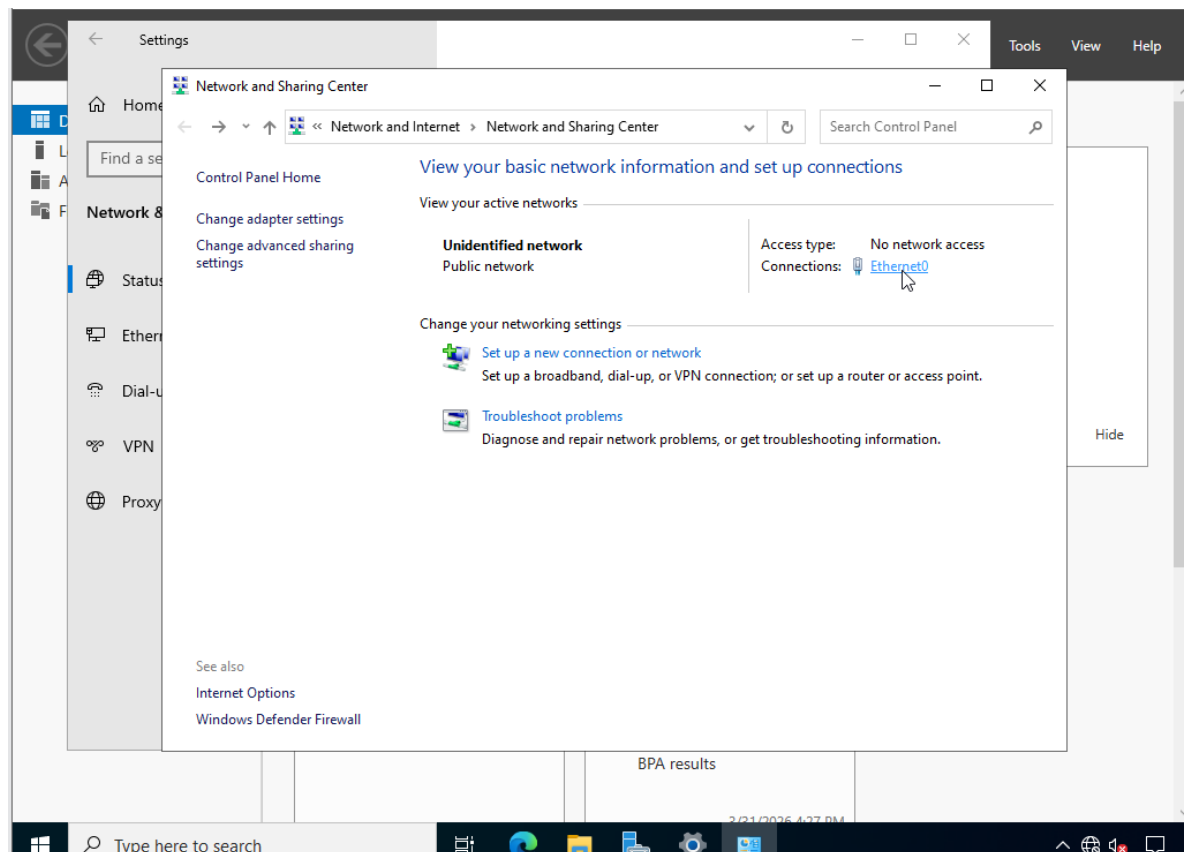
2) Click on Network & Internet



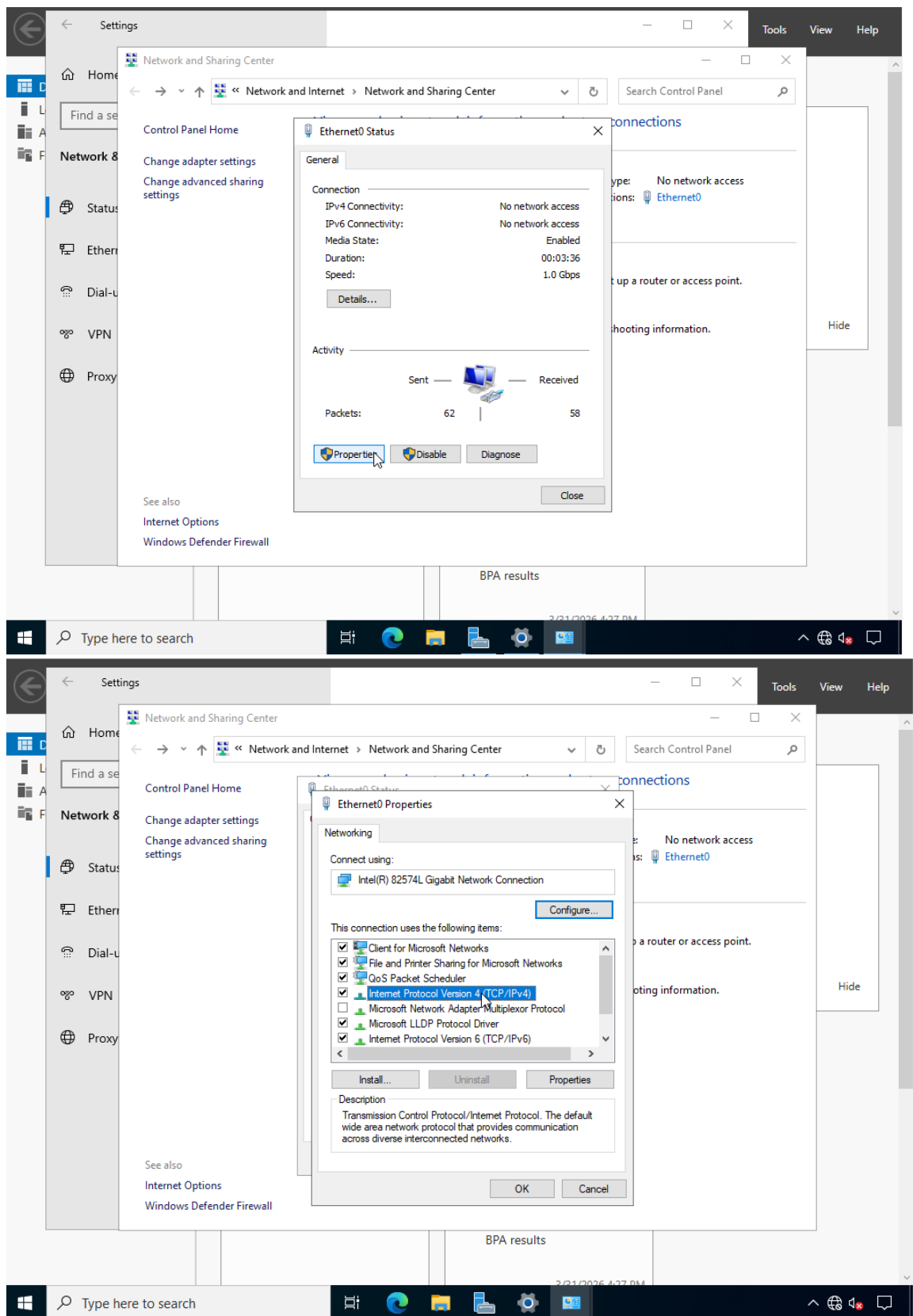
3) Scroll down and click on Network and Sharing Center



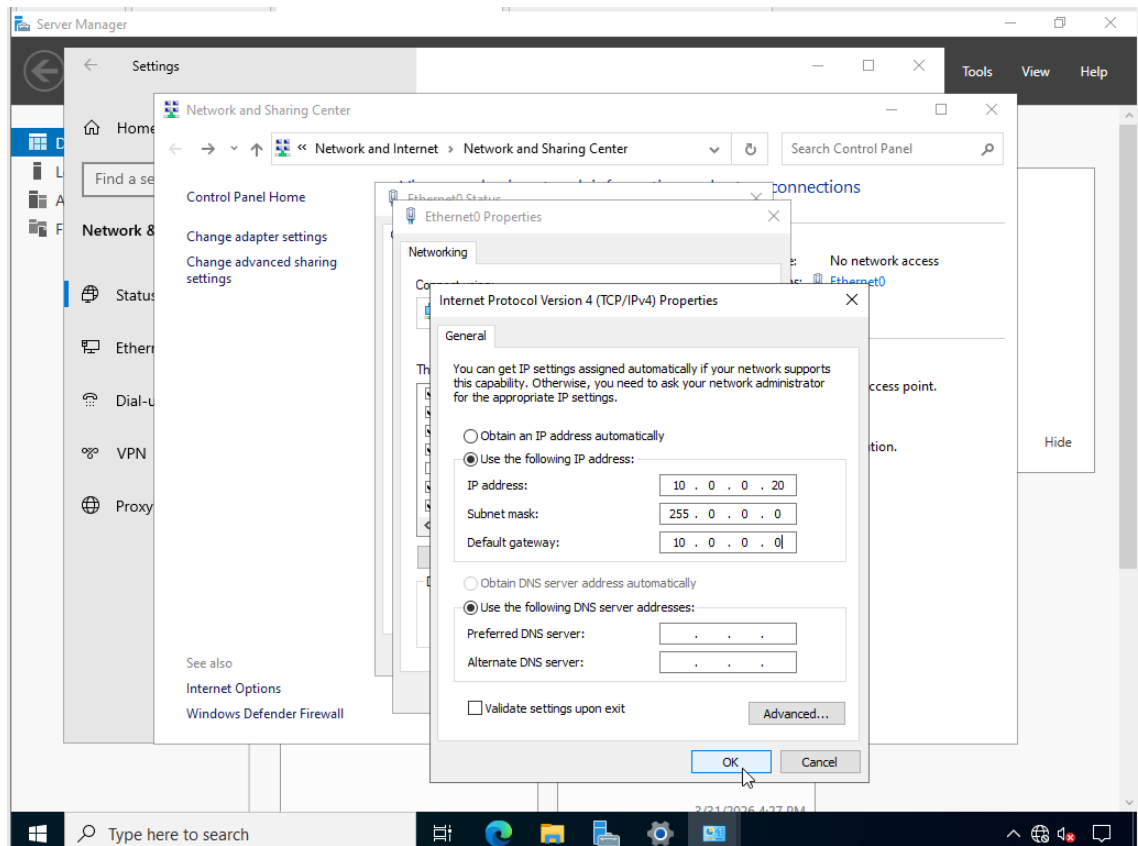
4) Click on Ethernet 0



5) Click on Properties, double click on Internet Protocol Version 4

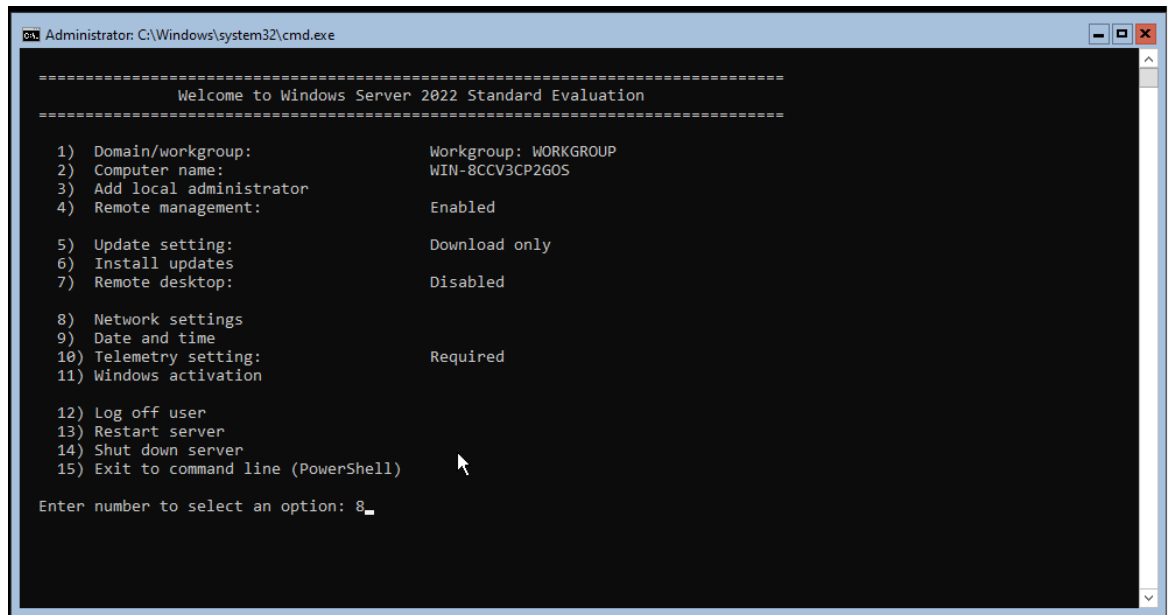


- 6) Select Use the following IP address option and assign Static IP address 10.0.0.20, Mask 255.0.0.0, Default Gateway 10.0.0.0



g. Assign Static IP to Windows Server Core

1) Type 8 to enter Network Settings



A screenshot of a Windows command prompt window titled "Administrator: C:\Windows\system32\cmd.exe". The window displays the "Welcome to Windows Server 2022 Standard Evaluation" menu. The menu lists 15 options, with option 8, "Network settings", highlighted by a mouse cursor. The prompt "Enter number to select an option: 8_" is visible at the bottom.

```
Administrator: C:\Windows\system32\cmd.exe

=====
Welcome to Windows Server 2022 Standard Evaluation
=====

1) Domain/workgroup:           Workgroup: WORKGROUP
2) Computer name:             WIN-8CCV3CP2G05
3) Add local administrator
4) Remote management:         Enabled

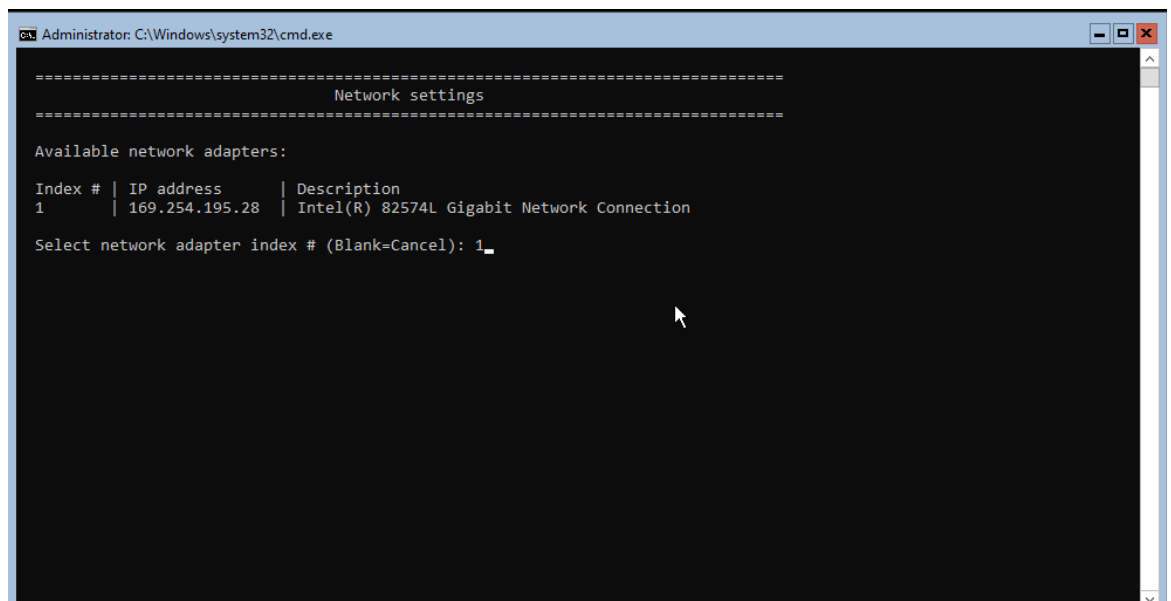
5) Update setting:             Download only
6) Install updates
7) Remote desktop:            Disabled

8) Network settings
9) Date and time
10) Telemetry setting:         Required
11) Windows activation

12) Log off user
13) Restart server
14) Shut down server
15) Exit to command line (PowerShell)

Enter number to select an option: 8_
```

2) Type 1 to configure the adapter



A screenshot of a Windows command prompt window titled "Administrator: C:\Windows\system32\cmd.exe". The window displays the "Network settings" menu. The menu lists "Available network adapters:" with a table showing one adapter. The prompt "Select network adapter index # (Blank=Cancel): 1_" is visible at the bottom, with a mouse cursor pointing at the number 1.

```
Administrator: C:\Windows\system32\cmd.exe

=====
Network settings
=====

Available network adapters:

Index # | IP address   | Description
-----|-----|-----
1      | 169.254.195.28 | Intel(R) 82574L Gigabit Network Connection

Select network adapter index # (Blank=Cancel): 1_
```

- 3) Set IP address to 10.0.0.30, Subnet Mask: 255.0.0.0, Default Gateway: 10.0.0.0

```
Administrator: C:\Windows\system32\cmd.exe

=====
Network adapter settings
=====

NIC index:      1
Description:    Intel(R) 82574L Gigabit Network Connection
IP address:     169.254.195.28,
                fe80::d437:fbd9:6792:c31c
Subnet mask:    255.255.0.0
DHCP enabled:   True

Default gateway:
Preferred DNS server:
Alternate DNS server:

1) Set network adapter address
2) Set DNS servers
3) Clear DNS server settings

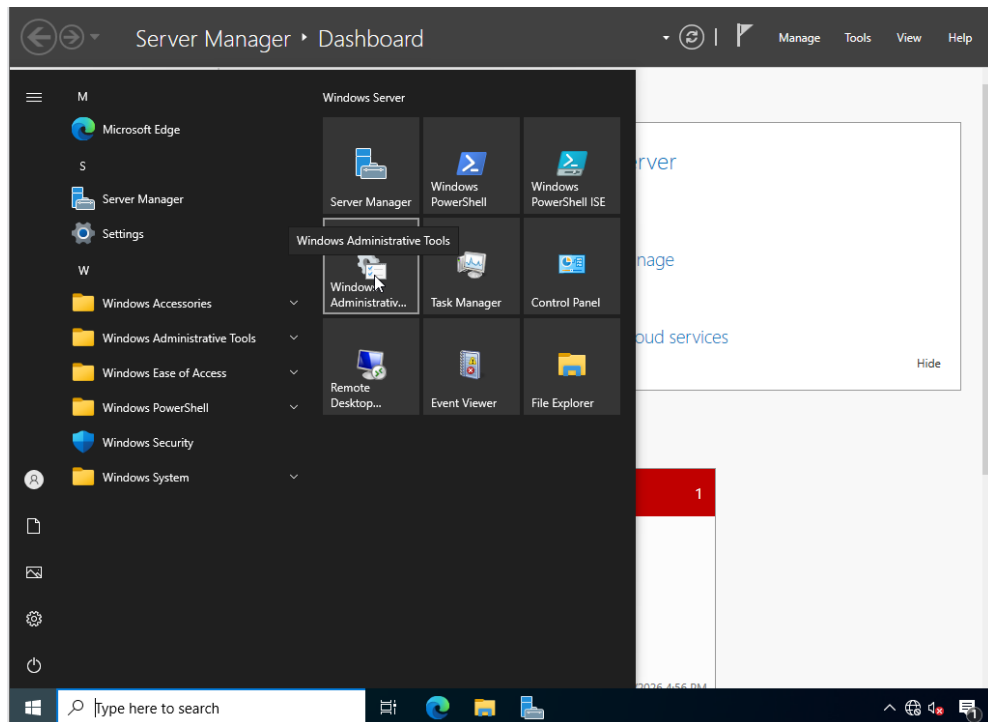
Enter selection (Blank=Cancel): 1
Select (D)HCP or (S)tatic IP address (Blank=Cancel): s
Enter static IP address (Blank=Cancel): 10.0.0.30
Enter subnet mask (Blank=255.255.255.0): 255.0.0.0
Enter default gateway (Blank=Cancel): 10.0.0.0
Setting NIC to static IP...
Successfully released DHCP lease.
Successfully enabled static addressing. DHCP for this network adapter is disabled.
Successfully set gateway.
Successfully set network adapter address.
(Press ENTER to continue):
```

h. Test connectivity between systems

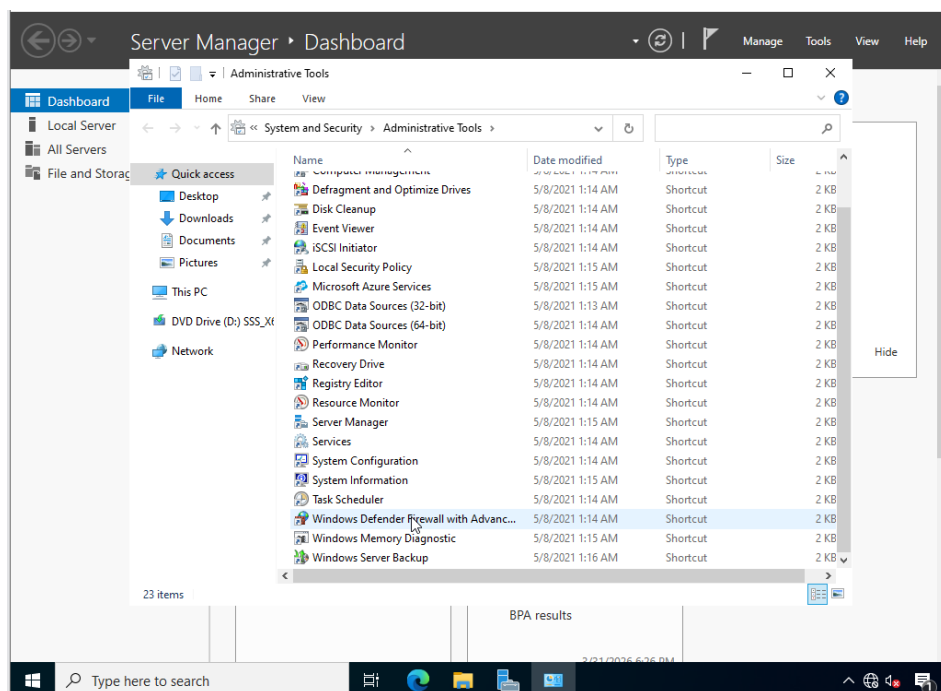
1) Before test connectivity between VMs, enable some inbound rules on the Windows firewall to allow ICMP requests

a) Windows Desktop

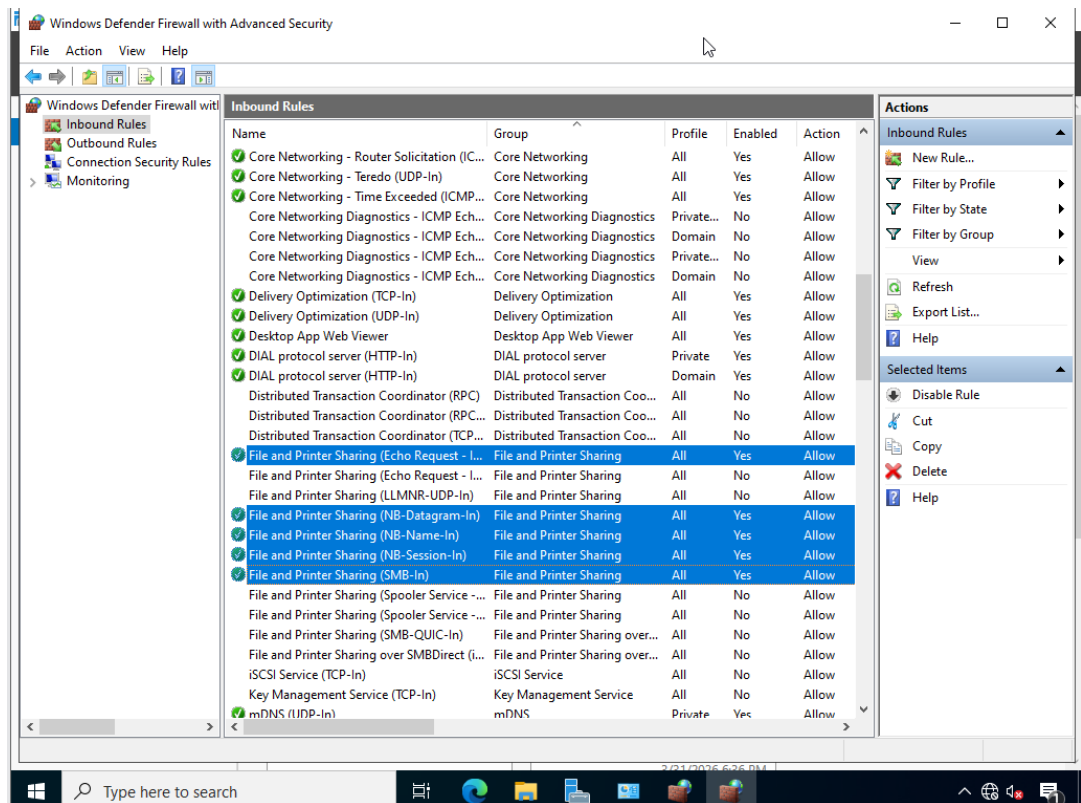
i. Go to Start Menu – Windows Administrative Tools



ii. Click on Windows Defender Firewall with Advanced Settings

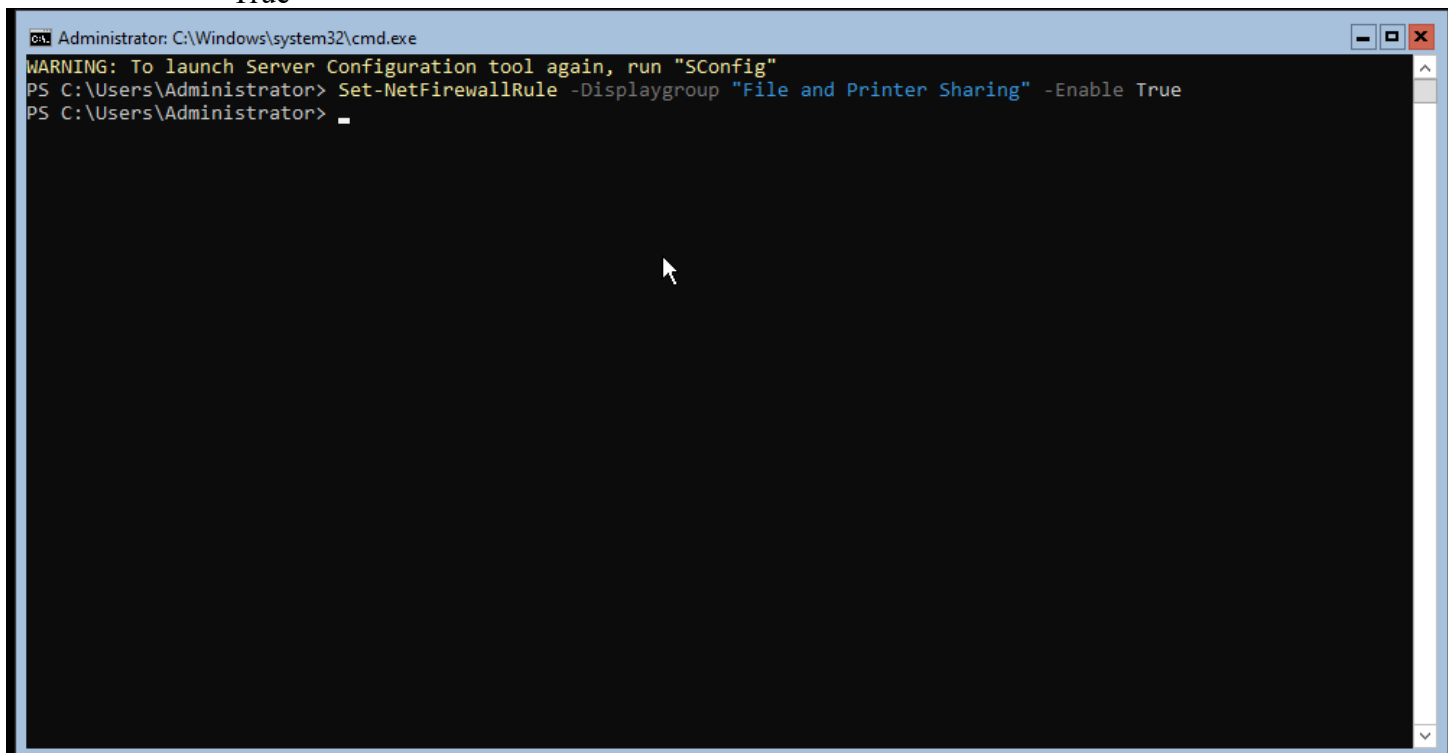


- iii. Click on Inbound Rules, enable following File and Printer Sharing rules



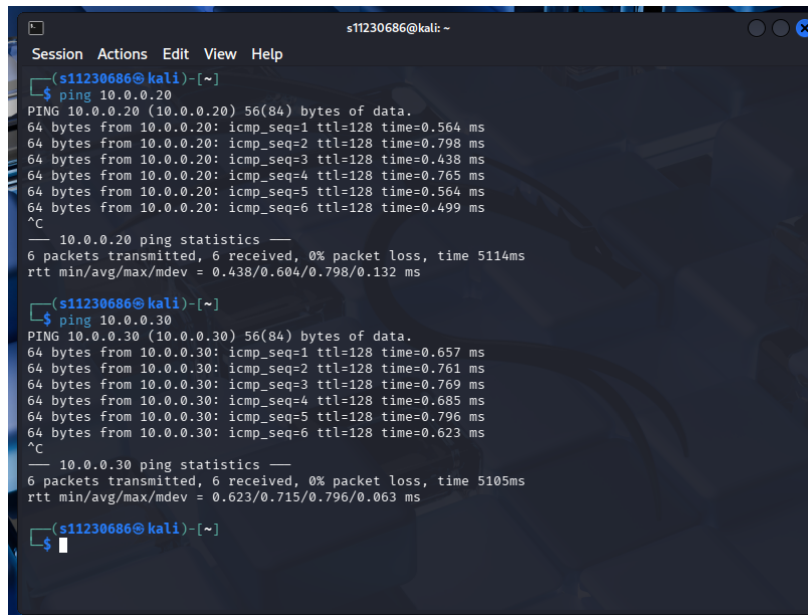
b). Windows Server Core

- i. Open terminal, enter command:
"Set-NetFirewallRule -DisplayGroup "File and Printer Sharing" -Enabled True"



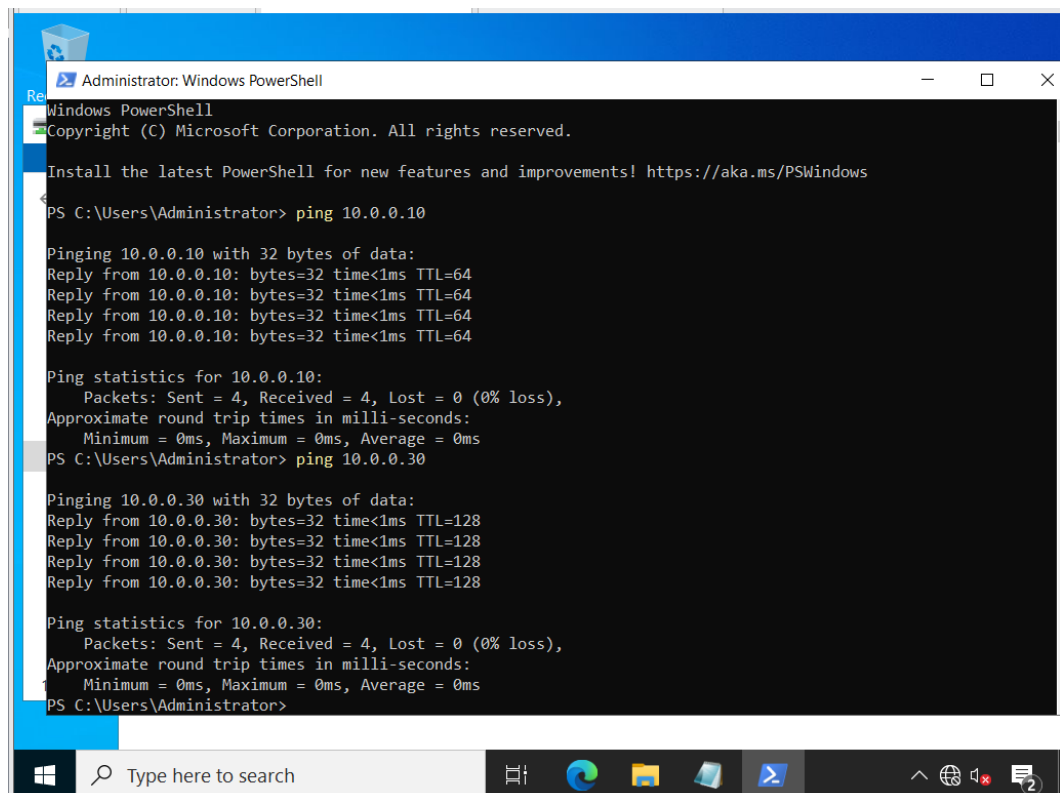
2) Test connectivity

a. From Linux to the two Windows Server



```
s11230686@kali: ~  
Session Actions Edit View Help  
$ ping 10.0.0.20  
PING 10.0.0.20 (10.0.0.20) 56(84) bytes of data:  
64 bytes from 10.0.0.20: icmp_seq=1 ttl=128 time=0.564 ms  
64 bytes from 10.0.0.20: icmp_seq=2 ttl=128 time=0.798 ms  
64 bytes from 10.0.0.20: icmp_seq=3 ttl=128 time=0.438 ms  
64 bytes from 10.0.0.20: icmp_seq=4 ttl=128 time=0.765 ms  
64 bytes from 10.0.0.20: icmp_seq=5 ttl=128 time=0.564 ms  
64 bytes from 10.0.0.20: icmp_seq=6 ttl=128 time=0.499 ms  
^C  
--- 10.0.0.20 ping statistics ---  
6 packets transmitted, 6 received, 0% packet loss, time 5114ms  
rtt min/avg/max/mdev = 0.438/0.604/0.798/0.132 ms  
$ ping 10.0.0.30  
PING 10.0.0.30 (10.0.0.30) 56(84) bytes of data:  
64 bytes from 10.0.0.30: icmp_seq=1 ttl=128 time=0.657 ms  
64 bytes from 10.0.0.30: icmp_seq=2 ttl=128 time=0.761 ms  
64 bytes from 10.0.0.30: icmp_seq=3 ttl=128 time=0.769 ms  
64 bytes from 10.0.0.30: icmp_seq=4 ttl=128 time=0.685 ms  
64 bytes from 10.0.0.30: icmp_seq=5 ttl=128 time=0.796 ms  
64 bytes from 10.0.0.30: icmp_seq=6 ttl=128 time=0.623 ms  
^C  
--- 10.0.0.30 ping statistics ---  
6 packets transmitted, 6 received, 0% packet loss, time 5105ms  
rtt min/avg/max/mdev = 0.623/0.715/0.796/0.063 ms  
$
```

b. From Windows Server Desktop to Core and Linux



```
Administrator: Windows PowerShell  
Windows PowerShell  
Copyright (C) Microsoft Corporation. All rights reserved.  
  
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows  
  
PS C:\Users\Administrator> ping 10.0.0.10  
  
Pinging 10.0.0.10 with 32 bytes of data:  
Reply from 10.0.0.10: bytes=32 time<1ms TTL=64  
Reply from 10.0.0.10: bytes=32 time<1ms TTL=64  
Reply from 10.0.0.10: bytes=32 time<1ms TTL=64  
Reply from 10.0.0.10: bytes=32 time<1ms TTL=64  
  
Ping statistics for 10.0.0.10:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 0ms, Maximum = 0ms, Average = 0ms  
PS C:\Users\Administrator> ping 10.0.0.30  
  
Pinging 10.0.0.30 with 32 bytes of data:  
Reply from 10.0.0.30: bytes=32 time<1ms TTL=128  
Reply from 10.0.0.30: bytes=32 time<1ms TTL=128  
Reply from 10.0.0.30: bytes=32 time<1ms TTL=128  
Reply from 10.0.0.30: bytes=32 time<1ms TTL=128  
  
Ping statistics for 10.0.0.30:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 0ms, Maximum = 0ms, Average = 0ms  
PS C:\Users\Administrator>
```

c. From Windows Server Core to Linux and Desktop

```
Administrator: C:\Windows\system32\cmd.exe
WARNING: To launch Server Configuration tool again, run "SConfig"
PS C:\Users\Administrator> ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:
Reply from 10.0.0.10: bytes=32 time<1ms TTL=64
Reply from 10.0.0.10: bytes=32 time<1ms TTL=64
Reply from 10.0.0.10: bytes=32 time<1ms TTL=64
Reply from 10.0.0.10: bytes=32 time<1ms TTL=64

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\Administrator> ping 10.0.0.20

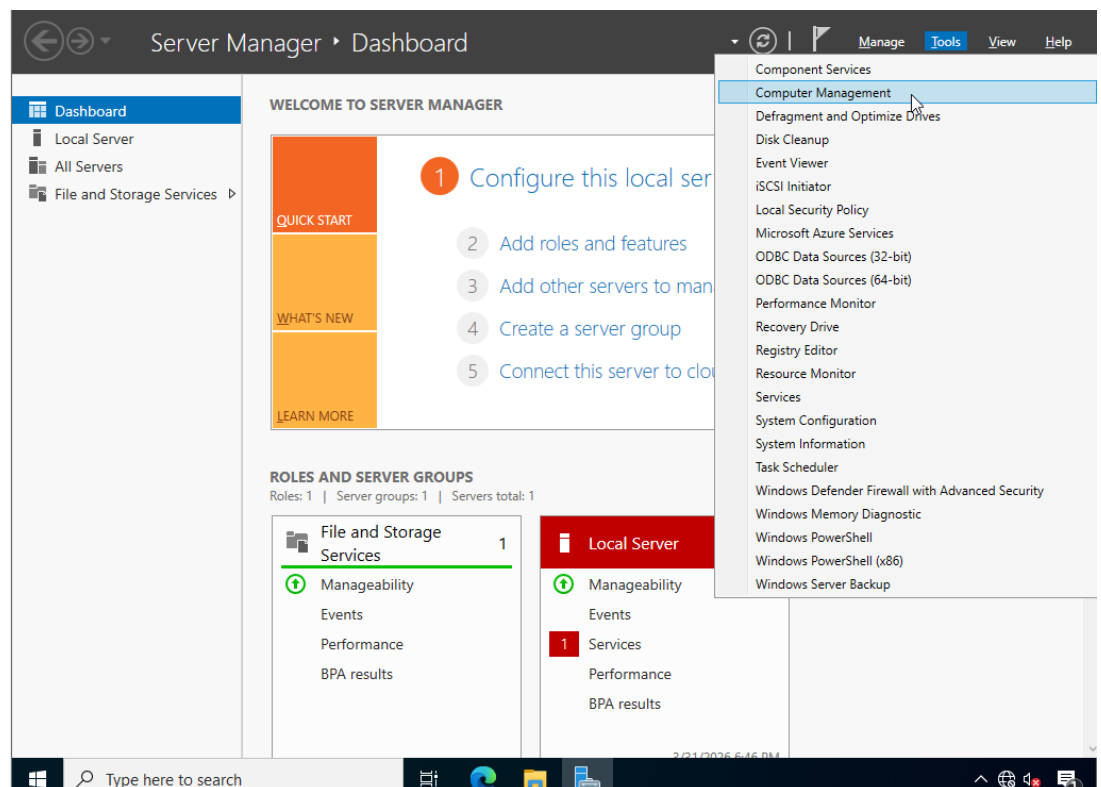
Pinging 10.0.0.20 with 32 bytes of data:
Reply from 10.0.0.20: bytes=32 time<1ms TTL=128
Reply from 10.0.0.20: bytes=32 time<1ms TTL=128
Reply from 10.0.0.20: bytes=32 time<1ms TTL=128
Reply from 10.0.0.20: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\Administrator>
```

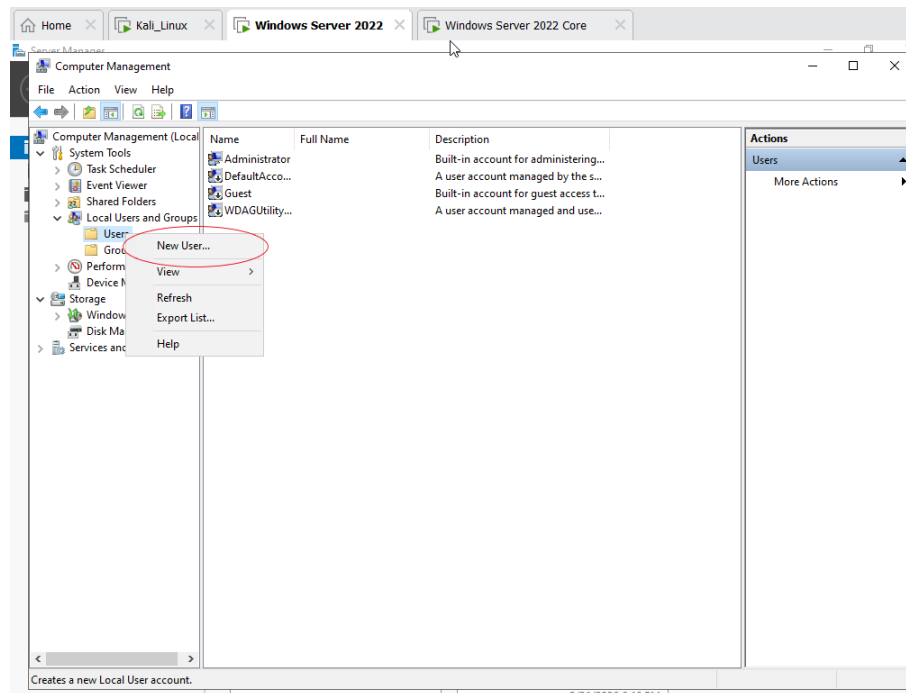
i. File Sharing

1) From Windows Server to Linux

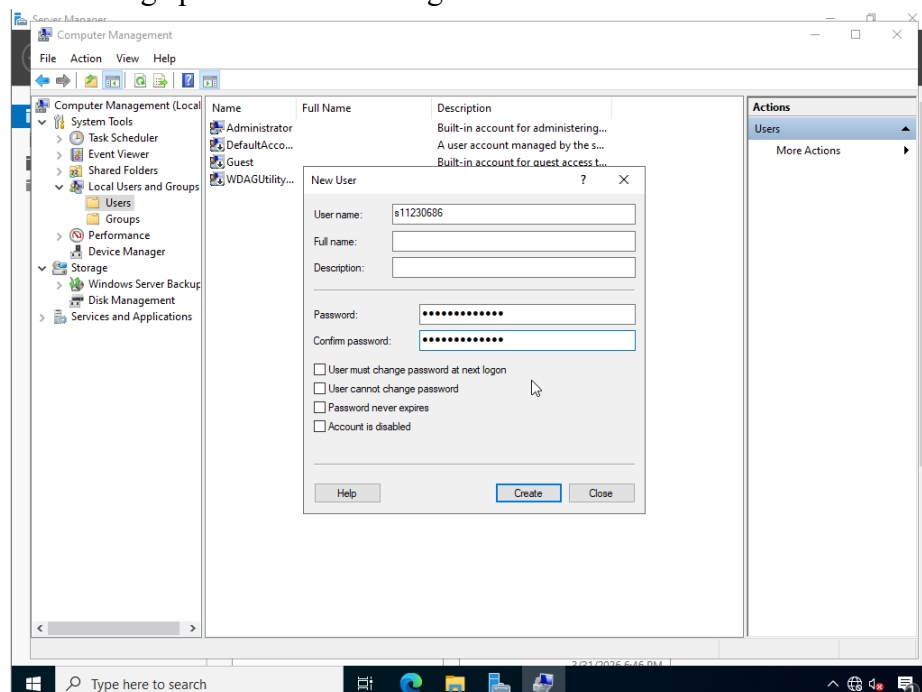
a) Go to server manager-Computer Management

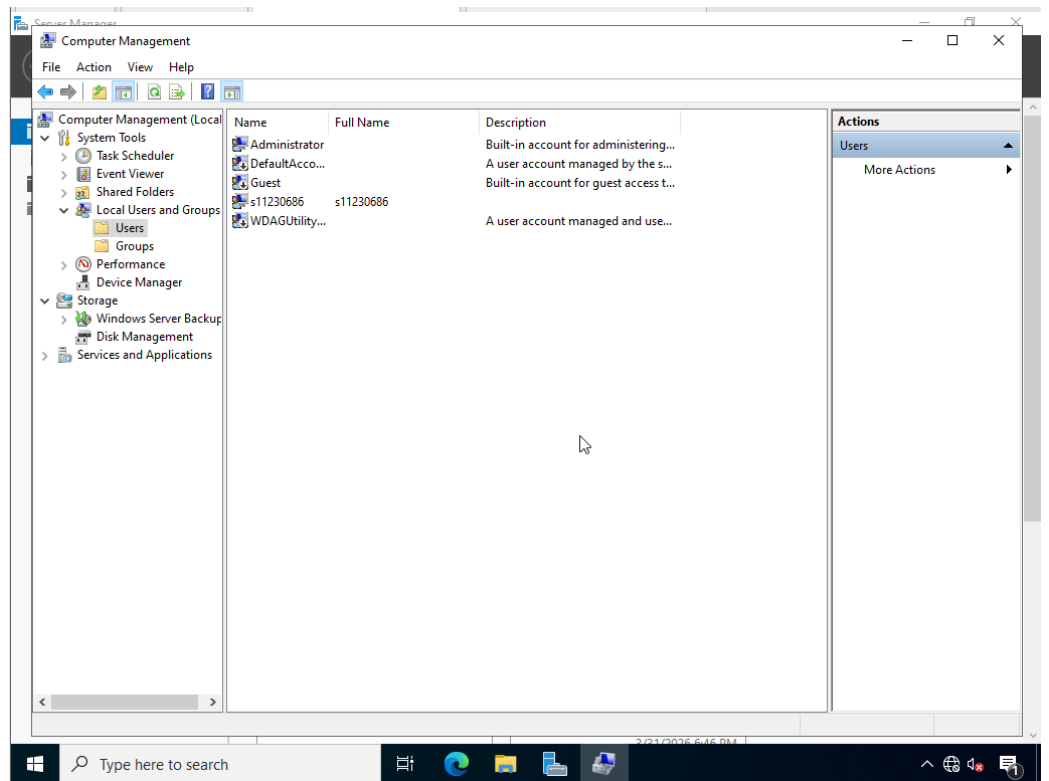


- b) Click on Local Users and Groups, Right click on New Users, create a new user

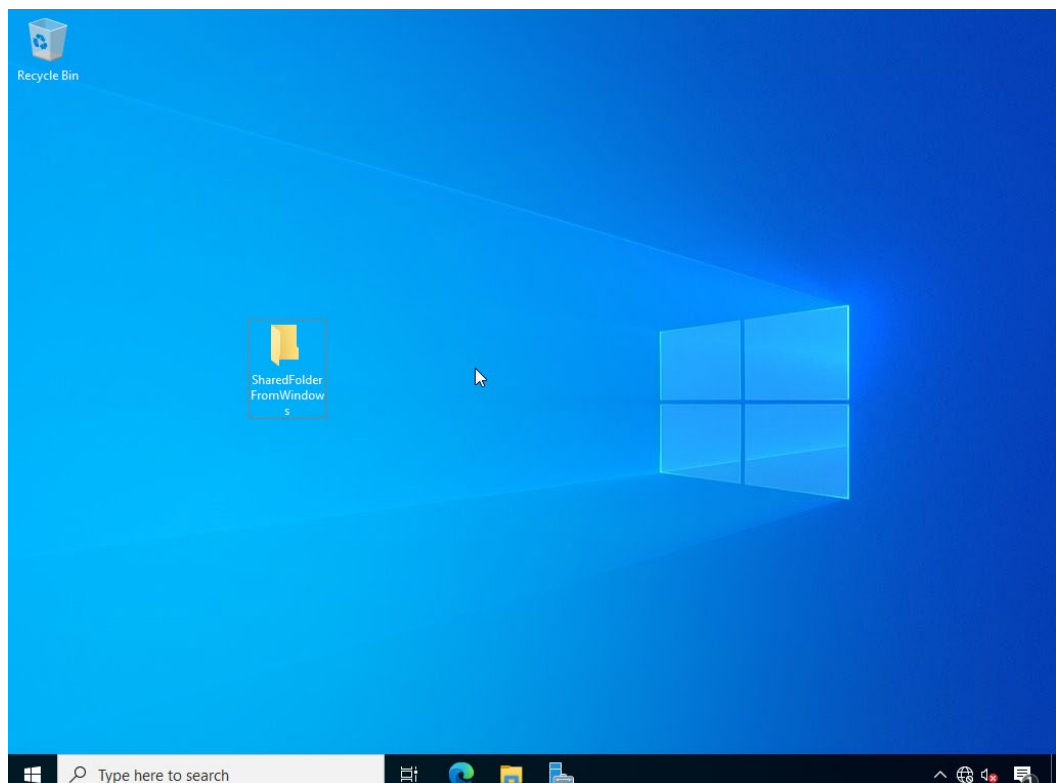


- c) Enter username and set password here, according to the assignment requirement, set student ID as the account number here. Uncheck user must change password at next login

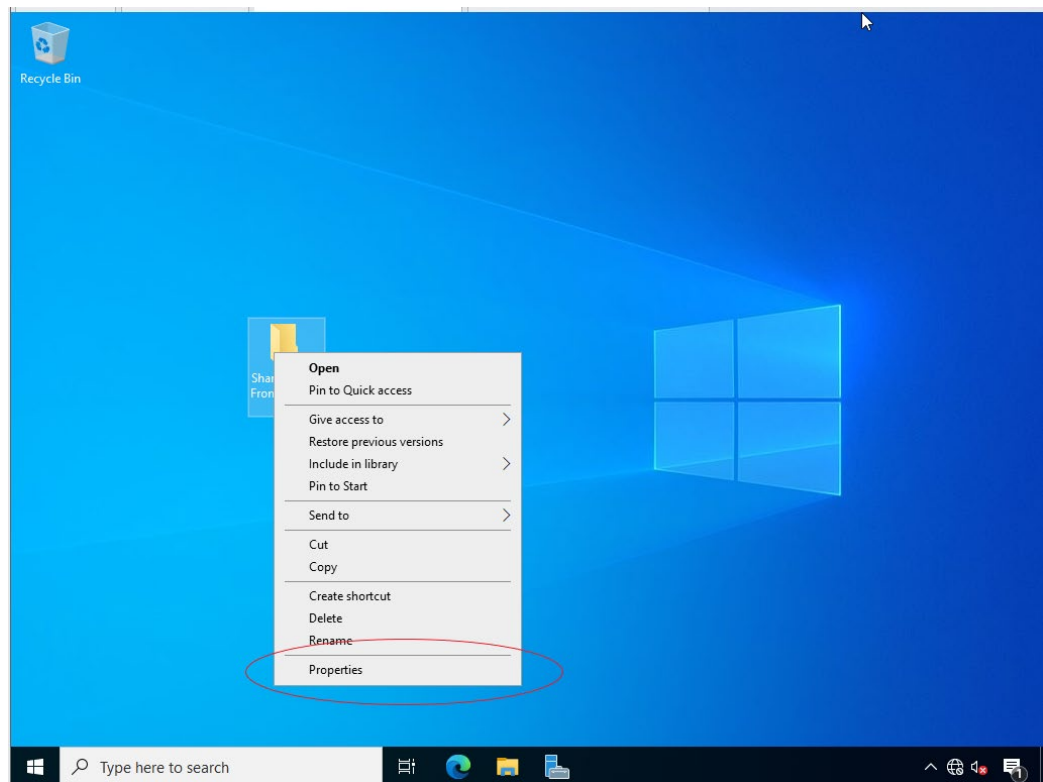




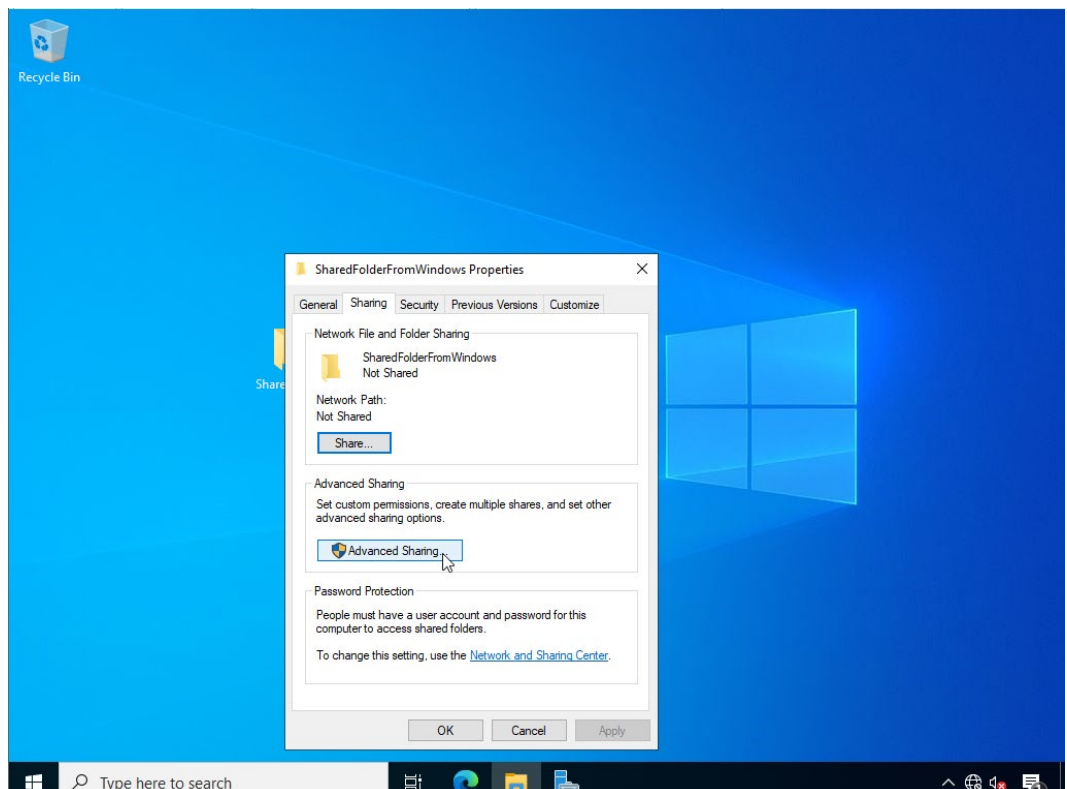
d) Create a new folder in the desired location



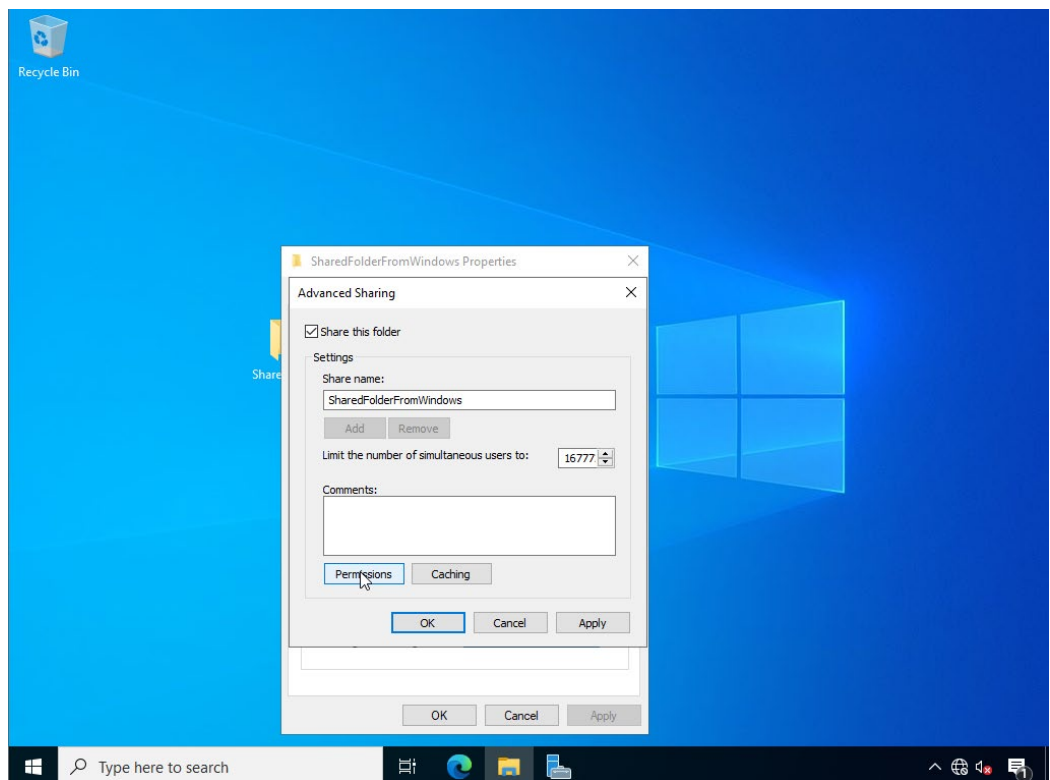
e) Right click on the folder, click on properties option



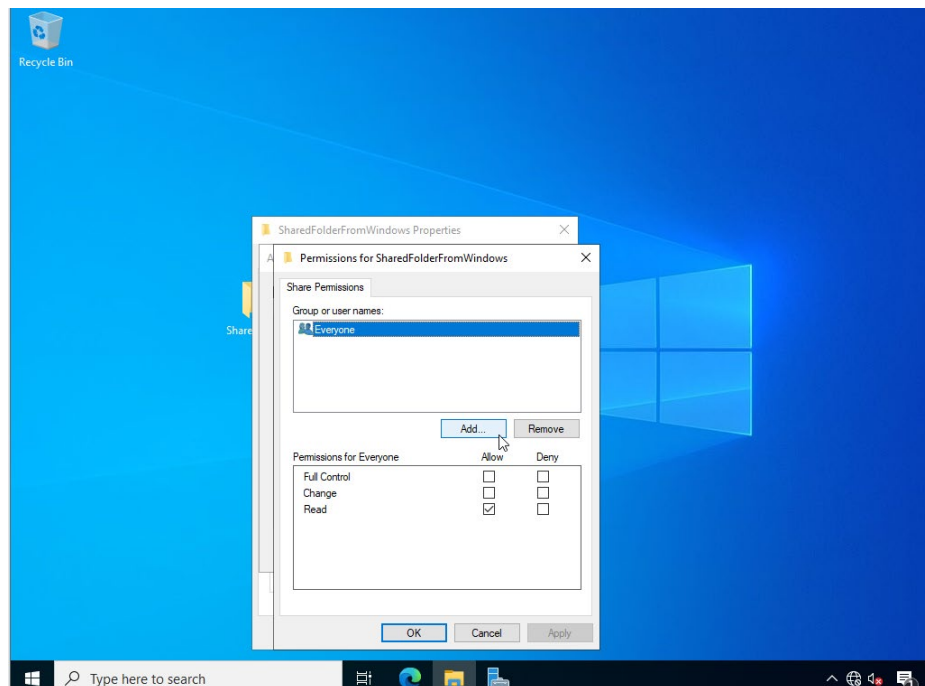
f) Click on Sharing, then click on Advanced Sharing



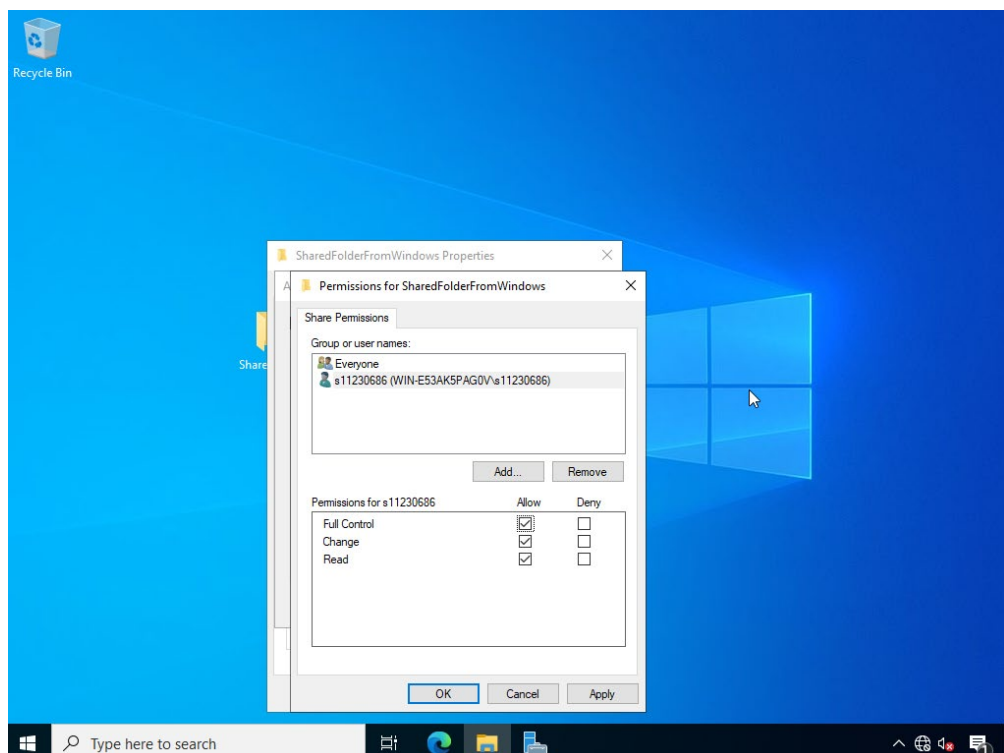
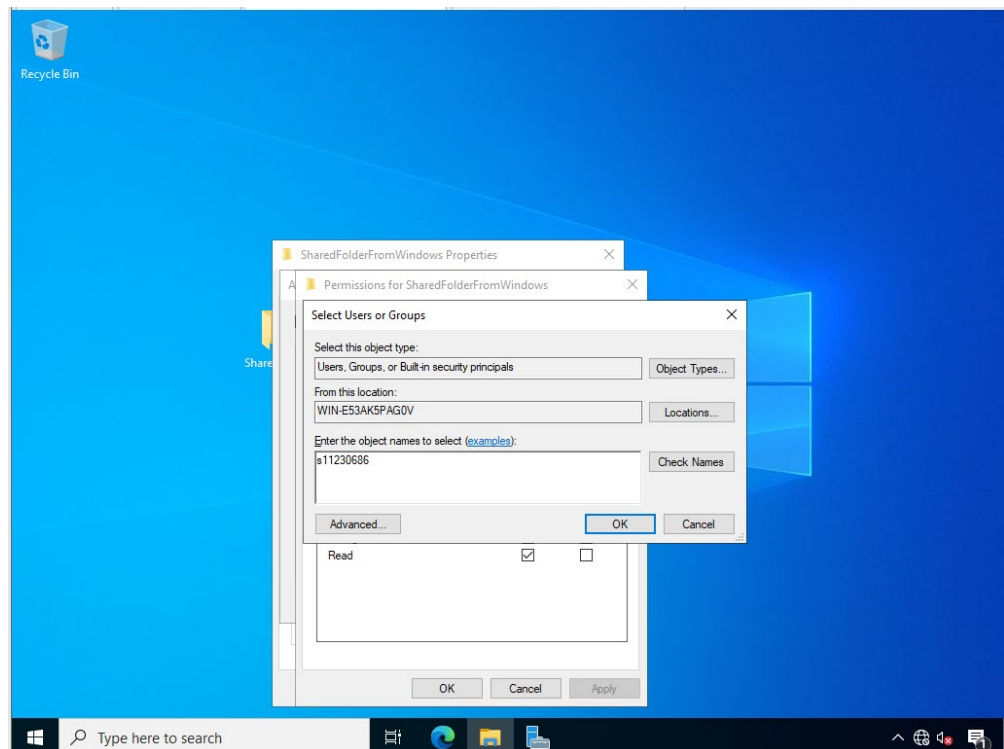
g) Check Share this folder, then click on Permission



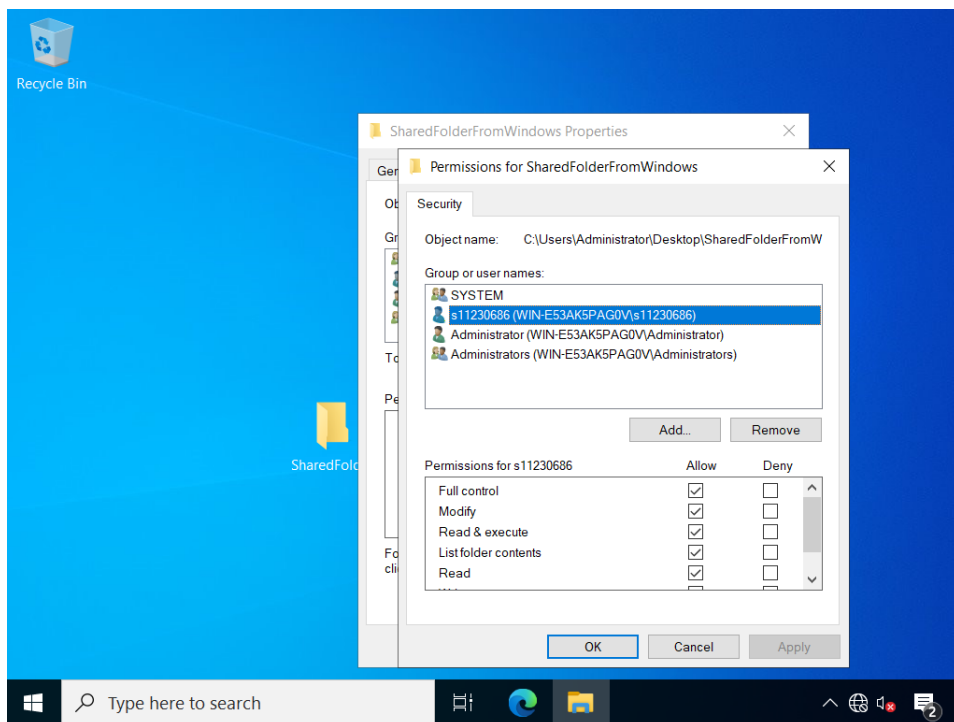
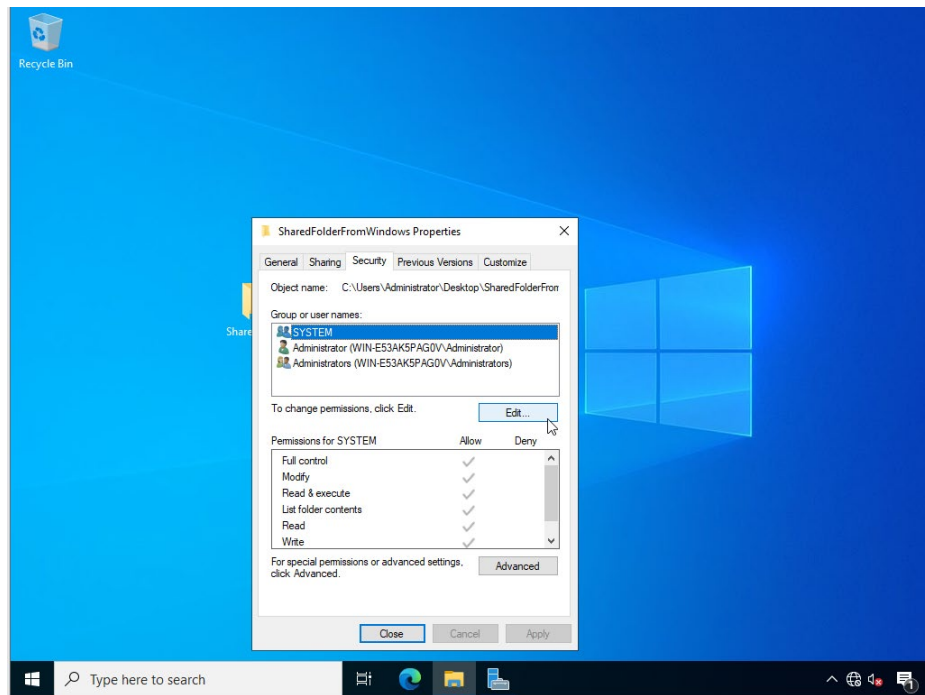
h) Click on Add



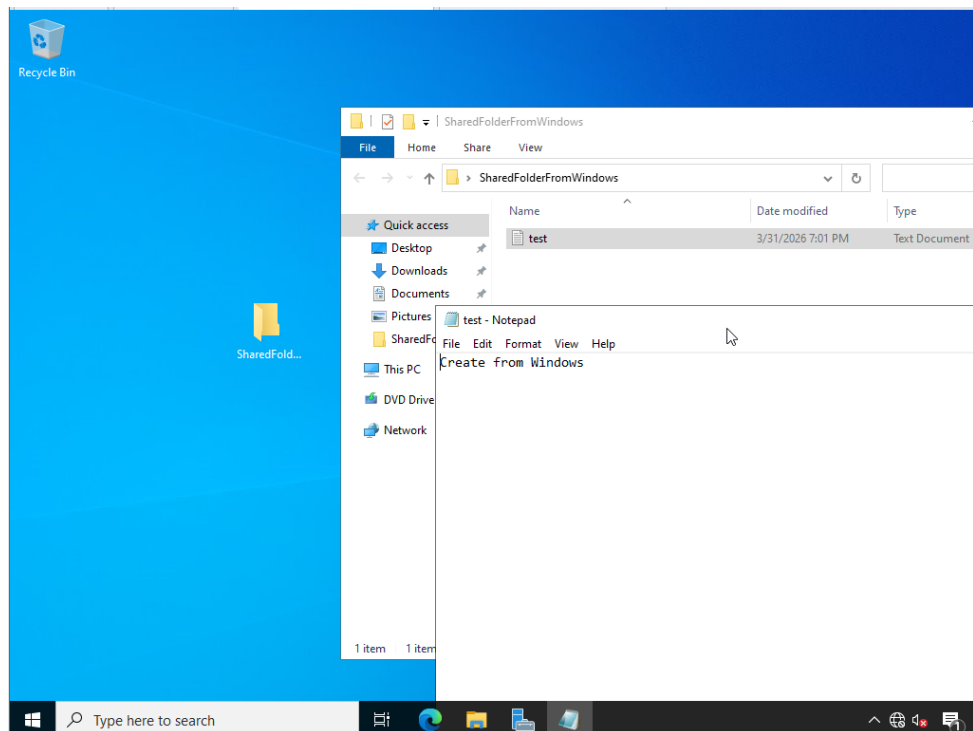
- i) Enter the user added before by type user account into the textbox, check Full Control to authorize the user full control of the shared folder



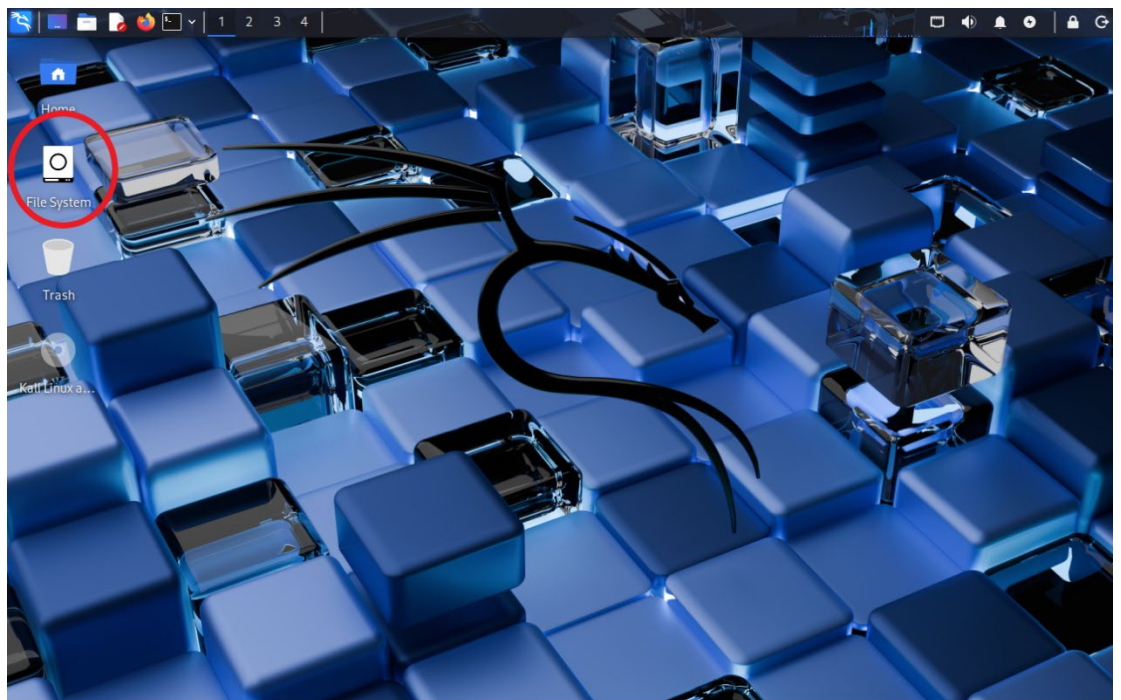
j) Go to Security, repeat step 8-10

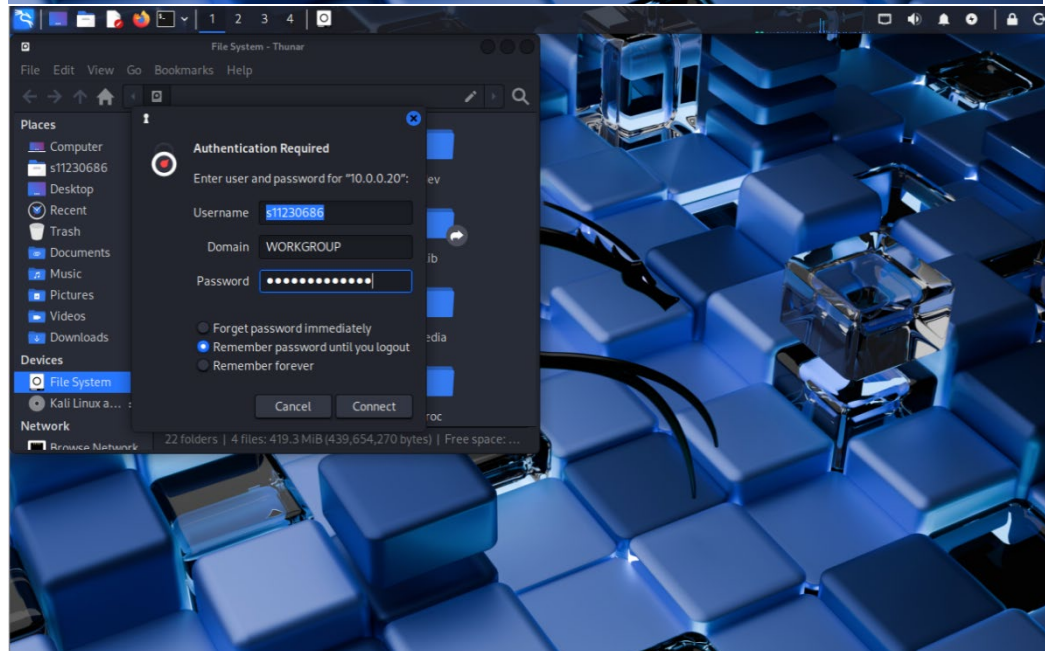
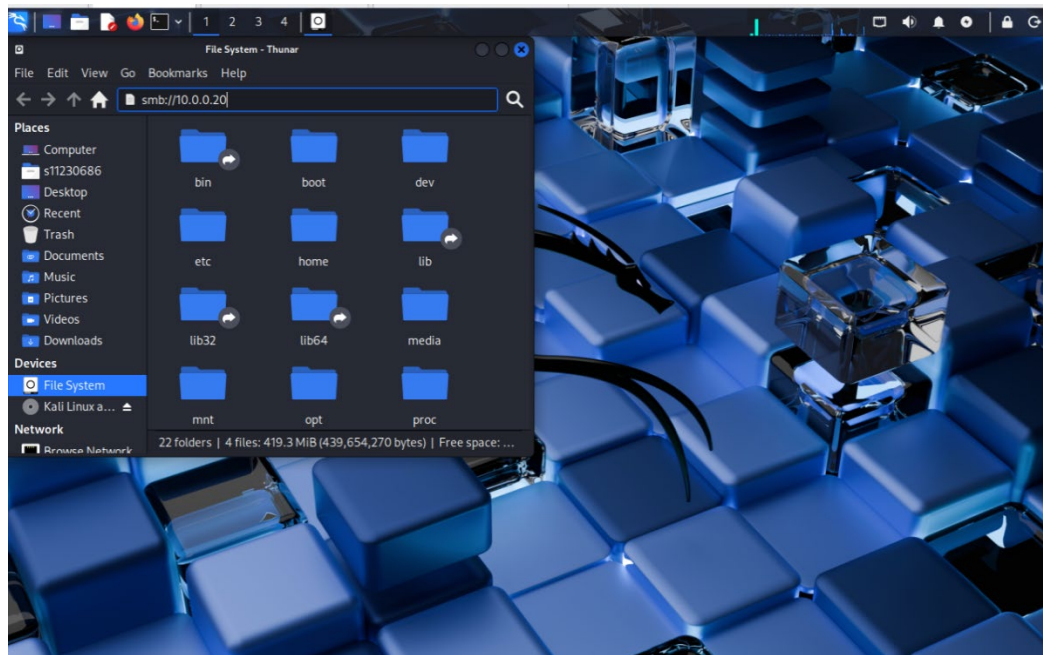


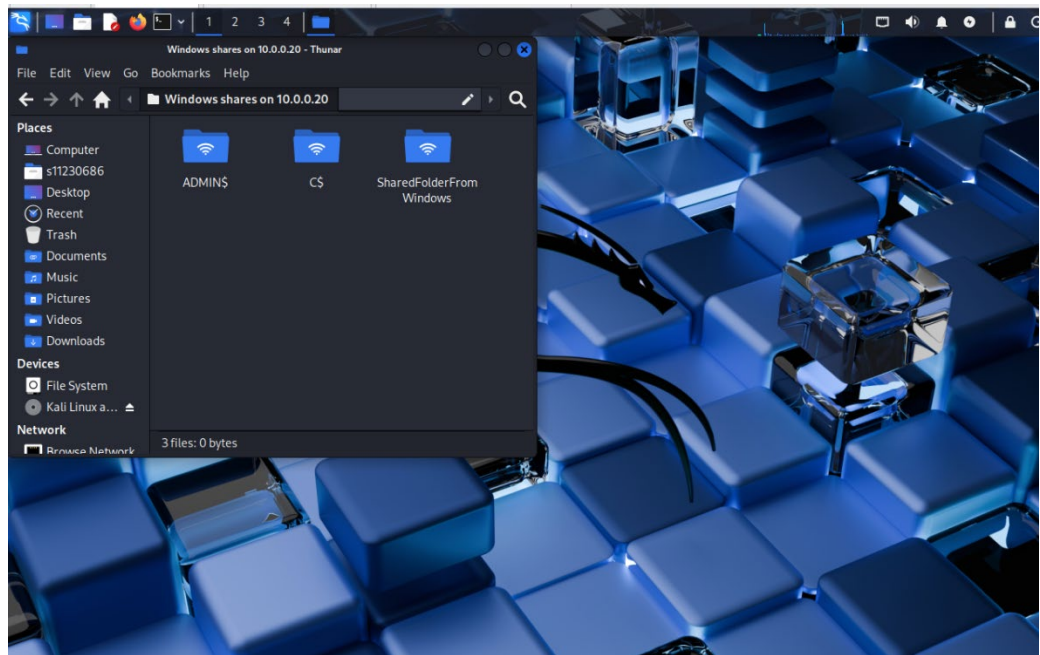
- k) Create a test.txt file to test authorizations of writing to file



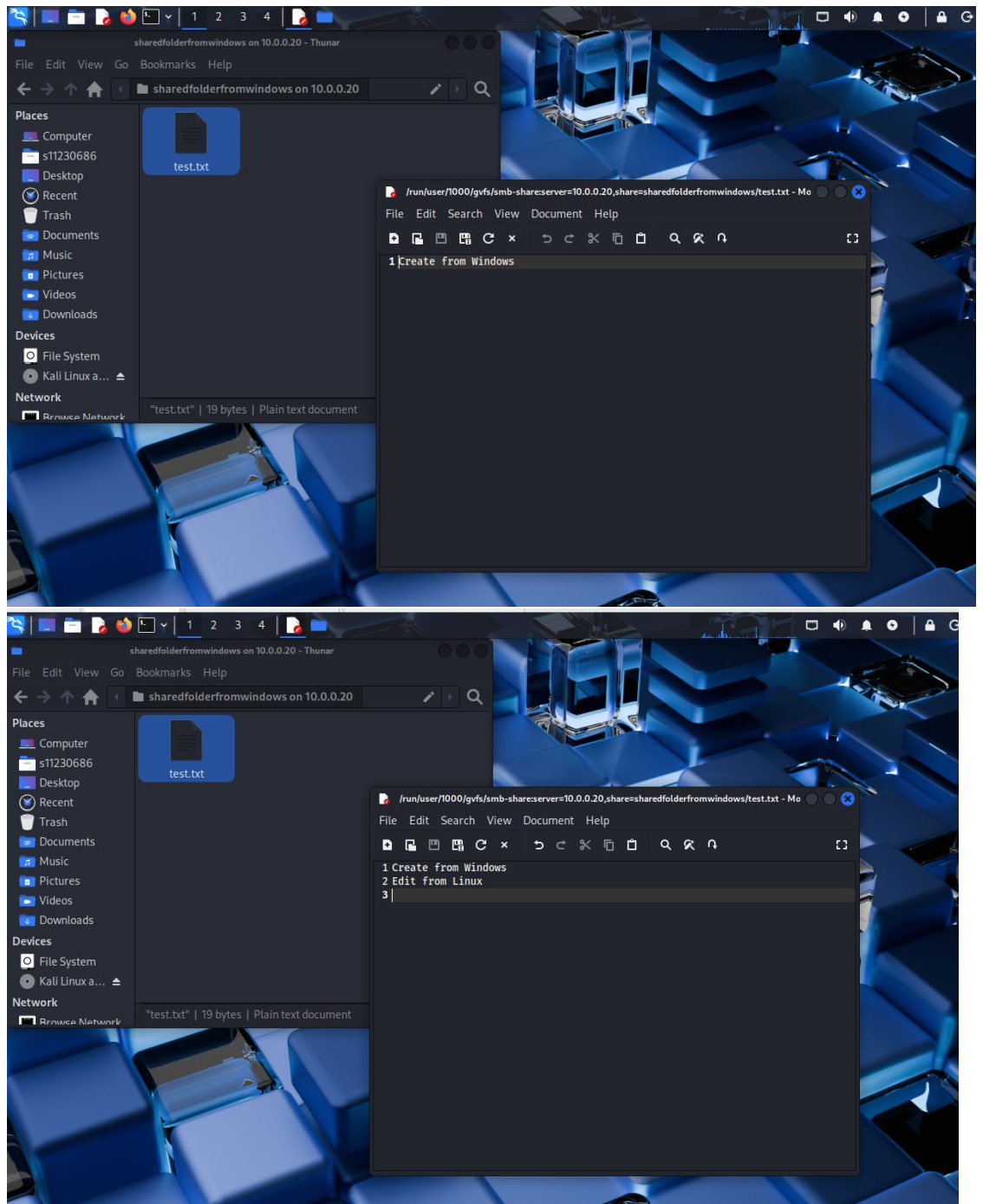
- l) Go to Kali Linux, check if file sharing is successful. Click on File System and enter `smb://10.0.0.20`, a login page will show up. Enter username and password, the shared folder should show up.



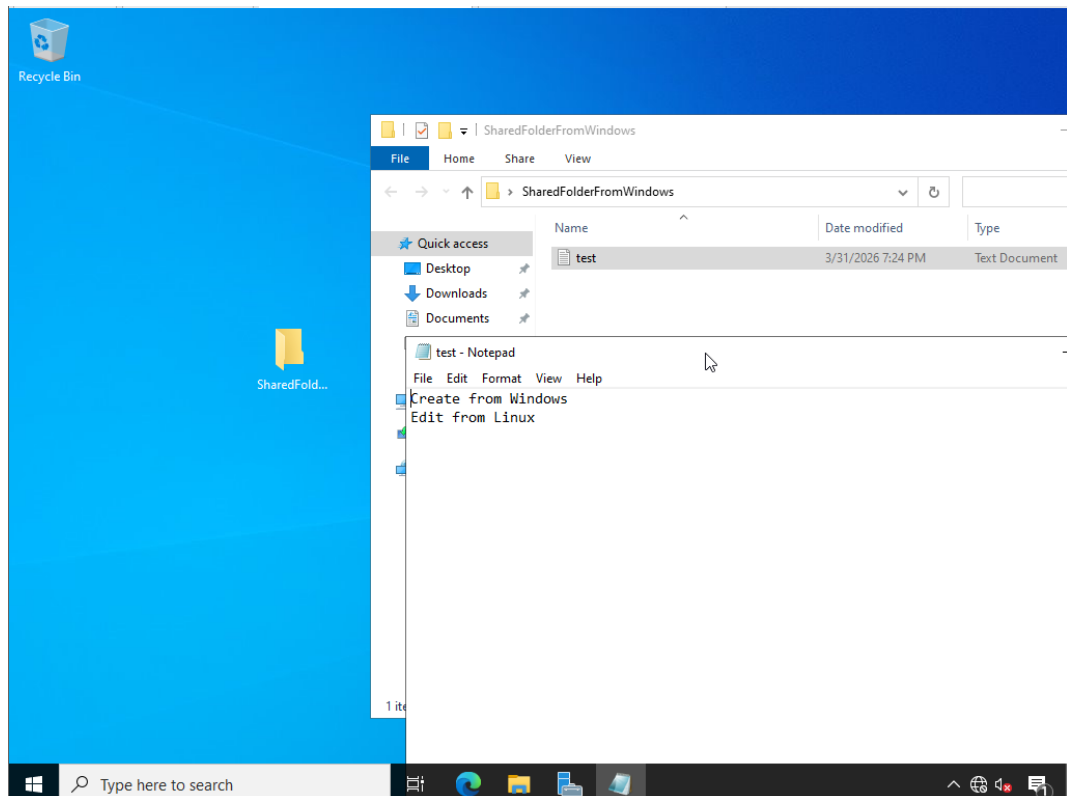




- m) Check writes permissions by editing the txt file, as shown in the figure, the txt file is editable from Linux



n) Crosscheck on Windows, the txt file has been successfully modified.



o) Go to Windows Server Core, open terminal, mount the path to Z by running command: `net use Z: \\10.0.0.x\sharedFolderName 'password' /user: username`. Enter Z: to access disk Z. Run `echo` command to test writes permissions from Windows Desktop Core.

```
Administrator: C:\Windows\system32\cmd.exe
WARNING: To launch Server Configuration tool again, run "SConfig"
PS C:\Users\Administrator> net use Z: \\10.0.0.20\SharedFolderFromWindows 'p@$w0rd12345' /user:s11230686
The command completed successfully.

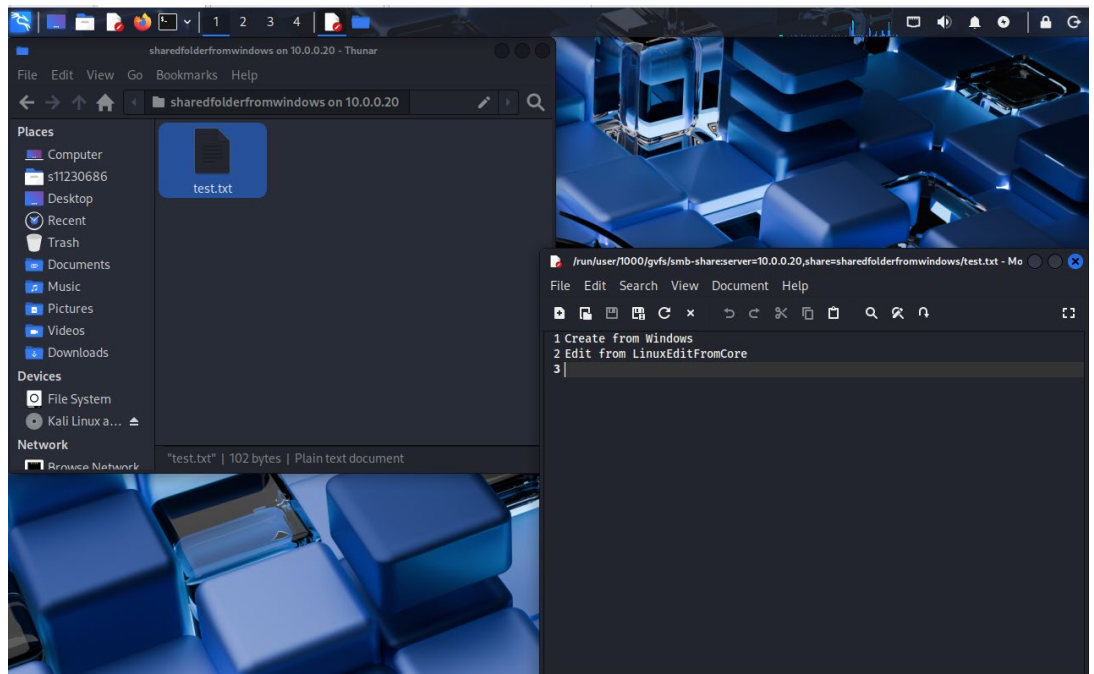
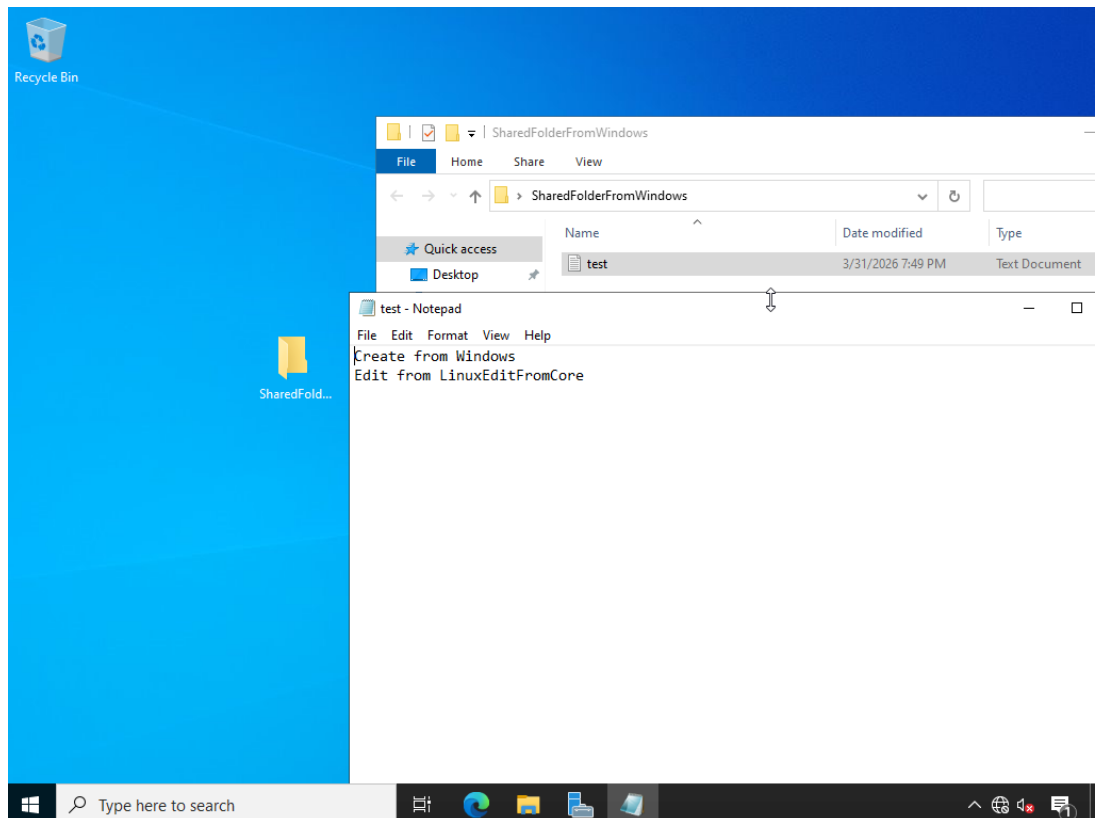
PS C:\Users\Administrator> Z:
PS Z:\> ls

Directory: Z:\

Mode                LastWriteTime         Length Name
----                -
-a----           3/31/2026   7:43 PM             74 test.txt

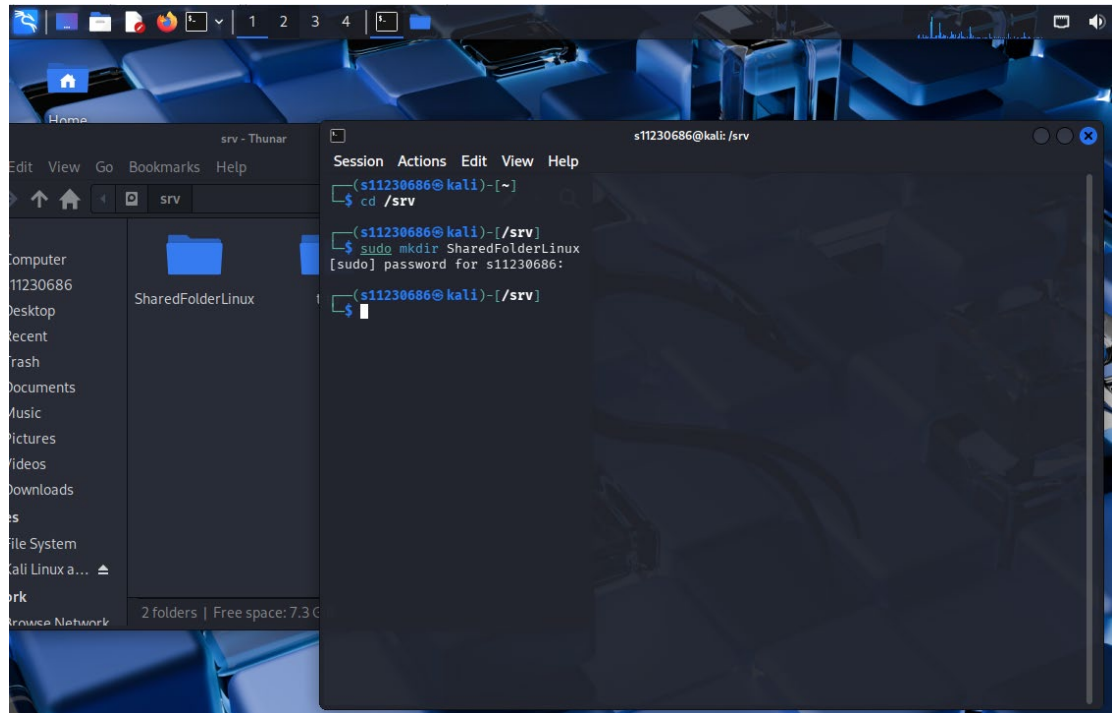
PS Z:\> cat test.txt
Create from Windows
Edit from Linux
PS Z:\> echo EditFromCore >> test.txt
PS Z:\>
```

p) Crosscheck with Windows Desktop and Linux

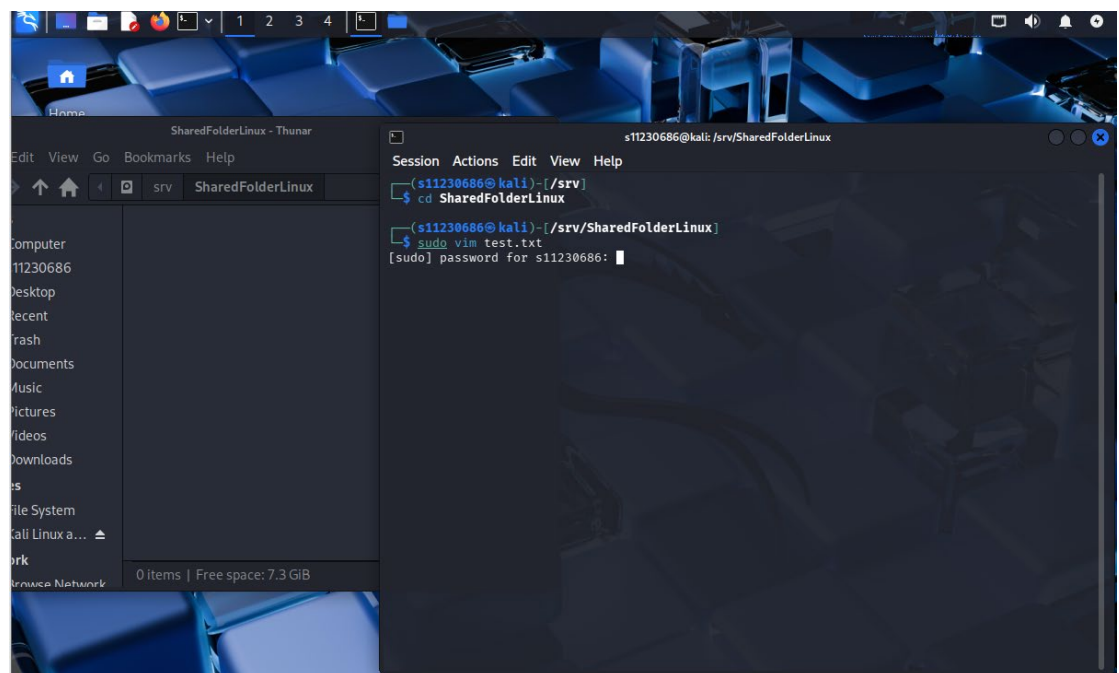


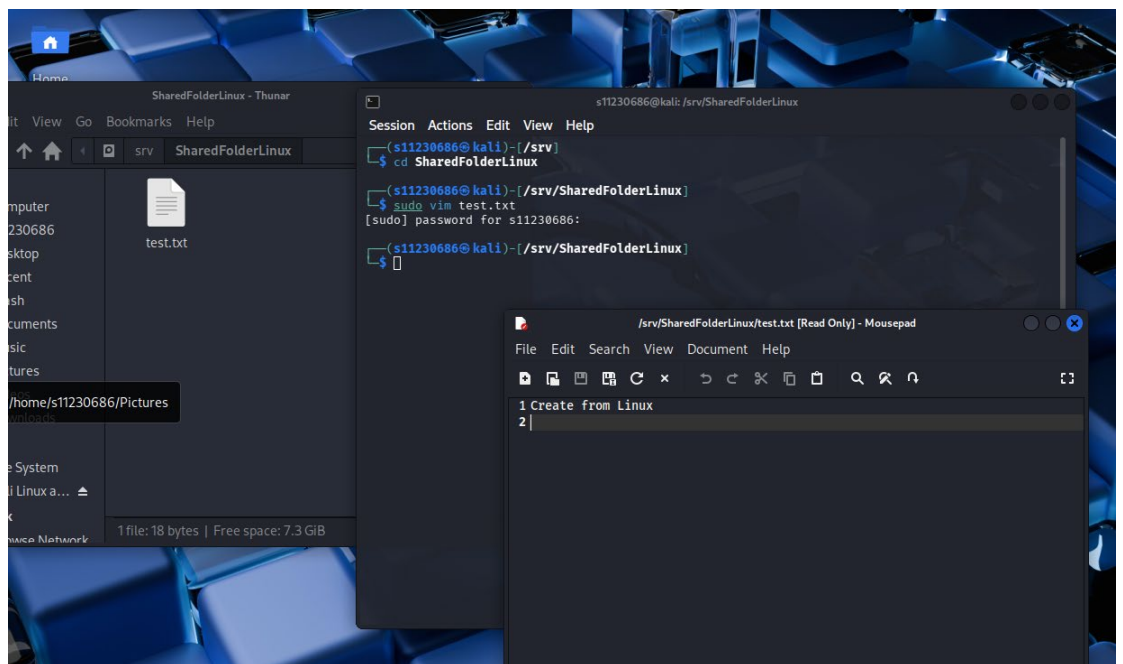
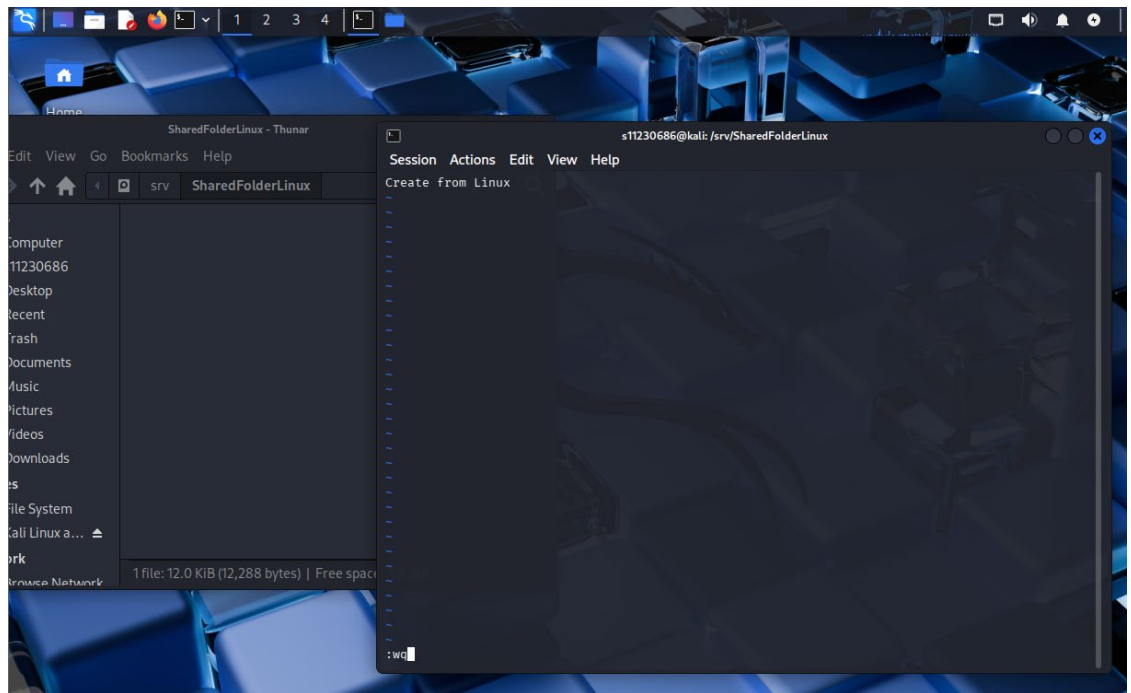
2) From Linux to Windows

- a) Open the terminal and navigate to the /srv directory by entering `cd /srv`. /srv is the default location for shared/service data of Linux system, the shared folder will be stored in here. Then use: `sudo mkdir FileName` to create a new folder under srv folder. The reason for using sudo will be discussed later



- b) Enter shared folder, create a txt file for the permission testing later





- c) Edit Samba configuration as figure below file to make the folder visible on the network. The name in square brackets will be used when mapping this path in Windows later

```
s11230686@kali: /srv/SharedFolderLinux
Session Actions Edit View Help
(s11230686@kali)-[/srv]
$ cd SharedFolderLinux

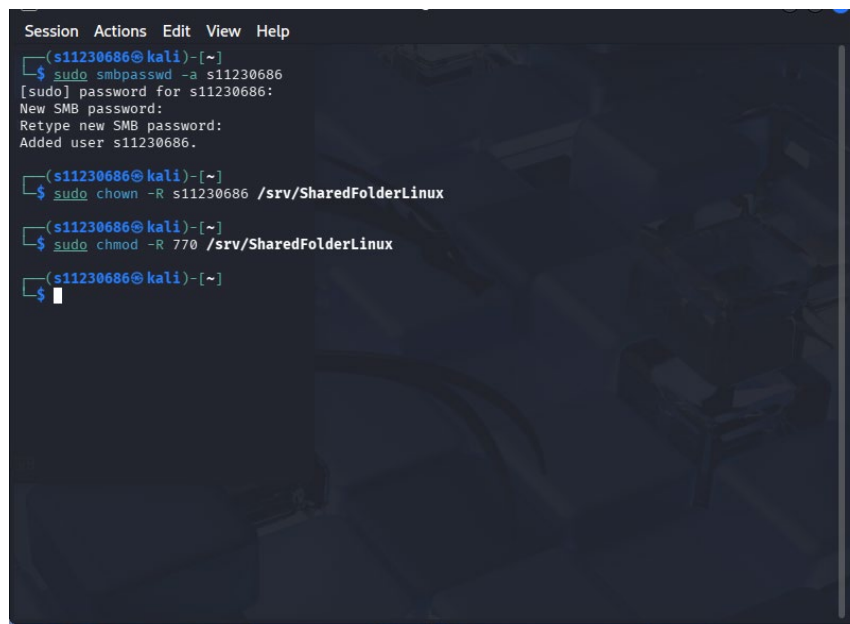
(s11230686@kali)-[/srv/SharedFolderLinux]
$ sudo vim test.txt
[sudo] password for s11230686:

(s11230686@kali)-[/srv/SharedFolderLinux]
$ sudo vim /etc/samba/smb.conf
```

```
s11230686@kali: /srv/SharedFolderLinux
Session Actions Edit View Help
[printers]
comment = All Printers
browseable = no
path = /var/spool/samba
printable = yes
guest ok = no
read only = yes
create mask = 0700

# Windows clients look for this share name as a source of downloadable
# printer drivers
[print$]
comment = Printer Drivers
path = /var/lib/samba/printers
browseable = yes
read only = yes
guest ok = no
# Uncomment to allow remote administration of Windows print drivers.
# You may need to replace 'lpadmin' with the name of the group your
# admin users are members of.
# Please note that you also need to set appropriate Unix permissions
# to the drivers directory for these users to have write rights in it
; write list = root, @lpadmin
[SharedFolderLinux]
path = /srv/SharedFolderLinux
browseable = yes
read only = no
guest ok = no
valid users = s11230686
~
~
~
248,23 Bot
```

- d) Samba has its own user system, the dedicated user should be added into the system manually by running: `sudo smbpasswd -a username`. After this, to ensure that specific users have permission to modify and access this file, execute these two commands:
1. `sudo chown -R username:workgroup /srv/SharedFolderName`
To assign ownership of the folder to the specific user in the specific workgroup, in this assignment, it's StudentID: StudentID
 2. `sudo chmod -R 770 /srv/SharedFolderName`
- e) To ensure the dedicated user (the owner of this folder) and other users in the same workgroup are able to write, modify and execute the file/folder



```
Session Actions Edit View Help
(s11230686@kali)-[~]
$ sudo smbpasswd -a s11230686
[sudo] password for s11230686:
New SMB password:
Retype new SMB password:
Added user s11230686.

(s11230686@kali)-[~]
$ sudo chown -R s11230686 /srv/SharedFolderLinux

(s11230686@kali)-[~]
$ sudo chmod -R 770 /srv/SharedFolderLinux

(s11230686@kali)-[~]
$
```

- f) Test whether file sharing works on the Windows Server desktop. Open powershell/terminal, run this command:
`net use Z: \\IP_Address\SharedFolderName /user:username`
Enter password, the command completed successfully message indicates the path has been mounted successfully.

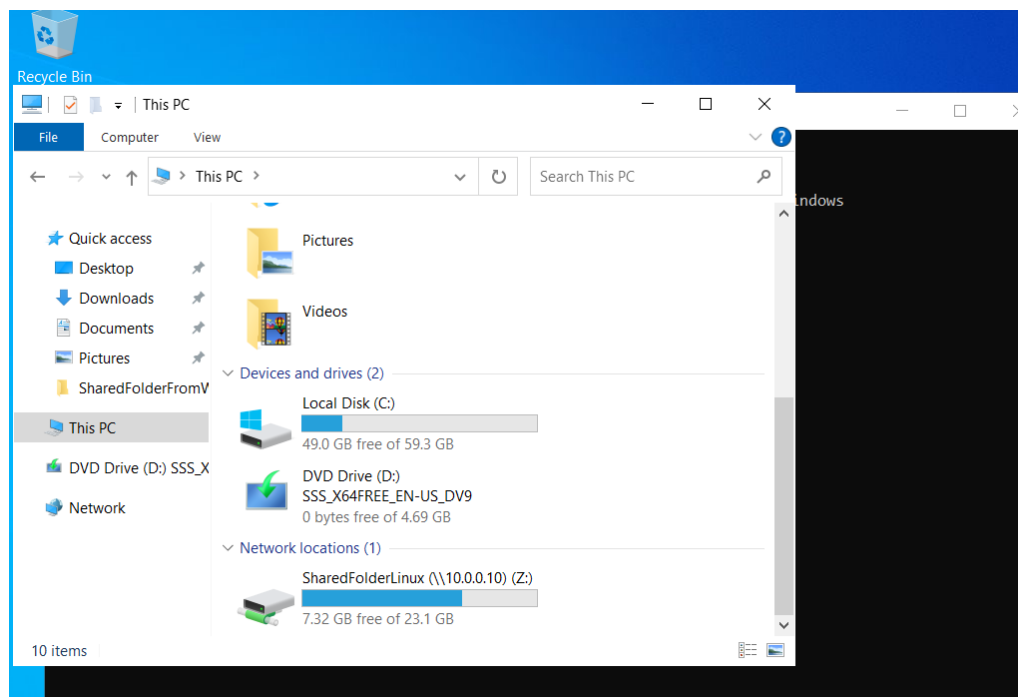
```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

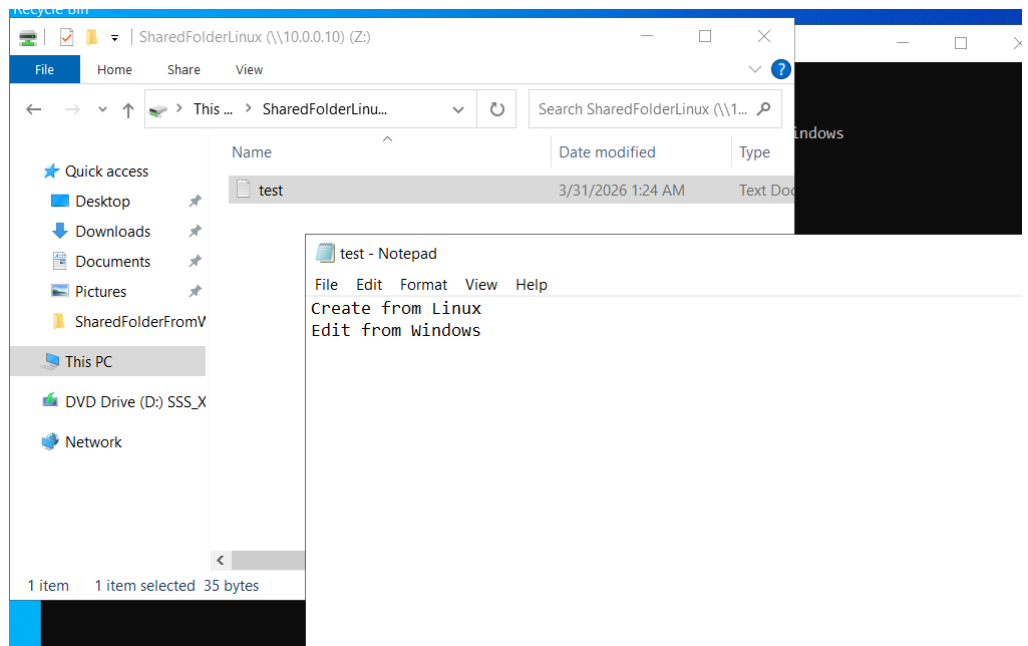
PS C:\Users\Administrator> net use Z: \\10.0.0.10\SharedFolderLinux /user:s11230686
Enter the password for 's11230686' to connect to '10.0.0.10':
The command completed successfully.

PS C:\Users\Administrator> _
```

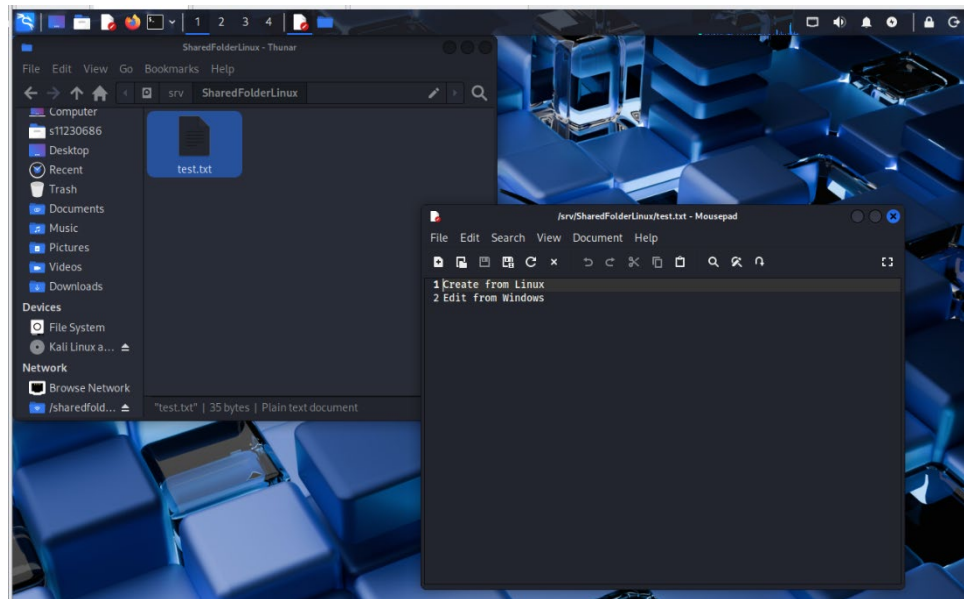
g) Open file explorer, the mounted disk should show up



h) Enter the shared folder, modify the txt file



i) Check from Linux



j) Go to Windows Desktop Core, mount the path of the shared folder, read it and modify it

```
Administrator: C:\Windows\system32\cmd.exe
WARNING: To launch Server Configuration tool again, run "SConfig"
PS C:\Users\Administrator> net use X: \\10.0.0.10\SharedFolderLinux /user:s11230686
Enter the password for 's11230686' to connect to '10.0.0.10':
The command completed successfully.

PS C:\Users\Administrator> X:
PS X:\> ls

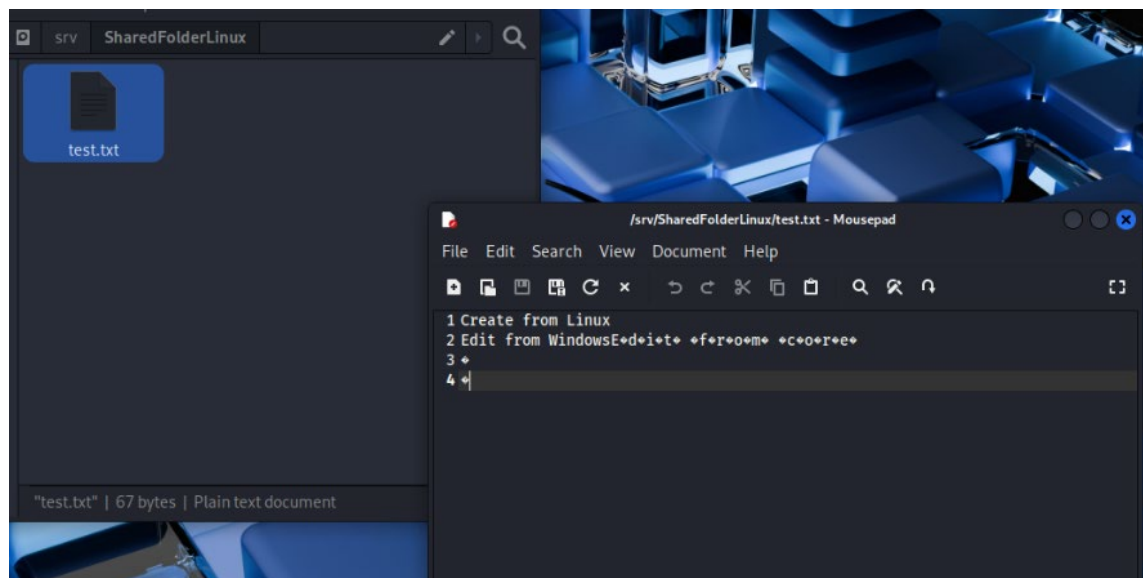
Directory: X:\

Mode                LastWriteTime         Length Name
----                -
-a----           3/31/2026  1:24 AM             35 test.txt

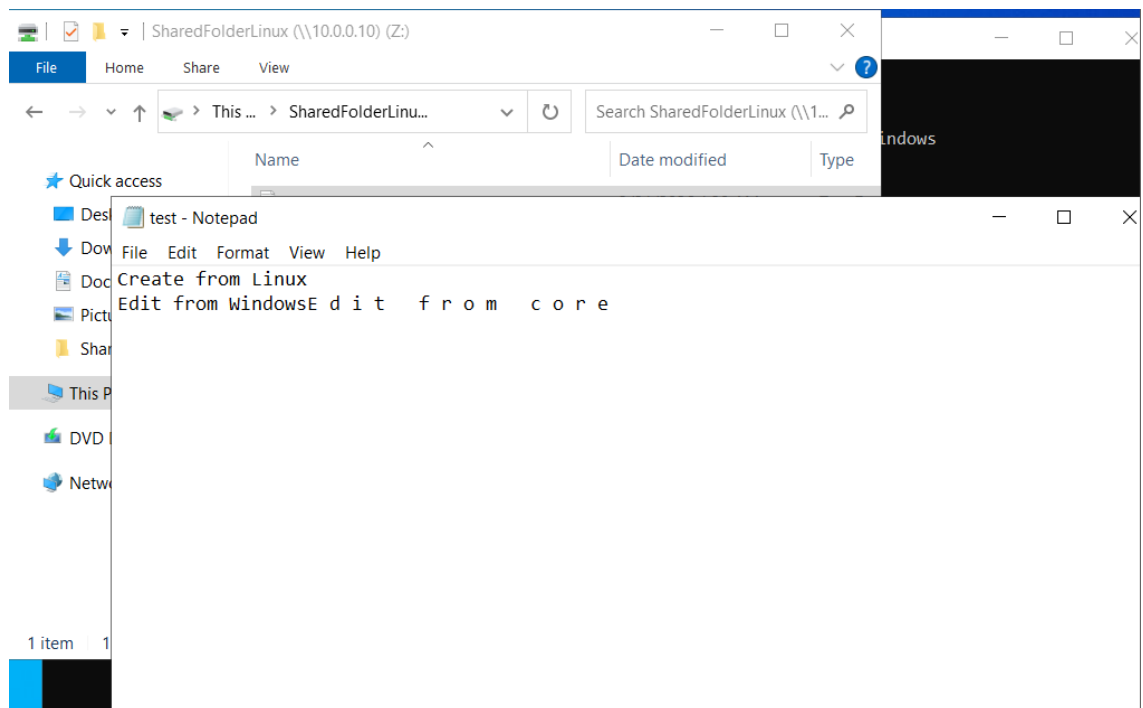
PS X:\> cat test.txt
Create from Linux
Edit from Windows
PS X:\> echo "Edit from core" >>test.txt
PS X:\> cat test.txt
Create from Linux
Edit from WindowsEdit from core

PS X:\> .
```

k) Check from Linux



l) Check from Windows Server Desktop



2. Differences between Windows and Linux

a. Security Features, User Management, and Access Control

1) Linux

In terms of security features, Linux manages network security by iptables or front-end wrappers like UFW. UFW is not a default system tool of Kali, however, for those distributions with UFW (Linux Mint), it is usually disabled by default. The iptables allow all traffic by default.

```
(s11230686@kali)-[~]
$ sudo iptables -S
[sudo] password for s11230686:
-P INPUT ACCEPT
-P FORWARD ACCEPT
-P OUTPUT ACCEPT
```

In terms of user management, user data is usually managed by /etc/passwd and /etc/group

```
(s11230686@kali)-[~]
$ cat /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
pcscd:x:983:983:PC/SC Smart Card Daemon:/:usr/sbin/nologin
s11230686:x:1000:1000:s11230686,,,:/home/s11230686:/usr/bin/zsh

(s11230686@kali)-[~]
$ cat /etc/group
root:x:0:
kali:x:1234:
s11230686:x:1000:
```

In addition, administrative privileges of Linux are strictly controlled by separating the super user from normal users. The super user/root user has full control over the system. Normal users are not able to perform system operations; they can only get limited privilege elevation temporarily through 'sudo'. For example, a normal user like the user s11230686 created before was not allowed to modify /srv directory which belongs to the root user unless using 'sudo' command. This is also a part of security features of Linux.

```
(s11230686@kali)-[~]
$ cd /srv

(s11230686@kali)-[/srv]
$ mkdir SharedFolderLinux
mkdir: cannot create directory 'SharedFolderLinux': Permission denied

(s11230686@kali)-[/srv]
$ sudo mkdir SharedFolderLinux
[sudo] password for s11230686:

(s11230686@kali)-[/srv]
$ ls
SharedFolderLinux  tftp
```

This leads to the discussion of access control, which is part of security features as well. Linux uses a user-group-others (UGO) model to control file/folder access. In this model, each file and folder belongs to an owner and a group. Other users are users other than 'Owner' and 'Group'. This is also a part of user management. The user management of Linux is highly combined with

access control. Basic permissions of file access are Read (symbolized as 'r'), Write (symbolized as 'w') and Execute (symbolized as 'x'). For example, as shown below, the SharedFolderLinux belongs to user root and group root, the owner of the file is allowed to read, write and execute this file, group users and other users are not allowed to write the file|

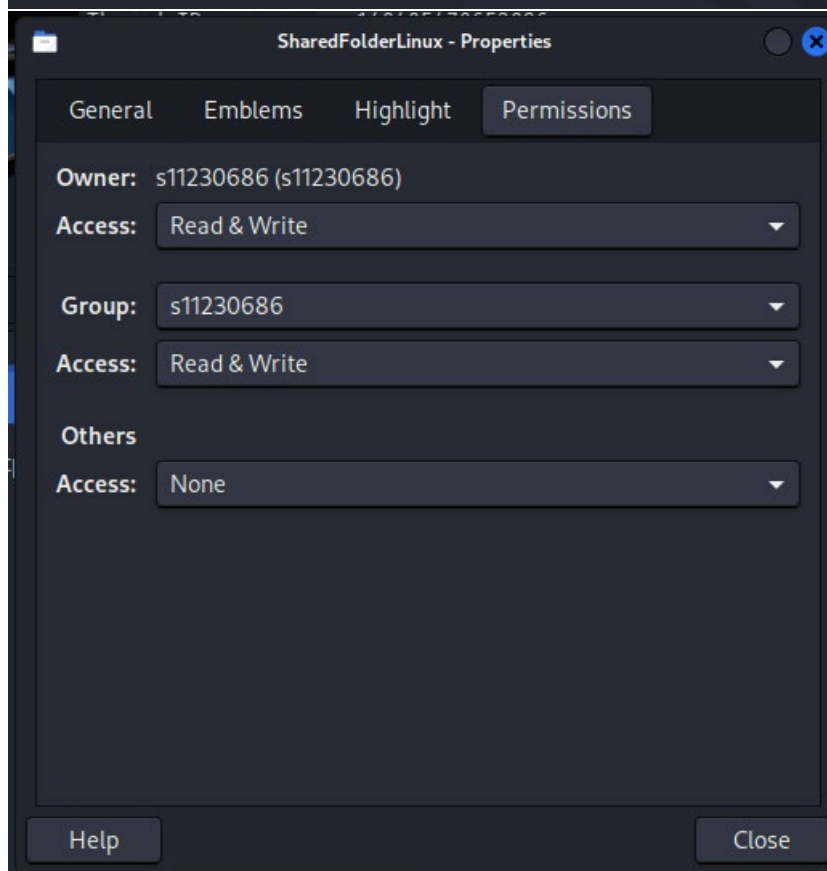
```
(s11230686@kali)-[/srv]
$ ls -l
total 8
drwxr-xr-x 2 root    root    4096 Mar 31 23:10 SharedFolderLinux
drwxr-xr-x 2 nobody nogroup 4096 Mar 28 20:19 tftp
```

Access permissions can be modified through 'chown' and 'chmod'. Use chown to change the owner of the file and 'chmod' to assign file access permissions to users.

```
(s11230686@kali)-[/srv]
$ sudo chown -R s11230686:s11230686 /srv/SharedFolderLinux
[sudo] password for s11230686:

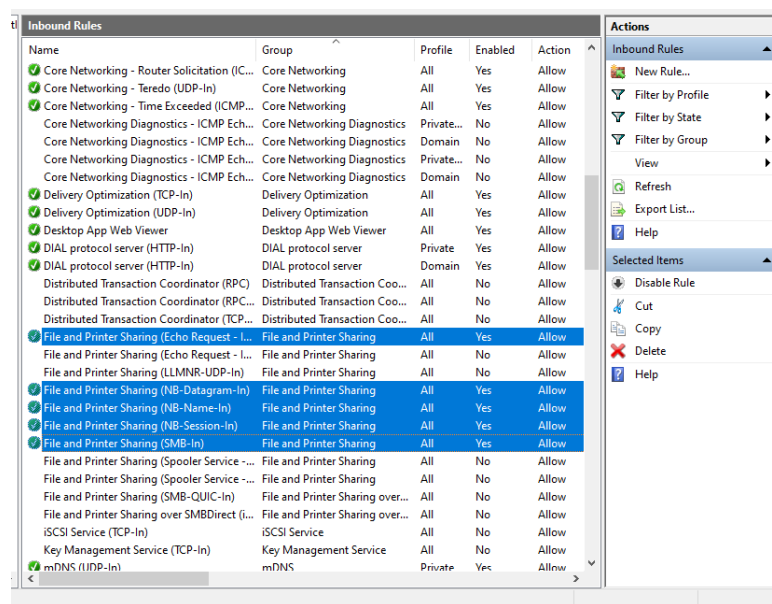
(s11230686@kali)-[/srv]
$ sudo chmod -R 770 /srv/SharedFolderLinux

(s11230686@kali)-[/srv]
$ ls -l
total 8
drwxrwx--- 2 s11230686 s11230686 4096 Mar 31 23:10 SharedFolderLinux
drwxr-xr-x 2 nobody    nogroup   4096 Mar 28 20:19 tftp
```

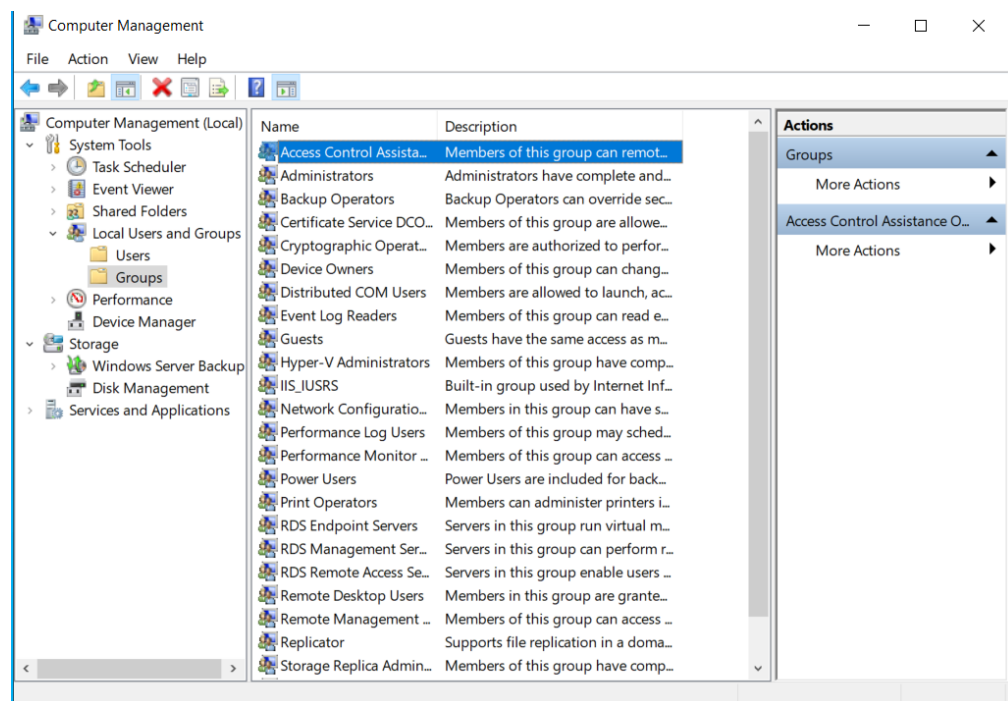


2) Windows

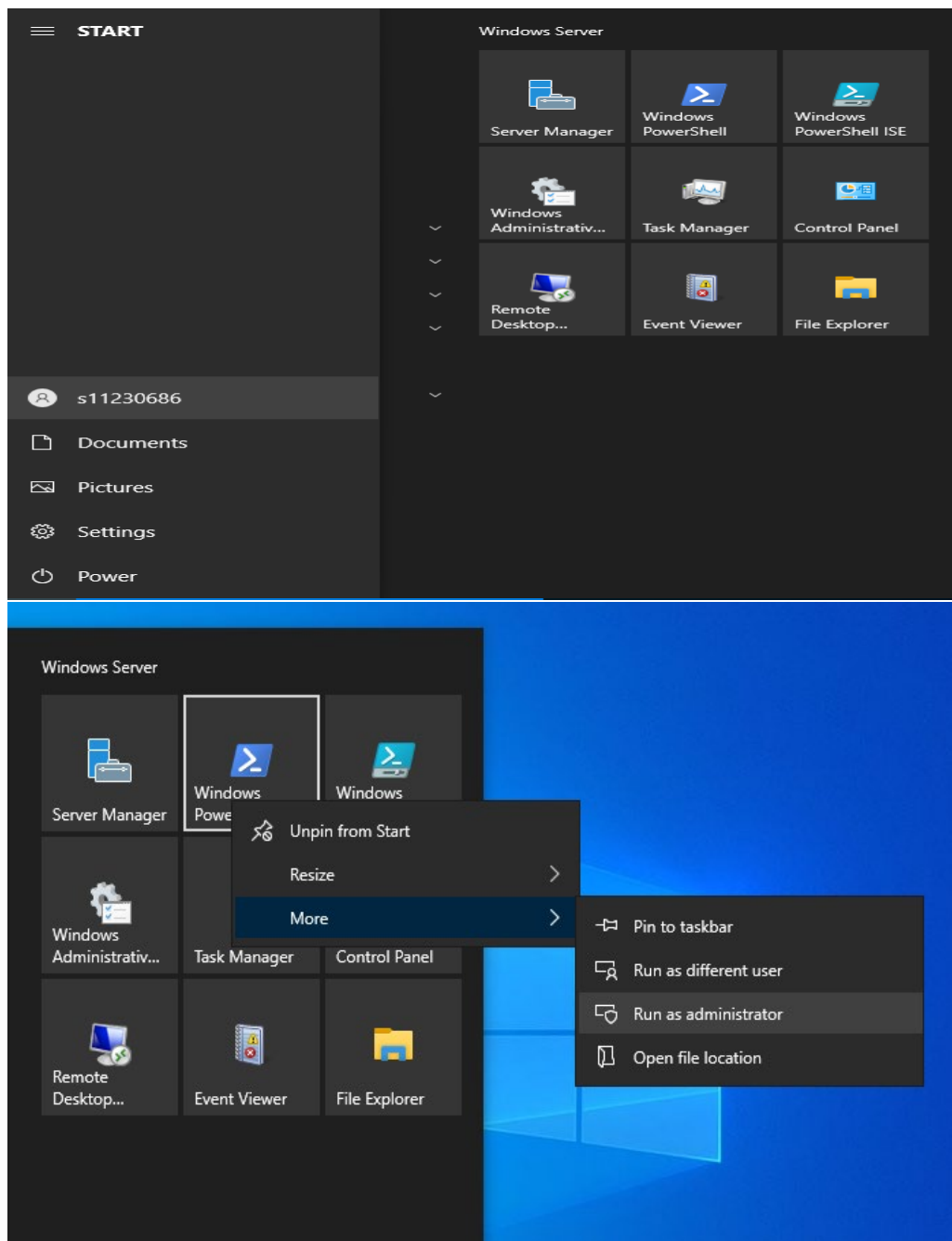
In terms of security features, Windows Server integrates Windows Defender Firewall into the system. All traffic through Windows is managed by profiles and filtering rules. Inbound rules include ICMP rules and the File and Printer Sharing rules, external ping requests and file sharing are controlled by these rules. This is the reason of enabling related inbound rules before sending ping requests and sharing folders.



In terms of user management, users and groups are managed through the Computer Management Microsoft Management Console (MMC), as shown below



Windows separates normal users from administrative users by group membership. Normal users are stored in user group by default with limited privileges, but they could get privilege elevation through User Access Control Lists (ACLs) to perform limited system-level operations or become an administrator account once added into Administrator group. For example, when the normal user created for this assignment before tried to open PowerShell with administrator privileges, a UAC prompt appears.



User Account Control



Do you want to allow this app to make changes to your device?



Windows PowerShell

Verified publisher: Microsoft Windows

[Show more details](#)

To continue, enter an admin user name and password.

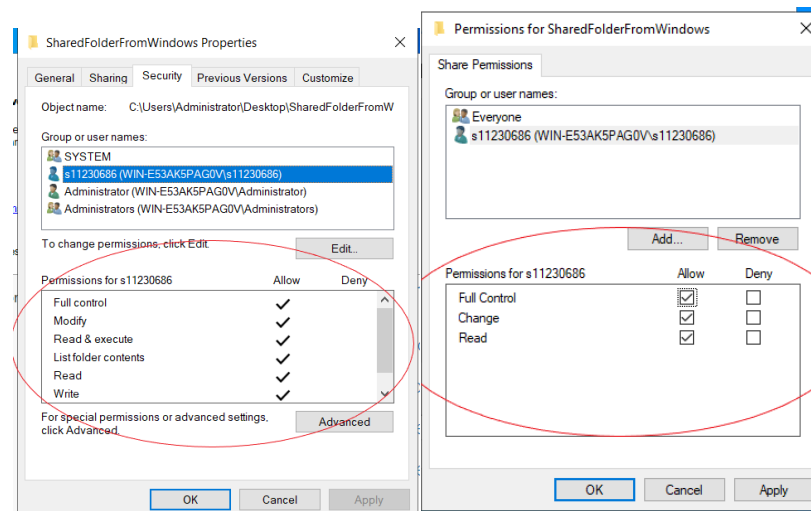
Administrator

WIN-E53AK5PAG0V\Administrator

Yes

No

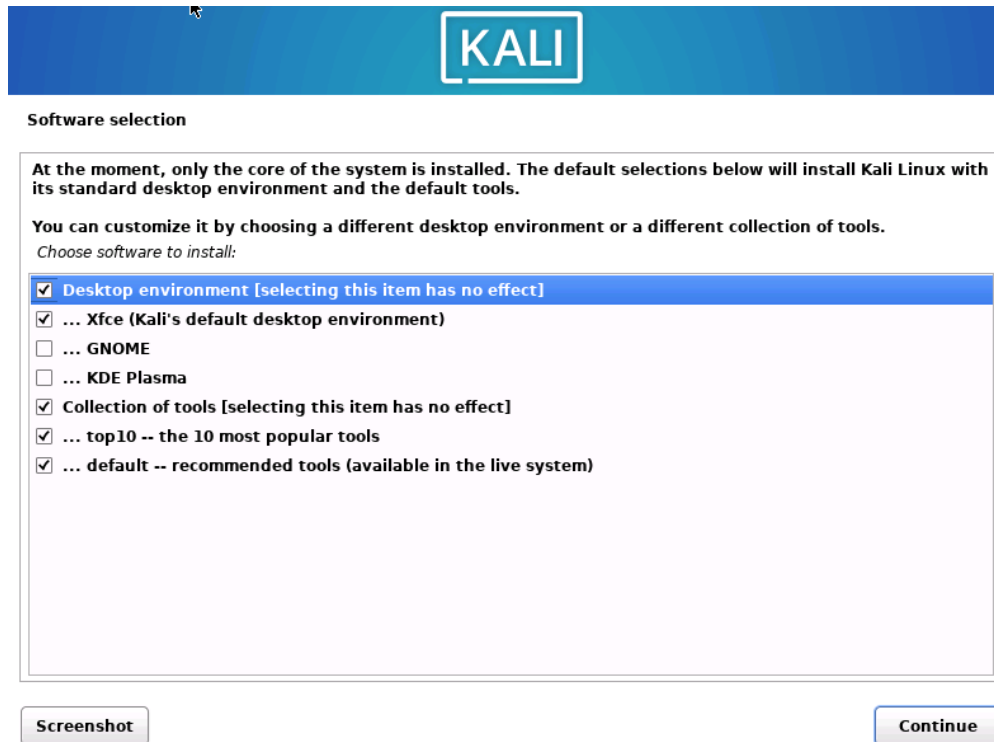
In terms of access control, New Technology File System (NTFS) controls file and folder access. NTFS enables detailed permissions of file/folder accessing to be assigned to users and groups using Access Control Lists. Share permissions is another access control layer of the shared folders. When a file is shared over the network through SMB protocol, the permissions for the shared folder will be isolated from the local rights to read or modify the file itself, therefore both Shared Permissions and Security Permissions need to be modified to share a folder



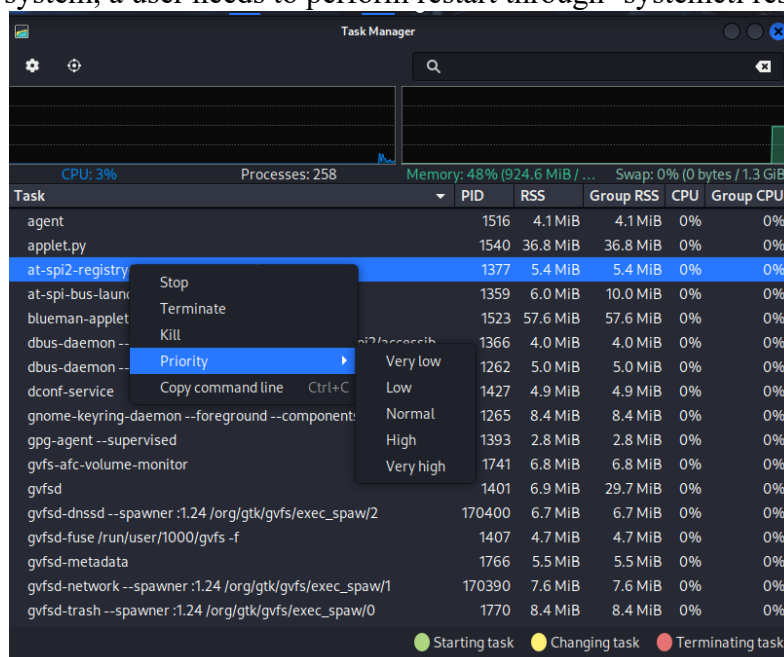
b. Design and User Interface

1) Linux

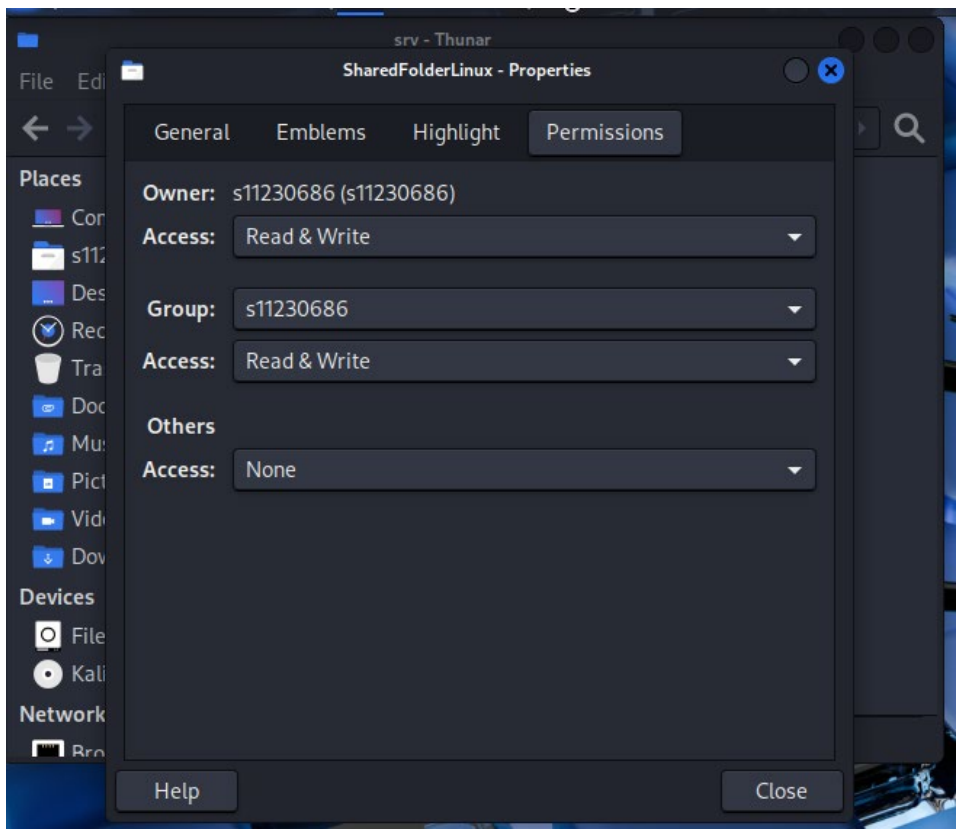
In Linux, graphic user interface is optional and customizable. For example, during the installation of Kali, desktop environment is optional. Xfce, KDE, Plasma are all desktop environments, a user can choose to install any of them or none of them.



However, due to the core design philosophy of Linux, which is deeply rooted in the terminal, administrative operations are completer and more precise. For example, Xfce, the default GUI of Kali doesn't provide restart option for progress running on the system, a user needs to perform restart through 'systemctl restart' command



Another example is the ownership of a file or folder is not able to be changed through Xfce GUI, in this assignment, the ownership of SharedFolderLinux is transferred to another user through ‘chown’

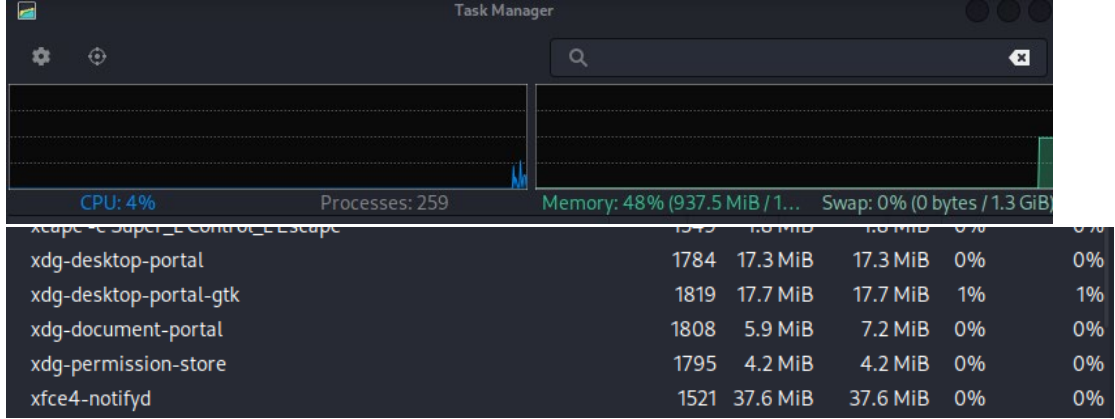


```
(s11230686@kali)-[/srv]
$ sudo chown -R s11230686:s11230686 /srv/SharedFolderLinux
[sudo] password for s11230686:

(s11230686@kali)-[/srv]
$ sudo chmod -R 770 /srv/SharedFolderLinux

(s11230686@kali)-[/srv]
$ ls -l
total 8
drwxrwx--- 2 s11230686 s11230686 4096 Mar 31 23:10 SharedFolderLinux
drwxr-xr-x 2 nobody    nogroup   4096 Mar 28 20:19 tftp
```

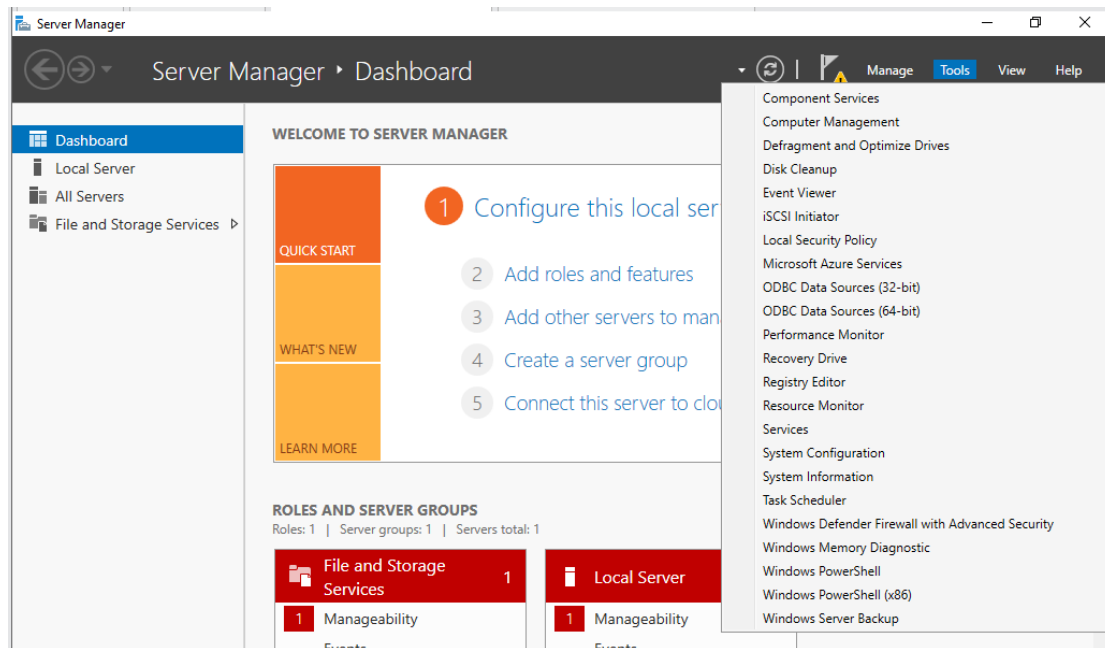
The design of Linux makes it more lightweighted, requiring less resources



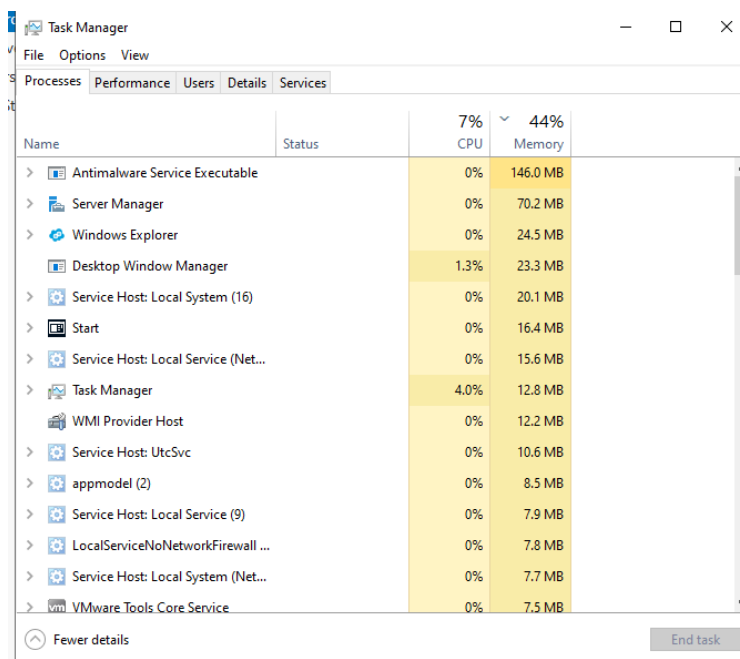
2) Windows Server

There are two different versions of Windows Server, Windows Server Desktop, which is based on GUI interfaces, and Windows Server Core, which is command-line based.

As demonstrated in the system installation session, the interaction logic of Windows Server is primarily rooted in GUI-based navigation. Users can perform most common administrative operations through graphical tools.



However, this also makes it heavy weighted, requiring more resources



In addition, although Windows Server Core is command line based, Microsoft still provides a user-friendly text-based management menu through SConfig

```
=====
Welcome to Windows Server 2022 Standard Evaluation
=====
```

- | | |
|---------------------------------------|----------------------|
| 1) Domain/workgroup: | Workgroup: WORKGROUP |
| 2) Computer name: | WIN-8CCV3CP2GOS |
| 3) Add local administrator | |
| 4) Remote management: | Enabled |
| 5) Update setting: | Download only |
| 6) Install updates | |
| 7) Remote desktop: | Disabled |
| 8) Network settings | |
| 9) Date and time | |
| 10) Telemetry setting: | Required |
| 11) Windows activation | |
| 12) Log off user | |
| 13) Restart server | |
| 14) Shut down server | |
| 15) Exit to command line (PowerShell) | |

Enter number to select an option:

c. Conclusive Comparison and Evaluation

Windows Server is more centralized, with more built-in features. Network sharing configuration and user access control are tightly integrated into system, with the user-friendly interaction experience. However, it is also heavy weighted and resource intensive. The combination of strengths and weaknesses makes the system suitable for beginners and enterprises.

Linux is highly modular, customizable and lightweight. System simplicity and precise system control make Linux more secure than Windows Server. However, Linux requires profound understanding of operating systems and memorizing a large number of commands, which make Linux less user-friendly. Combining advantages and disadvantages, Linux is highly compatible with servers and data centers.

Part B

1. Result screen of all test cases

a. Normal deposit

```
—— TRANSACTION LOG ——  
Thread ID       : 140316787381952  
Account Number  : S11221203  
Account Name    : Anav Chand  
Transaction Type : Deposit  
Transaction Amount : FJD 900.00  
Updated Balance : FJD 2400.00  
——
```

b. Normal withdrawal

```
—— TRANSACTION LOG ——  
Thread ID       : 140316778989248  
Account Number  : S11230430  
Account Name    : Vishant Kumar  
Transaction Type : Withdrawal  
Transaction Amount : FJD 900.00  
Updated Balance : FJD 1100.00  
——
```

c. Insufficient funds withdrawal

```
—— TRANSACTION LOG ——  
Thread ID       : 140316770596544  
Account Number  : S11234883  
Account Name    : Rohan Nandan  
Transaction Type : Withdrawal  
Transaction Amount : FJD 5000.00  
Status          : Insufficient Funds  
Updated Balance : FJD 3500.00  
——
```

d. Concurrent Withdrawal/Deposit on same account

<pre>—— TRANSACTION LOG —— Thread ID : 139676373673664 Account Number : S11221203 Account Name : Anav Chand Transaction Type : Withdrawal Transaction Amount : FJD 900.00 Updated Balance : FJD 600.00 ——</pre>	<pre>—— TRANSACTION LOG —— Thread ID : 140316787381952 Account Number : S11221203 Account Name : Anav Chand Transaction Type : Deposit Transaction Amount : FJD 900.00 Updated Balance : FJD 2400.00 ——</pre>
<pre>—— TRANSACTION LOG —— Thread ID : 139676365280960 Account Number : S11221203 Account Name : Anav Chand Transaction Type : Withdrawal Transaction Amount : FJD 200.00 Updated Balance : FJD 400.00 ——</pre>	<pre>—— TRANSACTION LOG —— Thread ID : 140316762203840 Account Number : S11221203 Account Name : Anav Chand Transaction Type : Deposit Transaction Amount : FJD 100.00 Updated Balance : FJD 2500.00 ——</pre>

2. Video and Source Code

https://drive.google.com/drive/folders/1ihnFGcsz_nNrfG5h0A6QTxcdVe3ZztVs?usp=sharing

3. POSIX threads, synchronization, and race conditions (brief explanation)

a. POSIX Threads (pthread)

POSIX threads allow a C program to execute multiple tasks in parallel. In this project, each bank transaction is handled by a separate thread so multiple users can transact at the same time.

b. Mutex Synchronization

A mutex is a lock used to ensure only one thread accesses critical shared data at a time.

- Each account has its own mutex (account.lock) to protect updates to that account balance.
- A separate global mutex protects totals (total_deposits, total_withdrawals).
- A print mutex is used to avoid mixed/garbled console logs from concurrent threads.

c. Race Condition

A race condition happens when two or more threads access and update shared data at the same time without synchronization.

Example: two threads withdraw from the same account simultaneously. Without a mutex, both could read the old balance before either writes, leading to incorrect final values.

Using `pthread_mutex_lock()` and `pthread_mutex_unlock()` prevents this by making the read-modify-write sequence atomic for each protected section.

Part C

1. $Q = 200\text{ms}$

Each process runs for at most 200ms before being preempted. The CPU cycles through the ready queue in FIFO order. Track remaining burst time for each process at each step.

Step	Process Running	Time (ms)	Duration (ms)	Remaining Burst	Queue After
1	P1	0 -> 200	200	$250-200 = 50$	P2 arrived at $t = 100$ Queue: [P2, P1(50)]
2	P2	200 -> 300	100	$100-100 = 0$	P2 finishes Queue: [P1(50)]
3	P1	300 -> 350	50	$50-50 = 0$	P1 finishes Queue: ->CPU IDLE
4	IDLE	350 -> 400	50	-	P3 arrives at $t = 400$ Queue: [P3]
5	P3	400 -> 600	200	$400-200 = 200$	P4 arrived at $t = 600$ Queue: [P3(200), P4]
6	P4	600 -> 800	200	$1000-200 = 800$	P5 arrived at $t = 700$ Queue: [P3(200),P5(600),P4(800)]
7	P3	800 -> 1000	200	$200-200 = 0$	P3 finishes Queue: [P5(600),P4(800)]
8	P5	1000 -> 1200	200	$600-200 = 400$	Queue: [P5(400),P4(800)]
9	P4	1200 -> 1400	200	$800-200 = 600$	Queue: [P5(400),P4(600)]

10	P5	1400 -> 1600	200	$400 - 200 = 200$	Queue: [P4(600),P5(200)]
11	P4	1600 -> 1800	200	$600 - 200 = 400$	Queue: [P5(200),P4(400)]
12	P5	1800 -> 2000	200	$200 - 200 = 0$	P5 finishes Queue: [P4(400)]
13	P4	2000 -> 2200	200	$400 - 200 = 200$	Queue: [P4(200)]
14	P4	2200 -> 2400	200	$200 - 200 = 0$	P4 finishes All processes

Gantt Charts

Each block represents one CPU time slice. Numbers below are time points in milliseconds.

	P1	P2	P1	IDLE	P3	P4	P3	P5	P4	P5	P4	P5	P4	P4	
0	200	300	350	400	600	800	1000	1200	1400	1600	1800	2000	2200	2400	

Calculate Turnaround Time and Waiting Time

Formulas:

Turnaround Time = Completion Time - Arrival Time

Waiting Time = Turnaround Time - Burst Time

Process	Arrival Time	Burst Time	Completion Time	Turn Around Time	Waiting Time
P1	0	250	350	350	100
P2	100	100	300	200	100
P3	400	400	1000	600	200
P4	600	1000	2400	1800	800
P5	700	600	2000	1300	700

Average Turn Around Time: $(350+200+600+1800+1300)/5 = \mathbf{850}$

Average Waiting Time: $(100+100+200+800+700)/5 = \mathbf{380}$

CPU UTILIZATION

Total Time = 2400ms

IDLE Time = 50ms (400-350)

CPU Busy Time = 2350ms

CPU Utilization = $(2350/2400)/100 = 97.9\%$

2. $Q = 500\text{ms}$

Each process runs for at most 500ms before being preempted. The CPU cycles through the ready queue in FIFO order. Track remaining burst time for each process at each step.

Step	Process Running	Time (ms)	Duration (ms)	Remaining Burst	Queue After
1	P1	0 -> 250	250	$250 - 250 = 0$	P2 arrived at $t = 100$ Queue: [P2]
2	P2	250 -> 350	100	$100 - 100 = 0$	P2 finishes Queue: []
3	IDLE	350 -> 400	50	-	P3 arrives at $t = 400$ Queue: [P3]
4	P3	400 -> 800	400	$400 - 400 = 0$	P4 arrives at $t = 600$ P5 arrives at $t = 700$ P3 finishes Queue: [P4(1000), P5(600)]
5	P4	800 -> 1300	500	$1000 - 500 = 500$	Queue: [P5(600), P4(500)]
6	P5	1300 -> 1800	500	$600 - 500 = 100$	Queue: [P4(500),P5(100)]
7	P4	1800 -> 2300	500	$500 - 500 = 0$	P4 finishes Queue: [P5(100)]
8	P5	2300 -> 2400	100	$100 - 100 = 0$	P5 finishes, All Processes

Gantt Charts

Each block represents one CPU time slice. Numbers below are time points in milliseconds.

	P1	P2	IDLE	P3	P4	P5	P4	P5
0	250	350	400	800	1300	1800	2300	2400

Calculate Turnaround Time and Waiting Time

Formulas:

Turnaround Time = Completion Time - Arrival Time

Waiting Time = Turnaround Time - Burst Time

Process	Arrival Time	Burst Time	Completion Time	Turn Around Time	Waiting Time
P1	0	250	250	250	0
P2	100	100	350	250	150
P3	400	400	800	400	0
P4	600	1000	2300	1700	700
P5	700	600	2400	1700	1100

Average Turn Around Time: $(250+250+400+1700+1700)/5 = 860$

Average Waiting Time: $(150+700+1100)/5 = 390$

CPU UTILIZATION

Total Time = 2400ms

IDLE Time = 50ms

CPU Busy Time = 2350ms

CPU Utilization = $(2350/2400)/100 = 97.9\%$

Effect of Changing Time Slice (Q) from 200 to 500 ms

For the increasing the time slice from 200ms to 500ms reduced the total number of contexts

switched from 14 steps to 8 steps. This occurs because the larger quantum allows each process to run longer before preemption and lowering CPU overhead from frequent switching.

Comparison Of Metrics

Metrics	Q=200ms	Q=500ms
Average Turnaround Time	850ms	860ms
Average Waiting Time	380ms	390ms
Number of Context Switches	14	8
CPU Utilization	97.9%	97.9%

3. Analysis

Increasing the time quantum from 200ms to 500ms reduces the number of context switches which lowers scheduling overhead, which is consistent with round-robin scheduling theory (Silberschatz, Galvin, & Gagne, 2018). However, this comes at a slight cost of performance as both average time and turnaround time is increasing (Silberschatz, Galvin, & Gagne, 2018).

Short processes (P1 and P3) benefit from the larger time quantum since they complete execution without preemption and result in zero waiting time. On the other hand, longer processes (P4 and P5) experience increased waiting time because they occupy the CPU for longer durations and delay other processes.

CPU utilization remains the same as 97.9% in both cases because the workload is entirely CPU bound and the only idle period occurs when no processes are available (350-400 ms). Therefore, CPU efficiency in this scenario.

The results demonstrate the classic trade off in Round Robin Scheduling Algorithm A smaller quantum 200ms improves fairness and responsiveness but increases context switching overhead. A larger time quantum of 500ms reduces overhead but slightly worsens waiting and turnaround time.

Overall, the 200ms quantum provides better scheduling performance for this workload while the 500ms quantum improves efficiency by reducing context switching. CPU utilization remains unchanged in both cases, indicating that time quantum mainly affects scheduling behavior than CPU usage.

References

Silberschatz, A., Galvin, P. B., & Gagne, G. (2018). *Operating system concepts* (10th ed.). Wiley.